The Road to

Gold

Mathematics for Competitions

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Preface

This book is intended to help students prepare for mathematical competitions, from middle- and high-school olympiads to undergraduate competitions. It brings together mathematical concepts, results, and problem-solving methods that are useful in approaching challenging competition problems.

The book is organized into two parts. *Mathematical Concepts* collects definitions, theorems, formulas, and other mathematical results that can be applied in competitions, from elementary foundations to more advanced theory. These are presented without proofs so that the essential statements and their conditions remain easy to consult. *Problem-Solving Strategies* is devoted to methods of proof, mathematical ideas, and techniques for solving difficult problems, including instructive proofs where appropriate.

This book is a companion to *Mathematics for Science Quiz*, which places greater emphasis on important results, memorable formulas, mathematical history, and interesting facts and connections. The two books may share fundamental definitions and theorems, but they emphasize different purposes: that volume offers broad mathematical knowledge, while this one focuses on the tools and reasoning used in mathematical competitions. Readers may consult either volume independently or use them together.

I hope this book helps readers develop mathematical insight, creativity, and perseverance, and encourages them to aim for the highest achievements in mathematical competitions.

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Part I

Mathematical Concepts

Chapter 1

Algebra

Formula 1.1. Basic Factorization Identities
For elements of a commutative ring,

$$a^2 - b^2 = (a - b)(a + b), \qquad a^3 \pm b^3 = (a \pm b)(a^2 \mp ab + b^2),$$

where the two signs in the second identity are paired consistently. More generally, for $n \geq 1$,

$$a^n - b^n = (a - b) \sum_{k=0}^{n-1} a^{n-1-k} b^k.$$

If n is odd, then

$$a^n + b^n = (a + b) \sum_{k=0}^{n-1} (-1)^k a^{n-1-k} b^k.$$

Formula 1.2. Three-Variable Cubic Identity
For a, b, c in a commutative ring,

$$a^3 + b^3 + c^3 - 3abc = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca).$$

In particular, $a + b + c = 0$ implies $a^3 + b^3 + c^3 = 3abc$.

Formula 1.3. Square and Fourth-Power Sums
For real a, b, c,

$$a^2 + b^2 + c^2 - ab - bc - ca = \frac{1}{2}((a - b)^2 + (b - c)^2 + (c - a)^2),$$

$$a^4 + b^4 + c^4 - \sum_{\text{cyc}} a^2 b^2 = \frac{1}{2}((a^2 - b^2)^2 + (b^2 - c^2)^2 + (c^2 - a^2)^2).$$

Here $\sum_{\text{cyc}} a^2 b^2 = a^2 b^2 + b^2 c^2 + c^2 a^2$.

Definition 1.4. Symmetric and Homogeneous Polynomials
A polynomial $P(x_1, \ldots, x_n)$ is *symmetric* if it is unchanged by every permutation of its variables. It is *homogeneous of degree d* if every nonzero monomial has total degree d, equivalently

$$P(tx_1, \ldots, tx_n) = t^d P(x_1, \ldots, x_n)$$

for every scalar t in an infinite coefficient field.

Formula 1.5. Elementary Symmetric Polynomials
For $x_1, \ldots, x_n$ and $0 \le k \le n$, put $e_0 = 1$ and

$$e_k = \sum_{1 \le i_1 < \cdots < i_k \le n} x_{i_1} \cdots x_{i_k}.$$

Then

$$\prod_{i=1}^{n} (t - x_i) = \sum_{k=0}^{n} (-1)^k e_k t^{n-k}.$$

Theorem 1.6. Fundamental Theorem of Symmetric Polynomials
Every symmetric polynomial in $x_1, \ldots, x_n$ over a commutative ring R can be expressed uniquely as a polynomial in $e_1, \ldots, e_n$ with coefficients in R.

Formula 1.7. Quadratic Equation
For $a, b, c \in \mathbb{C}$ with $a \neq 0$, the solutions of $ax^2 + bx + c = 0$ are

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}.$$

The discriminant is $\Delta = b^2 - 4ac$. For real coefficients, its sign determines whether there are two distinct, one repeated, or no real roots.

Formula 1.8. Two-Variable Symmetric Expressions
If $s = x + y$ and $p = xy$, then x, y are roots of $t^2 - st + p$, and

$$x^2 + y^2 = s^2 - 2p, \quad x^3 + y^3 = s^3 - 3ps, \quad (x - y)^2 = s^2 - 4p.$$

Formula 1.9. Sums of Powers of Integers
For $n \in \mathbb{N}$,

$$\sum_{k=1}^{n} k = \frac{n(n+1)}{2}, \qquad \sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6},$$

$$\sum_{k=1}^{n} k^3 = \left(\frac{n(n+1)}{2} \right)^2.$$

Definition 1.10. Polynomial, Degree, and Multiplicity
For a field F, a polynomial $f(x) = a_0 + \cdots + a_n x^n \in F[x]$ with $a_n \neq 0$ has degree n. A root $r \in F$ has multiplicity $m \ge 1$ if $(x - r)^m \mid f(x)$ but $(x - r)^{m+1} \nmid f(x)$.

Definition 1.11. Reciprocal Polynomial
For a polynomial $P(x)$ of degree n, its reciprocal polynomial is

$$P^*(x) = x^n P(1/x).$$

If $P^* = P$ or $P^* = -P$, the polynomial is called reciprocal or antireciprocal, respectively. Nonzero roots occur in reciprocal pairs for such polynomials. Substituting $t = x + x^{-1}$ reduces a reciprocal polynomial of even degree, after division by an appropriate power of x, to a polynomial in t.

Theorem 1.12. Polynomial Division and Remainder Theorems
For $f, g \in F[x]$ with $g \neq 0$, there are unique $q, r \in F[x]$ satisfying

$$f = qg + r, \qquad r = 0 \text{ or } \deg r < \deg g.$$

In particular, the remainder on division of $f(x)$ by $x - a$ is $f(a)$, and

$$f(a) = 0 \quad \Longleftrightarrow \quad (x - a) \mid f(x).$$

Definition 1.13. Rational Functions and Partial Fractions
A rational function over a field F is a quotient $P(x)/Q(x)$ with $P, Q \in F[x]$ and $Q \neq 0$. Over $\mathbb{C}$, every proper rational function has a unique partial fraction expansion

$$\frac{P(x)}{Q(x)} = \sum_{j=1}^{s} \sum_{k=1}^{m_j} \frac{c_{jk}}{(x-\alpha_j)^k},$$

where $Q(x) = a\prod_{j=1}^{s}(x-\alpha_j)^{m_j}$, the α_j are distinct, and $\deg P < \deg Q$.

Theorem 1.14. Polynomial Bézout Identity
Let F be a field and let $f, g \in F[x]$, not both zero. There are polynomials $u, v \in F[x]$ such that

$$uf + vg = \gcd(f, g),$$

where the greatest common divisor is chosen monic. In particular, f and g are coprime if and only if $uf + vg = 1$ for some $u, v \in F[x]$.

Proposition 1.15. Root-Counting Principle
A nonzero polynomial of degree d over a field has at most d distinct roots. Consequently, two polynomials of degree at most d that agree at $d+1$ distinct points are identical.

Theorem 1.16. Viète's Formulas
Let $f(x) = a_n x^n + \cdots + a_0 = a_n\prod_{i=1}^{n}(x - r_i)$ over $\mathbb{C}$, with $a_n \neq 0$, where the roots are counted with multiplicity. Then

$$e_k(r_1, \ldots, r_n) = (-1)^k \frac{a_{n-k}}{a_n} \qquad (1 \le k \le n).$$

In particular, $\sum_i r_i = -a_{n-1}/a_n$ and $\prod_i r_i = (-1)^n a_0/a_n$. For $t^3 - at^2 + bt - c$ with roots r_1, r_2, r_3, these give $r_1 + r_2 + r_3 = a$, $r_1r_2 + r_2r_3 + r_3r_1 = b$, and $r_1r_2r_3 = c$.

Formula 1.17. Newton's Identities
Let $p_k = \sum_{i=1}^{n} r_i^k$ and let e_k be the elementary symmetric polynomials in $r_1, \ldots, r_n$. For $1 \le k \le n$,

$$p_k - e_1 p_{k-1} + e_2 p_{k-2} - \cdots + (-1)^{k-1} e_{k-1} p_1 + (-1)^k k e_k = 0.$$

For $k > n$, the same identity holds with $e_j = 0$ for $j > n$.

Formula 1.18. Polynomial Discriminant
Let $P(x) = a_n\prod_{i=1}^{n}(x - r_i)$ over $\mathbb{C}$, with $a_n \neq 0$ and roots counted with multiplicity. Its discriminant is

$$\operatorname{disc}(P) = a_n^{2n-2} \prod_{1 \le i < j \le n} (r_i - r_j)^2.$$

It vanishes if and only if P has a repeated root. In particular, the discriminant of $ax^2 + bx + c$ is $b^2 - 4ac$.

Definition 1.19. Polynomial Resultant and Sylvester Matrix
For $P, Q \in F[x]$ of degrees $m, n \ge 1$, the resultant $\operatorname{Res}(P, Q)$ is the determinant of the $(m+n) \times (m+n)$ Sylvester matrix formed from the coefficient rows of $x^{n-1}P, \ldots, P, x^{m-1}Q, \ldots, Q$. If $P = a\prod_{i=1}^{m}(x - \alpha_i)$ in an algebraic closure, then

$$\operatorname{Res}(P, Q) = a^n \prod_{i=1}^{m} Q(\alpha_i).$$

Thus $\operatorname{Res}(P, Q) = 0$ precisely when P, Q have a common root in an algebraic closure, equivalently $\gcd(P, Q)$ has positive degree.

Formula 1.20. Discriminant via the Resultant
For $P \in F[x]$ of degree $n \geq 2$ and leading coefficient $a \neq 0$, where $\operatorname{char} F = 0$,

$$\operatorname{disc}(P) = (-1)^{n(n-1)/2} a^{-1} \operatorname{Res}(P, P').$$

Theorem 1.21. Repeated-Root Criterion
Let F be a field of characteristic zero and $P \in F[x]$ be nonconstant. The polynomial P has no repeated roots in an algebraic closure of F if and only if

$$\gcd(P, P') = 1.$$

More generally, a root r of multiplicity m satisfies $P^{(j)}(r) = 0$ for $0 \leq j < m$ and $P^{(m)}(r) \neq 0$.

Theorem 1.22. Sturm's Theorem
Let $P \in \mathbb{R}[x]$ be squarefree. Form $P_0 = P$, $P_1 = P'$, and recursively $P_{k+1} = -\operatorname{rem}(P_{k-1}, P_k)$, ending at the last nonzero remainder. For t where no $P_k(t)$ vanishes, let $V(t)$ be the number of sign changes in $(P_0(t), \ldots, P_r(t))$. If $a < b$ are not zeros of P, the number of distinct real zeros of P in (a, b) is $V(a) - V(b)$.

Theorem 1.23. Lagrange Interpolation
Let $x_0, \ldots, x_n$ be distinct elements of a field F. There is a unique polynomial $P \in F[x]$ of degree at most n with $P(x_i) = y_i$ for all i, namely

$$P(x) = \sum_{i=0}^{n} y_i \prod_{\substack{0 \leq j \leq n \\ j \neq i}} \frac{x - x_j}{x_i - x_j}.$$

Theorem 1.24. Rational Root Theorem
Let $f(x) = a_n x^n + \cdots + a_0 \in \mathbb{Z}[x]$ with $a_n \neq 0$. If u/v is a root with $u, v \in \mathbb{Z}$, $v > 0$, and $\gcd(u, v) = 1$, then $u \mid a_0$ and $v \mid a_n$.

Theorem 1.25. Conjugate Root Principles
For a polynomial with real coefficients, every nonreal complex root occurs together with its complex conjugate, with the same multiplicity. For a polynomial with rational coefficients, a root of the form $a + b\sqrt{d}$ with $a, b \in \mathbb{Q}$, $b \neq 0$, and $d \in \mathbb{Q}$ not a square in $\mathbb{Q}$ occurs together with $a - b\sqrt{d}$.

Theorem 1.26. Eisenstein's Irreducibility Criterion
Let $f(x) = a_n x^n + \cdots + a_0 \in \mathbb{Z}[x]$ be primitive. If a prime p satisfies

$$p \mid a_0, \ldots, a_{n-1}, \qquad p \nmid a_n, \qquad p^2 \nmid a_0,$$

then f is irreducible over $\mathbb{Q}$. The criterion can also be applied after a substitution $x \mapsto x + k$ with $k \in \mathbb{Z}$.

Theorem 1.27. Descartes' Rule of Signs
Let $f(x) \in \mathbb{R}[x]$ be a nonzero polynomial. The number of its positive real roots, counted with multiplicity, is at most the number of sign changes in the sequence of its nonzero coefficients and differs from that number by an even integer. Applying the same rule to $f(-x)$ bounds its negative real roots.

Theorem 1.28. Cauchy's Root Bound
If $f(z) = a_n z^n + \cdots + a_0$ with $a_n \neq 0$, then every complex root z satisfies

$$|z| \leq 1 + \max_{0 \leq k < n} \left| \frac{a_k}{a_n} \right|.$$

Definition 1.29. Irreducible Polynomial and Minimal Polynomial
A nonconstant $f \in F[x]$ is *irreducible over* F if it cannot be written as a product of two polynomials of positive degree in $F[x]$. If α is algebraic over F, its *minimal polynomial* $m_{\alpha,F} \in F[x]$ is the unique monic irreducible polynomial over F satisfying $m_{\alpha,F}(\alpha) = 0$.

Theorem 1.30. Fundamental Theorem of Algebra
Every nonconstant polynomial in $\mathbb{C}[x]$ has a root in $\mathbb{C}$. Equivalently, every polynomial of degree $n \geq 1$ over $\mathbb{C}$ factors into n linear factors, counted with multiplicity.

Theorem 1.31. Combinatorial Nullstellensatz, Coefficient Form
Let F be a field and let $P \in F[x_1, \ldots, x_n]$ have total degree $t_1 + \cdots + t_n$, where $t_i \in \mathbb{N}_0$. If the coefficient of $x_1^{t_1} \cdots x_n^{t_n}$ is nonzero, then for any sets $S_i \subseteq F$ with $|S_i| > t_i$, there exist $s_i \in S_i$ such that $P(s_1, \ldots, s_n) \neq 0$.

Theorem 1.32. Gauss's Lemma for Polynomials
If $f, g \in \mathbb{Z}[x]$ are primitive polynomials, then fg is primitive. In particular, a primitive polynomial in $\mathbb{Z}[x]$ is irreducible over $\mathbb{Q}$ if and only if it is irreducible over $\mathbb{Z}$.

Theorem 1.33. Irreducibility by Reduction Modulo a Prime
Let $f \in \mathbb{Z}[x]$ have positive degree and leading coefficient not divisible by a prime p. If its reduction $\overline{f} \in \mathbb{F}_p[x]$ has the same degree and is irreducible over $\mathbb{F}_p$, then f is irreducible over $\mathbb{Q}$ provided f is primitive.

Proposition 1.34. Derivative and Multiplicity of a Root
Let $f \in F[x]$, where F has characteristic 0. An element $a \in F$ is a root of multiplicity exactly m if and only if

$$f(a) = f'(a) = \cdots = f^{(m-1)}(a) = 0, \qquad f^{(m)}(a) \neq 0.$$

Theorem 1.35. Arithmetic–Geometric Mean Inequality
For $x_1, \ldots, x_n \geq 0$,

$$\frac{x_1 + \cdots + x_n}{n} \geq (x_1 \cdots x_n)^{1/n}.$$

Equality holds exactly when $x_1 = \cdots = x_n$. More generally, for positive weights w_i with $\sum_i w_i = 1$,

$$\sum_{i=1}^{n} w_i x_i \geq \prod_{i=1}^{n} x_i^{w_i},$$

with equality if and only if $x_1 = \cdots = x_n$, including the case when all $x_i = 0$.

Theorem 1.36. Cauchy–Schwarz Inequality
For real a_i, b_i,

$$\left(\sum_{i=1}^{n} a_i b_i\right)^2 \leq \left(\sum_{i=1}^{n} a_i^2\right)\left(\sum_{i=1}^{n} b_i^2\right).$$

Equality holds exactly when the two vectors are linearly dependent. For $x_i > 0$,

$$\sum_{i=1}^{n} \frac{a_i^2}{x_i} \geq \frac{(\sum_{i=1}^{n} a_i)^2}{\sum_{i=1}^{n} x_i}.$$

Theorem 1.37. Cauchy–Schwarz Inequality, Engel Form
For $x_1, \ldots, x_n \in \mathbb{R}$ and $a_1, \ldots, a_n > 0$,

$$\sum_{i=1}^{n} \frac{x_i^2}{a_i} \geq \frac{(x_1 + \cdots + x_n)^2}{a_1 + \cdots + a_n}.$$

Equality holds if and only if the ratios x_i/a_i are all equal.

Definition 1.38. Convex and Strictly Convex Functions
A function $f\colon I \to \mathbb{R}$ on an interval is convex if

$$f((1-t)x + ty) \le (1-t)f(x) + tf(y) \qquad (x, y \in I,\ 0 \le t \le 1).$$

It is strictly convex if the inequality is strict for $x \neq y$ and $0 < t < 1$. A twice differentiable function with $f'' \ge 0$ is convex; if $f'' > 0$, it is strictly convex.

Theorem 1.39. Tangent-Line Inequality for Convex Functions
If f is differentiable and convex on an open interval I, then for all $x, y \in I$,

$$f(y) \ge f(x) + f'(x)(y - x).$$

If f is strictly convex, equality holds only for $x = y$.

Theorem 1.40. Harmonic–Geometric–Arithmetic–Quadratic Means
For positive $x_1, \dots, x_n$,

$$\frac{n}{\sum_i 1/x_i} \le \left(\prod_i x_i\right)^{1/n} \le \frac{\sum_i x_i}{n} \le \sqrt{\frac{\sum_i x_i^2}{n}}.$$

Equality throughout holds exactly when all x_i are equal.

Theorem 1.41. Hölder's Inequality
Let $p, q > 1$ satisfy $1/p + 1/q = 1$. For nonnegative a_i, b_i,

$$\sum_{i=1}^{n} a_i b_i \le \left(\sum_{i=1}^{n} a_i^p\right)^{1/p} \left(\sum_{i=1}^{n} b_i^q\right)^{1/q}.$$

For nonzero sequences, equality holds exactly when (a_i^p) and (b_i^q) are proportional.

Theorem 1.42. Minkowski's Inequality
For $p \ge 1$ and real (or complex) sequences $(a_i), (b_i)$,

$$\left(\sum_i |a_i + b_i|^p\right)^{1/p} \le \left(\sum_i |a_i|^p\right)^{1/p} + \left(\sum_i |b_i|^p\right)^{1/p}.$$

Theorem 1.43. Jensen's Inequality
Let f be convex on an interval I, and let $x_i \in I$, $w_i \ge 0$, $\sum_i w_i = 1$. Then

$$f\left(\sum_i w_i x_i\right) \le \sum_i w_i f(x_i).$$

For a strictly convex f and positive weights, equality holds exactly when all x_i are equal.

Theorem 1.44. Bernoulli's Inequality
For $x \ge -1$ and real $r \ge 1$,

$$(1 + x)^r \ge 1 + rx.$$

For $0 \le r \le 1$ the inequality reverses when $x > -1$.

Theorem 1.45. Rearrangement Inequality
If $a_1 \le \cdots \le a_n$ and $b_1 \le \cdots \le b_n$ are real, then for any permutation $\sigma \in S_n$,

$$\sum_{i=1}^{n} a_i b_{n+1-i} \le \sum_{i=1}^{n} a_i b_{\sigma(i)} \le \sum_{i=1}^{n} a_i b_i.$$

Theorem 1.46. Chebyshev's Sum Inequality
If (a_i) and (b_i) are similarly sorted real sequences, then

$$\frac{1}{n}\sum_{i=1}^{n} a_i b_i \geq \left(\frac{1}{n}\sum_{i=1}^{n} a_i\right)\left(\frac{1}{n}\sum_{i=1}^{n} b_i\right).$$

The inequality reverses if the two sequences are oppositely sorted.

Theorem 1.47. Schur's Inequality
For $a, b, c \geq 0$ and $r > 0$ (or for $a, b, c > 0$ and $r = 0$),

$$\sum_{\text{cyc}} a^r (a-b)(a-c) \geq 0.$$

For $r = 1$, this becomes

$$a(a-b)(a-c) + b(b-c)(b-a) + c(c-a)(c-b) \geq 0.$$

Theorem 1.48. Nesbitt's Inequality
For $a, b, c > 0$,

$$\frac{a}{b+c} + \frac{b}{c+a} + \frac{c}{a+b} \geq \frac{3}{2}.$$

Equality holds exactly when $a = b = c$.

Theorem 1.49. Young's Inequality for Products
For $a, b \geq 0$ and $p, q > 1$ with $1/p + 1/q = 1$,

$$ab \leq \frac{a^p}{p} + \frac{b^q}{q}.$$

Equality holds exactly when $a^p = b^q$.

Definition 1.50. Majorization
For vectors $x, y \in \mathbb{R}^n$, let $x_{[1]} \geq \cdots \geq x_{[n]}$ and $y_{[1]} \geq \cdots \geq y_{[n]}$ be their coordinates in decreasing order. We say that x *majorizes* y if

$$\sum_{i=1}^{k} x_{[i]} \geq \sum_{i=1}^{k} y_{[i]} \quad (1 \leq k < n), \qquad \sum_i x_i = \sum_i y_i.$$

Theorem 1.51. Muirhead's Inequality
Let $a_1, \ldots, a_n \geq 0$. For a nonnegative integer n-tuple α, let $m_\alpha(a)$ denote the average of $a_1^{\alpha_{\sigma(1)}} \cdots a_n^{\alpha_{\sigma(n)}}$ over all $\sigma \in S_n$. If α majorizes β, then

$$m_\alpha(a) \geq m_\beta(a).$$

Theorem 1.52. Power Mean Inequality
For positive numbers $x_1, \ldots, x_n$ and real r, let

$$M_r(x) = \begin{cases} (\frac{1}{n}\sum_i x_i^r)^{1/r}, & r \neq 0, \\ (\prod_i x_i)^{1/n}, & r = 0. \end{cases}$$

If $r < s$, then $M_r(x) \leq M_s(x)$, with equality exactly when all x_i are equal.

Theorem 1.53. Maclaurin's Inequalities
For positive $x_1, \dots, x_n$, let e_k be the elementary symmetric sums and put

$$E_k = \frac{e_k}{\binom{n}{k}} \qquad (1 \le k \le n).$$

Then

$$E_1 \ge E_2^{1/2} \ge \cdots \ge E_n^{1/n},$$

with equality throughout exactly when $x_1 = \cdots = x_n$.

Theorem 1.54. Newton's Inequalities
For $x_1, \dots, x_n \ge 0$, let e_k be their elementary symmetric polynomials, with $e_0 = 1$. Then for $1 \le k < n$,

$$\left(\frac{e_k}{\binom{n}{k}}\right)^2 \ge \frac{e_{k-1}}{\binom{n}{k-1}} \frac{e_{k+1}}{\binom{n}{k+1}}.$$

In particular, the normalized symmetric means are log-concave.

Theorem 1.55. Karamata's Inequality
Let $x, y \in I^n$ for an interval I, and suppose x majorizes y. For a convex function $f\colon I \to \mathbb{R}$,

$$\sum_{i=1}^{n} f(x_i) \ge \sum_{i=1}^{n} f(y_i).$$

If f is strictly convex, equality holds exactly when x is a permutation of y.

Definition 1.56. Sequence and Linear Recurrence
A sequence (a_n) over a field F is a function from an index set such as $\mathbb{N}_0$ to F. It satisfies a linear recurrence of order k with constant coefficients if

$$a_{n+k} = c_1 a_{n+k-1} + \cdots + c_k a_n$$

for fixed $c_1, \dots, c_k \in F$, with $c_k \neq 0$.

Formula 1.57. Arithmetic and Geometric Progressions
For $n \ge 1$, an arithmetic progression $a_n = a_1 + (n-1)d$ satisfies

$$\sum_{k=1}^{n} a_k = \frac{n(a_1 + a_n)}{2}.$$

A geometric progression $a_n = a_1 r^{n-1}$ satisfies

$$\sum_{k=1}^{n} a_1 r^{k-1} = \begin{cases} a_1(1 - r^n)/(1 - r), & r \neq 1, \\ n a_1, & r = 1. \end{cases}$$

Theorem 1.58. Linear Recurrences and Characteristic Roots
Suppose a sequence over $\mathbb{C}$ satisfies

$$a_{n+k} = c_1 a_{n+k-1} + \cdots + c_k a_n$$

with $c_k \neq 0$. If the characteristic polynomial

$$P(t) = t^k - c_1 t^{k-1} - \cdots - c_k$$

has distinct roots r_i of multiplicities m_i, then every solution is of the form

$$a_n = \sum_i Q_i(n) r_i^n, \qquad \deg Q_i < m_i.$$

The constants in the polynomials Q_i are determined uniquely by k consecutive initial values.

Formula 1.59. Fibonacci Numbers and Binet's Formula
Let $F_0 = 0$, $F_1 = 1$, and $F_{n+2} = F_{n+1} + F_n$. Then

$$F_n = \frac{\varphi^n - \psi^n}{\sqrt{5}}, \qquad \varphi = \frac{1+\sqrt{5}}{2}, \quad \psi = \frac{1-\sqrt{5}}{2}.$$

For $m, n \geq 1$,

$$F_{m+n} = F_m F_{n+1} + F_{m-1} F_n, \qquad F_{n+1} F_{n-1} - F_n^2 = (-1)^n.$$

Formula 1.60. Telescoping Identities
For any sequence (a_k) and $m \leq n$,

$$\sum_{k=m}^{n} (a_{k+1} - a_k) = a_{n+1} - a_m.$$

If $a_k, a_{k+1} \neq 0$,

$$\frac{a_{k+1} - a_k}{a_k a_{k+1}} = \frac{1}{a_k} - \frac{1}{a_{k+1}}.$$

Formula 1.61. Finite Differences and Newton's Formula
For a sequence $(a_n)_{n \geq 0}$, define $(\Delta a)_n = a_{n+1} - a_n$. If $\Delta^{d+1} a = 0$, then

$$a_n = \sum_{k=0}^{d} \binom{n}{k} (\Delta^k a)_0 \qquad (n \in \mathbb{N}_0).$$

In particular, if $a_n = P(n)$ for a polynomial P of degree at most d, then $\Delta^{d+1} a = 0$.

Theorem 1.62. Integer-Valued Polynomials
Let $P \in \mathbb{Q}[x]$ have degree at most d. Then $P(k) \in \mathbb{Z}$ for every nonnegative integer k if and only if there are integers $c_0, \ldots, c_d$ such that

$$P(x) = \sum_{j=0}^{d} c_j \binom{x}{j}, \qquad c_j = \Delta^j P(0),$$

where $\Delta P(x) = P(x+1) - P(x)$. In this case $P(k) \in \mathbb{Z}$ for every $k \in \mathbb{Z}$.

Formula 1.63. First-Order Linear Recurrence
If $a_{n+1} = r a_n + b_n$ for $n \geq 0$, with fixed $r \in \mathbb{C}$, then

$$a_n = r^n a_0 + \sum_{j=0}^{n-1} r^{n-1-j} b_j \qquad (n \geq 1).$$

If $b_n = b$ is constant and $r \neq 1$, this becomes

$$a_n = r^n a_0 + b \frac{r^n - 1}{r - 1}.$$

Definition 1.64. Injective, Surjective, and Bijective Functions
A function $f \colon X \to Y$ is injective if $f(x) = f(y)$ implies $x = y$, surjective if every $y \in Y$ equals $f(x)$ for some $x \in X$, and bijective if both properties hold. A bijection has an inverse function $f^{-1} \colon Y \to X$.

Definition 1.65. Additive and Multiplicative Functions
A function $f\colon \mathbb{R} \to \mathbb{R}$ is additive if $f(x+y) = f(x)+f(y)$ for all $x, y \in \mathbb{R}$. A function $g\colon \mathbb{R} \to \mathbb{R}$ is multiplicative if $g(xy) = g(x)g(y)$ for all $x, y \in \mathbb{R}$. These notions are distinct from multiplicative arithmetic functions on $\mathbb{N}$.

Theorem 1.66. Cauchy's Additive Functional Equation
If $f\colon \mathbb{R} \to \mathbb{R}$ satisfies $f(x+y) = f(x) + f(y)$ for all $x, y \in \mathbb{R}$, then $f(q) = qf(1)$ for all $q \in \mathbb{Q}$. If, in addition, f is continuous at one point, monotone on an interval, Lebesgue measurable, or bounded on an interval of positive length, then

$$f(x) = cx \qquad (x \in \mathbb{R})$$

for some $c \in \mathbb{R}$.

Theorem 1.67. Jensen's Functional Equation
If $f\colon \mathbb{R} \to \mathbb{R}$ satisfies

$$f\left(\frac{x+y}{2}\right) = \frac{f(x)+f(y)}{2} \qquad (x, y \in \mathbb{R}),$$

then $f(x) = A(x)+b$ for an additive function $A\colon \mathbb{R} \to \mathbb{R}$ and a constant $b \in \mathbb{R}$. If f is continuous at one point, then $f(x) = cx + b$ for real b, c.

Theorem 1.68. Continuous Exponential Equation
If a continuous function $f\colon \mathbb{R} \to (0, \infty)$ satisfies

$$f(x+y) = f(x)f(y) \qquad (x, y \in \mathbb{R}),$$

then $f(x) = e^{cx}$ for some $c \in \mathbb{R}$.

Remark. In a functional equation, the domain, codomain, and any regularity assumptions are part of the statement. The same formal equation may have additional nonstandard solutions without continuity, measurability, or related hypotheses.

Theorem 1.69. Polynomial Identity Principle
If $P, Q \in F[x]$ over an infinite field F agree at every $t \in F$, then $P = Q$. More generally, a multivariate polynomial over an infinite field which vanishes at every point of F^n is the zero polynomial.

Formula 1.70. Trigonometric Addition and Multiple-Angle Identities
For real x, y,

$$\sin(x \pm y) = \sin x \cos y \pm \cos x \sin y,$$
$$\cos(x \pm y) = \cos x \cos y \mp \sin x \sin y.$$

Consequently,

$$\sin(2x) = 2\sin x \cos x, \quad \cos(2x) = 1 - 2\sin^2 x = 2\cos^2 x - 1,$$

$$\sin(3x) = 3\sin x - 4\sin^3 x, \quad \cos(3x) = 4\cos^3 x - 3\cos x.$$

Formula 1.71. Product-to-Sum and Sum-to-Product Identities
For real x, y,

$$2\cos x \cos y = \cos(x+y) + \cos(x-y),$$
$$2\sin x \sin y = \cos(x-y) - \cos(x+y),$$
$$\sin x + \sin y = 2\sin\frac{x+y}{2}\cos\frac{x-y}{2},$$
$$\cos x + \cos y = 2\cos\frac{x+y}{2}\cos\frac{x-y}{2}.$$

Definition 1.72. Chebyshev Polynomials
The Chebyshev polynomials of the first kind (T_n) are defined by

$$T_0(x) = 1, \quad T_1(x) = x, \quad T_{n+1}(x) = 2xT_n(x) - T_{n-1}(x).$$

They satisfy $T_n(\cos\theta) = \cos(n\theta)$. For $n \geq 1$, the n distinct roots of T_n are

$$\cos\frac{(2k-1)\pi}{2n}, \qquad 1 \leq k \leq n.$$

Formula 1.73. Polar Form and De Moivre's Formula
For $z = r(\cos\theta + i\sin\theta)$ with $r \geq 0$ and $n \in \mathbb{Z}$ (when $n < 0$, assume $r > 0$),

$$z^n = r^n(\cos(n\theta) + i\sin(n\theta)).$$

The nth roots of a nonzero complex number $re^{i\theta}$ are

$$r^{1/n}e^{i(\theta+2\pi k)/n}, \qquad k = 0, \ldots, n-1.$$

Definition 1.74. Roots of Unity
For $n \geq 1$, the nth roots of unity are the solutions of $z^n = 1$. They form a cyclic group of order n, generated by

$$\zeta_n = e^{2\pi i/n}.$$

An nth root of unity is *primitive* if its multiplicative order is n.

Formula 1.75. Sums of Roots of Unity
For $n \geq 2$ and any integer m,

$$\sum_{k=0}^{n-1} \zeta_n^{mk} = \begin{cases} n, & n \mid m, \\ 0, & n \nmid m. \end{cases}$$

The same identity holds for any primitive nth root in place of ζ_n.

Formula 1.76. Roots-of-Unity Filter
If $P(x) = \sum_{k=0}^{N} a_k x^k \in \mathbb{C}[x]$ and $r \in \mathbb{Z}$, then

$$\sum_{\substack{0 \leq k \leq N \\ k \equiv r \pmod{n}}} a_k = \frac{1}{n}\sum_{j=0}^{n-1} \zeta_n^{-rj} P(\zeta_n^j).$$

Theorem 1.77. Cyclotomic Factorization
For $n \geq 1$, let $\Phi_n(x)$ be the monic polynomial whose roots are the primitive nth roots of unity. Then $\Phi_n(x) \in \mathbb{Z}[x]$, $\deg\Phi_n = \varphi(n)$, and

$$x^n - 1 = \prod_{d \mid n} \Phi_d(x).$$

Each Φ_n is irreducible over $\mathbb{Q}$. Here $\varphi(n)$ denotes the number of integers $1 \leq k \leq n$ relatively prime to n.

Definition 1.78. Group and Subgroup
A group $(G, *)$ is a set with an associative binary operation, an identity e, and a two-sided inverse for each element. A subset $H \subseteq G$ is a subgroup, written $H \leq G$, if it is a group under the restricted operation. The group is *abelian* if $a * b = b * a$ for all $a, b \in G$.

Definition 1.79. Order of an Element and Cyclic Group
The order $\operatorname{ord}(g)$ of an element g of a group G is the least positive integer m for which $g^m = e$, if it exists, and is infinite otherwise. A group is cyclic if $G = \langle g \rangle$ for some $g \in G$.

Definition 1.80. Permutation and Dihedral Groups
The symmetric group S_n consists of all permutations of $[n]$, under composition. The cyclic group of order n is denoted by C_n; the dihedral group D_n is the symmetry group of a regular n-gon and has order $2n$.

Theorem 1.81. Lagrange's Theorem
If $H \leq G$ and G is finite, then

$$|G| = [G : H]|H|.$$

Consequently, $|H| \mid |G|$ and the order $\operatorname{ord}(g)$ of any $g \in G$ divides $|G|$; in particular, $g^{|G|} = e$.

Theorem 1.82. Cauchy's Theorem for Finite Groups
If a prime p divides the order of a finite group G, then G has an element of order p.

Theorem 1.83. Structure of Finite Cyclic Groups
If $G = \langle g \rangle$ has order n, then

$$\operatorname{ord}(g^k) = \frac{n}{\gcd(n, k)}.$$

For each positive divisor $d \mid n$ there is exactly one subgroup of G of order d, and G has exactly $\varphi(n)$ generators, where φ is Euler's totient function.

Definition 1.84. Homomorphism, Kernel, and Quotient Group
A group homomorphism $\varphi\colon G \to H$ satisfies $\varphi(xy) = \varphi(x)\varphi(y)$. Its kernel is $\ker\varphi = \{g \in G \mid \varphi(g) = e_H\}$. A subgroup $N \leq G$ is normal if $gNg^{-1} = N$ for all $g \in G$; in that case G/N is the quotient group of cosets.

Theorem 1.85. First Isomorphism Theorem for Groups
For any group homomorphism $\varphi\colon G \to H$,

$$G/\ker\varphi \cong \operatorname{im}\varphi.$$

Definition 1.86. Group Action, Orbit, and Stabilizer
A left action of G on X is a map $(g, x) \mapsto g \cdot x$ satisfying $e \cdot x = x$ and $(gh) \cdot x = g \cdot (h \cdot x)$. The orbit of x is $G \cdot x = \{g \cdot x \mid g \in G\}$, and the stabilizer is $\operatorname{Stab}_G(x) = \{g \in G \mid g \cdot x = x\}$.

Theorem 1.87. Orbit–Stabilizer Theorem
If a finite group G acts on a set X, then for each $x \in X$,

$$|G \cdot x| = [G : \operatorname{Stab}_G(x)] = \frac{|G|}{|\operatorname{Stab}_G(x)|}.$$

Theorem 1.88. Burnside's Lemma; Cauchy–Frobenius Formula
If a finite group G acts on a finite set X, the number of orbits is

$$|G\backslash X| = \frac{1}{|G|}\sum_{g \in G} |\operatorname{Fix}(g)|,$$

where $\operatorname{Fix}(g) = \{x \in X \mid g \cdot x = x\}$.

Theorem 1.89. Class Equation
For a finite group G, choose representatives $x_1, \ldots, x_r$ of the conjugacy classes not contained in its center $Z(G)$. Then

$$|G| = |Z(G)| + \sum_{i=1}^{r} [G : C_G(x_i)].$$

Theorem 1.90. Sylow Theorems
Let $|G| = p^k m$ where p is prime and $p \nmid m$. Then G has a subgroup of order p^k, all such subgroups are conjugate, and their number n_p satisfies

$$n_p \equiv 1 \pmod{p}, \qquad n_p \mid m.$$

Theorem 1.91. Fundamental Theorem of Finite Abelian Groups
Every finite abelian group is isomorphic to a direct product of cyclic groups of prime-power order. Equivalently, it is isomorphic to

$$C_{d_1} \times \cdots \times C_{d_r}, \qquad d_1 \mid d_2 \mid \cdots \mid d_r,$$

with the invariant factors $d_i > 1$ uniquely determined (the trivial group corresponds to an empty product).

Definition 1.92. Permutation Inversions and Sign
For a permutation $\sigma \in S_n$, its inversion number is

$$\operatorname{inv}(\sigma) = \#\{(i,j) \mid 1 \le i < j \le n,\ \sigma(i) > \sigma(j)\}.$$

Its sign is $\operatorname{sgn}(\sigma) = (-1)^{\operatorname{inv}(\sigma)}$. A permutation is even if its sign is 1, and odd otherwise.

Theorem 1.93. Sign and Cycle Decomposition
Every permutation $\sigma \in S_n$ is a product of disjoint cycles, uniquely up to their ordering and cyclic rotation. If its disjoint-cycle decomposition has c cycles, including fixed points, then

$$\operatorname{sgn}(\sigma) = (-1)^{n-c}.$$

Definition 1.94. Ring, Integral Domain, and Field
Unless stated otherwise, rings are associative with an identity and need not be commutative. An integral domain is a nonzero commutative ring with identity and without zero divisors. A field is a nonzero commutative ring in which every nonzero element is invertible.

Definition 1.95. Ideal, Quotient Ring, and Characteristic
An ideal I of a ring R means a two-sided ideal. The quotient R/I is a ring. The characteristic $\operatorname{char} R$ is the additive order of 1_R, or 0 if it has infinite additive order. Ring homomorphisms preserve identities.

Definition 1.96. Polynomial Ring and Irreducibility over a Field
For a field F, the polynomial ring $F[x]$ consists of polynomials with coefficients in F. A nonconstant polynomial $f \in F[x]$ is irreducible over F if it cannot be written as $f = gh$ with both $g, h \in F[x]$ of positive degree. Irreducibility depends on the coefficient field.

Theorem 1.97. Finite Integral Domains
Every finite integral domain is a field. A finite field has cardinality p^k for some prime p and $k \ge 1$, and for each such prime power q there is, up to isomorphism, a unique field $\mathbb{F}_q$.

Theorem 1.98. Lagrange's Theorem on Finite Subgroups of a Field
Every finite subgroup G of the multiplicative group $F^\times$ of a field F is cyclic. In particular, the multiplicative group of each finite field is cyclic.

Theorem 1.99. Multiplicative Group of a Finite Field
The group $\mathbb{F}_q^\times$ is cyclic of order $q-1$. In particular, each $a \in \mathbb{F}_q$ satisfies $a^q = a$, and for $m \in \mathbb{N}$ the equation $x^m = 1$ in $\mathbb{F}_q$ has exactly $\gcd(m, q-1)$ solutions.

Proposition 1.100. Irreducible Polynomial Quotients
For a field F and nonconstant $f \in F[x]$, the quotient ring $F[x]/(f)$ is a field if and only if f is irreducible over F. If $F = \mathbb{F}_q$ and $\deg f = d$, the quotient has q^d elements.

Theorem 1.101. Polynomial Method over Finite Fields
For a nonzero polynomial $f \in \mathbb{F}_q[x]$ of degree d, the equation $f(x) = 0$ has at most d solutions in $\mathbb{F}_q$. A polynomial function on $\mathbb{F}_q$ has a unique representative of degree at most $q - 1$ modulo $x^q - x$.

Chapter 2

Number Theory

Definition 2.1. Divisibility, Primes, and Coprimality
For $a, b \in \mathbb{Z}$, write $a \mid b$ if $b = ak$ for some $k \in \mathbb{Z}$. A prime is an integer $p > 1$ whose only positive divisors are 1 and p. Two integers a, b, not both zero, are coprime if $\gcd(a, b) = 1$.

Definition 2.2. Floor, Ceiling, and Fractional Part
For $x \in \mathbb{R}$, the floor $\lfloor x \rfloor$ is the greatest integer at most x, the ceiling $\lceil x \rceil$ is the least integer at least x, and

$$\{x\} = x - \lfloor x \rfloor \in [0, 1)$$

is its fractional part. For every $m \in \mathbb{Z}$, $\lfloor x + m \rfloor = \lfloor x \rfloor + m$ and $\lceil x + m \rceil = \lceil x \rceil + m$.

Proposition 2.3. Floor and Fractional-Part Identities
For $x \in \mathbb{R}$ and every positive integer m,

$$\lfloor x \rfloor + \lfloor -x \rfloor = \begin{cases} -1, & x \notin \mathbb{Z}, \\ 0, & x \in \mathbb{Z}, \end{cases} \qquad \lceil x \rceil = -\lfloor -x \rfloor,$$

$$\sum_{j=0}^{m-1} \left\lfloor x + \frac{j}{m} \right\rfloor = \lfloor mx \rfloor.$$

For all real x, y,

$$\lfloor x \rfloor + \lfloor y \rfloor \le \lfloor x + y \rfloor \le \lfloor x \rfloor + \lfloor y \rfloor + 1.$$

Theorem 2.4. Division Algorithm
For $a \in \mathbb{Z}$ and $b \in \mathbb{N}$, there exist unique $q, r \in \mathbb{Z}$ such that

$$a = bq + r, \qquad 0 \le r < b.$$

Proposition 2.5. Euclidean Algorithm
For integers a, b with $b \neq 0$,

$$\gcd(a, b) = \gcd(b, a \bmod |b|).$$

Repeated Euclidean division produces the greatest common divisor as the last nonzero remainder.

Theorem 2.6. Fundamental Theorem of Arithmetic
Every integer $n > 1$ has a factorization

$$n = p_1^{\alpha_1} \cdots p_k^{\alpha_k},$$

where the p_i are distinct primes and $\alpha_i \in \mathbb{N}$. It is unique up to the order of the prime factors.

Theorem 2.7. Euclid's Lemma
If a prime p divides ab, then $p \mid a$ or $p \mid b$. More generally, if $\gcd(a,b) = 1$ and $a \mid bc$, then $a \mid c$.

Theorem 2.8. Bézout's Identity
For integers a, b, not both zero, there exist $x, y \in \mathbb{Z}$ such that

$$ax + by = \gcd(a,b).$$

Moreover, the set of integer linear combinations $ax + by$ equals the set of multiples of $\gcd(a,b)$.

Formula 2.9. GCD and LCM from Prime Factorizations
For positive integers a, b and prime-power exponents $a = \prod_p p^{\alpha_p}$, $b = \prod_p p^{\beta_p}$ (allowing zero exponents),

$$\gcd(a,b) = \prod_p p^{\min(\alpha_p,\beta_p)}, \qquad \operatorname{lcm}(a,b) = \prod_p p^{\max(\alpha_p,\beta_p)}.$$

Consequently,

$$\gcd(a,b)\operatorname{lcm}(a,b) = ab.$$

Theorem 2.10. GCD of Differences of Powers
For integers $a > 1$ and $m, n \in \mathbb{N}$,

$$\gcd(a^m - 1, a^n - 1) = a^{\gcd(m,n)} - 1.$$

In particular, if $m \mid n$, then $a^m - 1 \mid a^n - 1$.

Theorem 2.11. Euclid's Theorem
There are infinitely many prime numbers.

Theorem 2.12. Bertrand's Postulate; Chebyshev's Theorem
For every integer $n > 1$, there is a prime p satisfying

$$n < p < 2n.$$

Definition 2.13. Congruence and Residue Class
For $n \in \mathbb{N}$,

$$a \equiv b \pmod{n} \iff n \mid (a - b).$$

The residue class of a modulo n is $[a]_n \in \mathbb{Z}/n\mathbb{Z}$. The group $(\mathbb{Z}/n\mathbb{Z})^\times$ consists of the classes $[a]_n$ with $\gcd(a,n) = 1$.

Proposition 2.14. Algebra of Congruences
If $a \equiv b \pmod{n}$ and $c \equiv d \pmod{n}$, then

$$a + c \equiv b + d \pmod{n}, \qquad ac \equiv bd \pmod{n}.$$

If $ac \equiv bc \pmod{n}$ and $g = \gcd(c,n)$, then $a \equiv b \pmod{n/g}$; cancellation modulo n is valid when $\gcd(c,n) = 1$.

Theorem 2.15. Linear Congruence Criterion
The congruence $ax \equiv b \pmod{n}$, with $n \geq 1$, has an integer solution if and only if $d = \gcd(a,n)$ divides b. In that case there are exactly d incongruent solutions modulo n.

Theorem 2.16. Chinese Remainder Theorem
If $m_1, \ldots, m_k \in \mathbb{N}$ are pairwise coprime, then for arbitrary integers $a_1, \ldots, a_k$, the system

$$x \equiv a_i \pmod{m_i} \qquad (1 \leq i \leq k)$$

has a unique solution modulo $M = m_1 \cdots m_k$.

Proposition 2.17. Chinese Remainder Theorem for General Moduli
For positive moduli $m_1, \dots, m_k$, the congruences $x \equiv a_i \pmod{m_i}$ are simultaneously solvable if and only if

$$a_i \equiv a_j \pmod{\gcd(m_i, m_j)} \qquad \text{for all } i, j.$$

When solvable, the solution is unique modulo $\operatorname{lcm}(m_1, \dots, m_k)$.

Formula 2.18. Decimal Divisibility Tests
For decimal digits $d_0, \dots, d_m$ and $N = \sum_{k=0}^{m} d_k 10^k$,

$$N \equiv \sum_{k=0}^{m} d_k \pmod 9, \qquad N \equiv \sum_{k=0}^{m} (-1)^k d_k \pmod{11}.$$

The last r digits determine $N \pmod{2^r}$ and $N \pmod{5^r}$.

Definition 2.19. Positional Numeral Systems
Let $b \geq 2$ be an integer. Every nonnegative integer N has a unique base-b expansion

$$N = d_0 + d_1 b + \cdots + d_k b^k, \quad 0 \leq d_i < b, \quad d_k \neq 0 \ (N > 0).$$

The base-b digit sum is $s_b(N) = \sum_{i=0}^{k} d_i$.

Formula 2.20. Digit Sums and Repeated Digits
For integers $b \geq 2$ and $N \geq 0$,

$$N \equiv s_b(N) \pmod{b-1}, \qquad N \equiv \sum_{i=0}^{k} (-1)^i d_i \pmod{b+1}.$$

The base-b integer consisting of m digits all equal to 1 is

$$1 + b + \cdots + b^{m-1} = \frac{b^m - 1}{b - 1}.$$

In base 10, the last k digits of N are determined by $N \bmod 10^k$.

Theorem 2.21. Rational and Eventually Periodic Expansions
A real number has an eventually periodic base-b expansion, for any fixed integer $b \geq 2$, if and only if it is rational. A rational number in lowest terms has a terminating base-b expansion if and only if every prime divisor of its denominator divides b.

Formula 2.22. Polynomial Congruences
If $f \in \mathbb{Z}[x]$ and $a \equiv b \pmod n$, then $f(a) \equiv f(b) \pmod n$. In particular, the integral roots of $f(x) = 0$ satisfy every reduction of that equation modulo n.

Theorem 2.23. Wilson's Theorem
An integer $n > 1$ is prime if and only if

$$(n-1)! \equiv -1 \pmod n.$$

Proposition 2.24. Units and Inverses Modulo n
An integer a has a multiplicative inverse modulo n if and only if $\gcd(a, n) = 1$. *If integers u, v satisfy $au + nv = 1$, then u represents an inverse of a modulo n.*

Definition 2.25. Arithmetic and Multiplicative Functions
An arithmetic function is a map $f \colon \mathbb{N} \to \mathbb{C}$. It is multiplicative if $f(1) = 1$ and $f(mn) = f(m)f(n)$ whenever $\gcd(m, n) = 1$. It is completely multiplicative if this identity holds for all $m, n \in \mathbb{N}$.

Definition 2.26. Euler's Totient Function
The Euler totient function is

$$\varphi(n) = |\{1 \le k \le n \mid \gcd(k,n) = 1\}|.$$

It equals the order of the multiplicative group $(\mathbb{Z}/n\mathbb{Z})^\times$.

Formula 2.27. Euler's Totient Formula
For $n = \prod_{i=1}^{r} p_i^{\alpha_i}$ with $\alpha_i \ge 1$,

$$\varphi(n) = n \prod_{p \mid n} \left(1 - \frac{1}{p}\right) = \prod_{i=1}^{r} p_i^{\alpha_i - 1}(p_i - 1).$$

The empty product gives $\varphi(1) = 1$. Also

$$\sum_{d \mid n} \varphi(d) = n.$$

Formula 2.28. Divisor Functions
For $n = \prod_{i=1}^{r} p_i^{\alpha_i}$, the number $\tau(n)$ and sum $\sigma(n)$ of positive divisors satisfy

$$\tau(n) = \prod_{i=1}^{r} (\alpha_i + 1), \qquad \sigma(n) = \prod_{i=1}^{r} \frac{p_i^{\alpha_i+1} - 1}{p_i - 1}.$$

More generally, $\sigma_k(n) = \sum_{d \mid n} d^k = \prod_i (1 + p_i^k + \cdots + p_i^{k\alpha_i})$ for $k \in \mathbb{N}_0$.

Definition 2.29. Möbius Function
The Möbius function $\mu \colon \mathbb{N} \to \{-1, 0, 1\}$ is given by

$$\mu(n) = \begin{cases} 1, & n = 1, \\ (-1)^r, & n \text{ is a product of } r \text{ distinct primes}, \\ 0, & n \text{ is divisible by a square greater than 1.} \end{cases}$$

Theorem 2.30. Möbius Inversion Formula
For arithmetic functions f, g, if

$$g(n) = \sum_{d \mid n} f(d),$$

then

$$f(n) = \sum_{d \mid n} \mu(d) g(n/d) = \sum_{d \mid n} \mu(n/d) g(d).$$

In particular,

$$\sum_{d \mid n} \mu(d) = \begin{cases} 1, & n = 1, \\ 0, & n > 1. \end{cases}$$

Formula 2.31. GCD Sum and Totient Identity
For every $n \in \mathbb{N}$,

$$\sum_{k=1}^{n} \gcd(k,n) = \sum_{d \mid n} d\, \varphi(n/d).$$

More generally, grouping integers by their gcd with n gives

$$\#\{1 \le k \le n \mid \gcd(k,n) = d\} = \varphi(n/d) \qquad (d \mid n).$$

Definition 2.32. Dirichlet Convolution
For arithmetic functions f, g, their Dirichlet convolution is

$$(f * g)(n) = \sum_{d|n} f(d)g(n/d).$$

It is associative and commutative; the identity is $\varepsilon(1) = 1$, $\varepsilon(n) = 0$ for $n > 1$. The constant function $\mathbf{1}(n) = 1$ satisfies $\mu * \mathbf{1} = \varepsilon$.

Proposition 2.33. Multiplicative Arithmetic Functions
If f, g are multiplicative arithmetic functions, then their Dirichlet convolution $f*g$ is multiplicative. Moreover, τ, σ_k, φ, and μ are multiplicative.

Theorem 2.34. Fermat's Little Theorem
If p is prime and $p \nmid a$, then

$$a^{p-1} \equiv 1 \pmod{p}.$$

Equivalently, $a^p \equiv a \pmod{p}$ for every integer a.

Theorem 2.35. Euler's Theorem
For $n \in \mathbb{N}$ and $\gcd(a, n) = 1$,

$$a^{\varphi(n)} \equiv 1 \pmod{n}.$$

Theorem 2.36. Even Perfect Numbers; Euclid–Euler Theorem
A positive even integer is perfect if and only if it has the form

$$2^{p-1}(2^p - 1),$$

where p and $2^p - 1$ are both prime.

Definition 2.37. Multiplicative Order
For $\gcd(a, n) = 1$, the multiplicative order $\mathrm{ord}_n(a)$ is the least positive integer r such that $a^r \equiv 1 \pmod{n}$. It divides $\varphi(n)$, and

$$a^k \equiv 1 \pmod{n} \quad \Longleftrightarrow \quad \mathrm{ord}_n(a) \mid k$$

for $k \in \mathbb{Z}$.

Proposition 2.38. Repetend Length and Multiplicative Order
Let a/b be a reduced rational number with $b > 1$ and $\gcd(b, B) = 1$ for an integer base $B \geq 2$. Its base-B fractional expansion is purely periodic when $0 \leq a < b$, and its least positive period length is $\mathrm{ord}_b(B)$.

Definition 2.39. Primitive Root
An integer g is a primitive root modulo n if $\gcd(g, n) = 1$ and

$$\mathrm{ord}_n(g) = \varphi(n).$$

Equivalently, $[g]_n$ generates $(\mathbb{Z}/n\mathbb{Z})^\times$.

Theorem 2.40. Primitive Root Theorem
For $n > 1$, primitive roots modulo n exist exactly when

$$n \in \{2, 4, p^k, 2p^k\},$$

where p is an odd prime and $k \in \mathbb{N}$.

Proposition 2.41. Power Congruences in a Cyclic Group
Suppose $(\mathbb{Z}/n\mathbb{Z})^\times$ is cyclic of order $N = \varphi(n)$. For an integer $k \geq 1$ and $\gcd(a, n) = 1$, the congruence

$$x^k \equiv a \pmod{n}$$

has solutions if and only if $a^{N/d} \equiv 1 \pmod{n}$, where $d = \gcd(k, N)$. When solvable, it has exactly d solutions modulo n.

Definition 2.42. Carmichael Function
For $n \geq 1$, the Carmichael function $\lambda(n)$ is the exponent of $(\mathbb{Z}/n\mathbb{Z})^\times$: the least positive integer m such that $a^m \equiv 1 \pmod{n}$ for all a coprime to n. For odd primes p,

$$\lambda(p^k) = p^{k-1}(p-1),$$

while $\lambda(2) = 1$, $\lambda(4) = 2$, and $\lambda(2^k) = 2^{k-2}$ for $k \geq 3$. For pairwise coprime a, b, $\lambda(ab) = \operatorname{lcm}(\lambda(a), \lambda(b))$.

Theorem 2.43. Primitive Root Test
Let p be an odd prime and $g \in \mathbb{Z}$ with $p \nmid g$. Then g is a primitive root modulo p if and only if

$$g^{(p-1)/\ell} \not\equiv 1 \pmod{p}$$

for every distinct prime divisor ℓ of $p - 1$.

Proposition 2.44. The Group of Units Modulo a Power of Two
For $k \geq 3$,

$$(\mathbb{Z}/2^k\mathbb{Z})^\times \cong C_2 \times C_{2^{k-2}}.$$

In particular, there is no primitive root modulo 2^k when $k \geq 3$.

Definition 2.45. p-Adic Valuation
For a prime p and nonzero integer n, $v_p(n)$ is the largest $r \in \mathbb{N}_0$ such that $p^r \mid n$. For nonzero integers a, b,

$$v_p(ab) = v_p(a) + v_p(b), \qquad v_p(a+b) \geq \min(v_p(a), v_p(b))$$

when $a + b \neq 0$. The second inequality is an equality if $v_p(a) \neq v_p(b)$.

Proposition 2.46. Valuation of a Product and a Sum
Let p be prime and $x, y \in \mathbb{Z} \setminus \{0\}$. Then

$$v_p(xy) = v_p(x) + v_p(y), \qquad v_p(x+y) \geq \min(v_p(x), v_p(y))$$

when $x + y \neq 0$. If $v_p(x) \neq v_p(y)$, equality holds in the second inequality.

Theorem 2.47. Legendre's Formula
For a prime p and $n \in \mathbb{N}$,

$$v_p(n!) = \sum_{k \geq 1} \left\lfloor \frac{n}{p^k} \right\rfloor = \frac{n - s_p(n)}{p-1},$$

where $s_p(n)$ is the sum of the base-p digits of n.

Formula 2.48. Legendre's Formula via Digit Sums
For a prime p and a nonnegative integer n,

$$v_p(n!) = \sum_{j \geq 1} \left\lfloor \frac{n}{p^j} \right\rfloor = \frac{n - s_p(n)}{p-1}.$$

Consequently, for nonnegative $r \leq n$,

$$v_p\binom{n}{r} = \frac{s_p(r) + s_p(n-r) - s_p(n)}{p-1}.$$

Theorem 2.49. Lifting the Exponent Lemma; LTE
Let p be an odd prime, $n \in \mathbb{N}$, and $x, y \in \mathbb{Z}$ with $x \neq y$, $p \mid x - y$, and $p \nmid xy$. Then

$$v_p(x^n - y^n) = v_p(x - y) + v_p(n).$$

If instead $p \mid x + y$, $p \nmid xy$, n is odd, and $x^n + y^n \neq 0$, then

$$v_p(x^n + y^n) = v_p(x + y) + v_p(n).$$

For distinct odd integers x, y and even n,

$$v_2(x^n - y^n) = v_2(x - y) + v_2(x + y) + v_2(n) - 1.$$

For odd n, $v_2(x^n - y^n) = v_2(x - y)$.

Theorem 2.50. Lucas's Theorem
Let p be prime and write $n = \sum_{i \geq 0} n_i p^i$, $k = \sum_{i \geq 0} k_i p^i$, with $0 \leq n_i, k_i < p$. For $0 \leq k \leq n$,

$$\binom{n}{k} \equiv \prod_{i \geq 0} \binom{n_i}{k_i} \pmod{p},$$

where $\binom{a}{b} = 0$ if $b > a$.

Theorem 2.51. Kummer's Theorem
For a prime p and integers $0 \leq k \leq n$, the valuation $v_p\left(\binom{n}{k}\right)$ equals the number of carries when k and $n - k$ are added in base p.

Formula 2.52. Valuation of a Binomial Coefficient
For a prime p and $0 \leq k \leq n$,

$$v_p\left(\binom{n}{k}\right) = \sum_{r \geq 1} \left(\left\lfloor \frac{n}{p^r} \right\rfloor - \left\lfloor \frac{k}{p^r} \right\rfloor - \left\lfloor \frac{n-k}{p^r} \right\rfloor \right).$$

Formula 2.53. Powers of Two in Factorials
For $n \in \mathbb{N}$,

$$v_2(n!) = n - s_2(n),$$

where $s_2(n)$ is the number of ones in the binary expansion of n.

Proposition 2.54. Valuations of Products
For nonzero integers $a_1, \ldots, a_m$,

$$v_p\left(\prod_{i=1}^{m} a_i\right) = \sum_{i=1}^{m} v_p(a_i).$$

For nonzero integers a, b,

$$v_p(\gcd(a, b)) = \min(v_p(a), v_p(b)),$$

where $\gcd(a, b)$ is positive.

Definition 2.55. Quadratic Residue and Legendre Symbol
For an odd prime p and $a \in \mathbb{Z}$, a is a quadratic residue modulo p if $x^2 \equiv a \pmod{p}$ for some integer x. The Legendre symbol is

$$\left(\frac{a}{p}\right) = \begin{cases} 0, & p \mid a, \\ 1, & a \text{ is a nonzero quadratic residue modulo } p, \\ -1, & \text{otherwise.} \end{cases}$$

Theorem 2.56. Euler's Criterion
For an odd prime p and any integer a,

$$a^{(p-1)/2} \equiv \left(\frac{a}{p}\right) \pmod{p}.$$

If $p \nmid a$, there are exactly two solutions modulo p to $x^2 \equiv a \pmod{p}$ when the Legendre symbol is 1, and none when it is -1.

Proposition 2.57. Multiplicativity of the Legendre Symbol
For an odd prime p and integers a, b,

$$\left(\frac{ab}{p}\right) = \left(\frac{a}{p}\right)\left(\frac{b}{p}\right).$$

Theorem 2.58. Gauss's Lemma for Quadratic Residues
Let p be an odd prime, $a \in \mathbb{Z}$ with $p \nmid a$, and let m be the number of integers $k \in \{1, \ldots, (p-1)/2\}$ for which the least positive residue of ak modulo p exceeds $p/2$. Then

$$\left(\frac{a}{p}\right) = (-1)^m.$$

Theorem 2.59. Law of Quadratic Reciprocity
For distinct odd primes p, q,

$$\left(\frac{p}{q}\right)\left(\frac{q}{p}\right) = (-1)^{(p-1)(q-1)/4}.$$

The supplementary laws are

$$\left(\frac{-1}{p}\right) = (-1)^{(p-1)/2}, \qquad \left(\frac{2}{p}\right) = (-1)^{(p^2-1)/8}.$$

Definition 2.60. Jacobi Symbol
For a positive odd integer $n = \prod_i p_i^{\alpha_i}$ and $a \in \mathbb{Z}$, define

$$\left(\frac{a}{n}\right) = \prod_i \left(\frac{a}{p_i}\right)^{\alpha_i}.$$

By convention $\left(\frac{a}{1}\right) = 1$. For composite n, Jacobi symbol 1 does not necessarily imply that a is a quadratic residue modulo n.

Proposition 2.61. Jacobi Symbol Identities
For odd positive integers m, n and integers a, b,

$$\left(\frac{ab}{n}\right) = \left(\frac{a}{n}\right)\left(\frac{b}{n}\right), \quad \left(\frac{a}{mn}\right) = \left(\frac{a}{m}\right)\left(\frac{a}{n}\right).$$

If $\gcd(a, n) = 1$, then $\left(\frac{a}{n}\right) \in \{-1, 1\}$; otherwise it is 0. For odd n,

$$\left(\frac{-1}{n}\right) = (-1)^{(n-1)/2}, \qquad \left(\frac{2}{n}\right) = (-1)^{(n^2-1)/8}.$$

For coprime odd positive integers m, n,

$$\left(\frac{m}{n}\right)\left(\frac{n}{m}\right) = (-1)^{\frac{m-1}{2}\frac{n-1}{2}}.$$

Proposition 2.62. Solutions of $x^2 \equiv 1$ Modulo Prime Powers
For an odd prime p and $k \geq 1$, the solutions of $x^2 \equiv 1 \pmod{p^k}$ are exactly $x \equiv \pm 1 \pmod{p^k}$. Modulo 2^k, there is one solution for $k = 1$, two for $k = 2$, and four for $k \geq 3$.

Theorem 2.63. Quadratic Residues of -1 and 2
For an odd prime p,

$$-1 \text{ is a quadratic residue modulo } p \iff p \equiv 1 \pmod{4},$$

$$2 \text{ is a quadratic residue modulo } p \iff p \equiv \pm 1 \pmod{8}.$$

Proposition 2.64. Square and Cube Valuation Criteria
An integer $N \geq 1$ is a perfect square if and only if $v_p(N)$ is even for every prime p. More generally, N is a positive perfect kth power if and only if $k \mid v_p(N)$ for every prime p. In particular, every integer square is congruent to 0 or 1 modulo 4, and to 0, 1, or 4 modulo 8.

Definition 2.65. Diophantine Equation
A Diophantine equation is an equation in which solutions are required to be integers (or, when explicitly specified, natural numbers or rational numbers). The domain restriction is part of the problem.

Theorem 2.66. Linear Diophantine Equations
Let $a, b, c \in \mathbb{Z}$, with a, b not both zero. The equation $ax + by = c$ has integer solutions if and only if $d = \gcd(a, b)$ divides c. If (x_0, y_0) is one solution, all solutions are

$$x = x_0 + \frac{b}{d}t, \qquad y = y_0 - \frac{a}{d}t, \qquad t \in \mathbb{Z}.$$

Theorem 2.67. Primitive Pythagorean Triples
All triples of positive integers a, b, c satisfying $a^2 + b^2 = c^2$ and $\gcd(a, b, c) = 1$ arise, after possibly interchanging a, b, as

$$a = m^2 - n^2, \quad b = 2mn, \quad c = m^2 + n^2,$$

where $m > n \geq 1$, $\gcd(m, n) = 1$, and m, n have opposite parity.

Theorem 2.68. Frobenius Coin Theorem for Two Denominations
If $a, b \geq 2$ are coprime integers, the greatest nonnegative integer not expressible as $ax + by$ with $x, y \in \mathbb{N}_0$ is

$$ab - a - b.$$

The number of nonnegative integers not expressible in this form is $(a - 1)(b - 1)/2$.

Theorem 2.69. Fermat's Two-Square Theorem
An odd prime p can be written as a sum of two integer squares if and only if $p \equiv 1 \pmod{4}$. The prime $2 = 1^2 + 1^2$ is also such a sum.

Theorem 2.70. Sum of Two Squares Criterion
A positive integer n is a sum of two integer squares if and only if every prime $p \equiv 3 \pmod{4}$ appears to an even exponent in the prime factorization of n.

Formula 2.71. Brahmagupta–Fibonacci Identity
For integers a, b, c, d,

$$(a^2 + b^2)(c^2 + d^2) = (ac - bd)^2 + (ad + bc)^2.$$

The companion expression $(ac + bd)^2 + (ad - bc)^2$ has the same value.

Theorem 2.72. Lagrange's Four-Square Theorem
Every nonnegative integer is a sum of four integer squares.

Theorem 2.73. Legendre's Three-Square Theorem
A nonnegative integer n is a sum of three integer squares if and only if it is not of the form

$$n = 4^a(8b + 7), \qquad a, b \in \mathbb{N}_0.$$

Proposition 2.74. Solutions of Homogeneous Linear Diophantine Equations
If $\gcd(a, b) = d$ and a, b are not both zero, then every integer solution of $ax + by = 0$ has the form

$$(x, y) = \left(\frac{b}{d}t, -\frac{a}{d}t\right), \qquad t \in \mathbb{Z}.$$

Theorem 2.75. Zsigmondy's Theorem, Integer Form
Let $a > b \geq 1$ be coprime integers and $n > 1$. Except when $n = 2$ and $a + b$ is a power of 2, or when $(a, b, n) = (2, 1, 6)$, the integer $a^n - b^n$ has a prime divisor which divides none of $a^k - b^k$ for $1 \leq k < n$.

Theorem 2.76. Dirichlet's Approximation Theorem
For any irrational $\alpha \in \mathbb{R}$ and integer $N \geq 1$, there exist integers p and q with $1 \leq q \leq N$ such that

$$|q\alpha - p| < \frac{1}{N}.$$

In particular, infinitely many reduced fractions p/q satisfy $|\alpha - p/q| < 1/q^2$.

Definition 2.77. Farey Sequences
The Farey sequence $\mathcal{F}_N$ lists in increasing order all reduced fractions a/b with $0 \leq a \leq b \leq N$. Consecutive terms $a/b < c/d$ satisfy $bc - ad = 1$. Conversely, reduced fractions $a/b < c/d$ are neighbors in $\mathcal{F}_N$ if and only if $bc - ad = 1$ and $b + d > N$.

Definition 2.78. Simple Continued Fraction
For integers $a_0 \in \mathbb{Z}$, $a_i \in \mathbb{N}$ for $i \geq 1$, the simple continued fraction

$$[a_0; a_1, \ldots, a_n] = a_0 + \cfrac{1}{a_1 + \cfrac{1}{a_2 + \cdots + \cfrac{1}{a_n}}}$$

has a reduced form p_n/q_n with $q_n > 0$, called its nth convergent.

Proposition 2.79. Convergent Recurrences
With initial values $p_{-2} = 0$, $p_{-1} = 1$, $q_{-2} = 1$, $q_{-1} = 0$, the convergents satisfy

$$p_n = a_n p_{n-1} + p_{n-2}, \qquad q_n = a_n q_{n-1} + q_{n-2},$$

and

$$p_n q_{n-1} - p_{n-1} q_n = (-1)^{n-1}.$$

Theorem 2.80. Approximation by Continued Fractions
If $\alpha \in \mathbb{R} \setminus \mathbb{Q}$ has convergents p_n/q_n, then

$$\left|\alpha - \frac{p_n}{q_n}\right| < \frac{1}{q_n^2}.$$

Theorem 2.81. Legendre's Criterion for Convergents
Let α be irrational and p/q a reduced rational number with $q > 0$. If

$$\left|\alpha - \frac{p}{q}\right| < \frac{1}{2q^2},$$

then p/q is a convergent of the simple continued fraction of α.

Theorem 2.82. Best Approximation by Convergents
Let p_n/q_n be a convergent of an irrational number α. For integers p, q with $0 < q < q_{n+1}$, one has

$$|q_n\alpha - p_n| \leq |q\alpha - p|,$$

with equality only when $(p, q) = (p_n, q_n)$ up to the possible sign convention for q. In particular, convergents are best approximations of the second kind.

Definition 2.83. Pell Equation
For a fixed positive nonsquare integer D, Pell's equation is

$$x^2 - Dy^2 = 1, \qquad x, y \in \mathbb{Z}.$$

Theorem 2.84. Solutions of Pell's Equation
For every positive nonsquare integer D, there is a least solution (x_1, y_1) in positive integers to $x^2 - Dy^2 = 1$ with $x_1 > 1$. Every positive-integer solution (x_n, y_n) is determined by

$$x_n + y_n\sqrt{D} = (x_1 + y_1\sqrt{D})^n, \qquad n \in \mathbb{N}.$$

Theorem 2.85. Negative Pell Equation
For a positive nonsquare integer D, the equation

$$x^2 - Dy^2 = -1$$

has an integer solution if and only if the period length of the simple continued fraction of $\sqrt{D}$ is odd. If a least positive solution (x_1, y_1) exists, all positive solutions are given by

$$x + y\sqrt{D} = (x_1 + y_1\sqrt{D})^{2k+1}, \qquad k \in \mathbb{N}_0.$$

Theorem 2.86. Periodicity of Continued Fractions of Quadratic Irrationals
A real irrational number has an eventually periodic simple continued fraction if and only if it is a quadratic irrational, that is, a root of an irreducible quadratic polynomial over $\mathbb{Q}$.

Definition 2.87. Cyclotomic Polynomial
For $n \geq 1$, the nth cyclotomic polynomial is

$$\Phi_n(x) = \prod_{\substack{1 \leq k \leq n \\ \gcd(k,n)=1}} \left(x - e^{2\pi ik/n}\right).$$

It has integer coefficients, is monic of degree $\varphi(n)$, and satisfies

$$x^n - 1 = \prod_{d|n} \Phi_d(x).$$

For a prime p, $\Phi_p(x) = 1 + x + \cdots + x^{p-1}$.

Definition 2.88. Gaussian Integers and Norm
The Gaussian integers form the ring $\mathbb{Z}[i] = \{a + bi \mid a, b \in \mathbb{Z}\}$. The norm is

$$N(a + bi) = a^2 + b^2 = (a + bi)(a - bi).$$

It is multiplicative: $N(zw) = N(z)N(w)$. The units of $\mathbb{Z}[i]$ are $\pm 1, \pm i$.

Theorem 2.89. Unique Factorization in Gaussian Integers
The ring $\mathbb{Z}[i]$ is a Euclidean domain with respect to its norm and hence a unique factorization domain. Every nonzero nonunit factors into Gaussian primes, uniquely up to order and multiplication by units.

Theorem 2.90. Finite Fields and Their Multiplicative Groups
For each prime power q, there is up to isomorphism a unique field $\mathbb{F}_q$ with q elements. Its nonzero elements form a cyclic group of order $q - 1$, so every $a \in \mathbb{F}_q$ satisfies $a^q = a$.

Theorem 2.91. Hensel's Lemma, Simple-Root Form
Let p be prime and $f \in \mathbb{Z}[x]$. If an integer a satisfies

$$f(a) \equiv 0 \pmod{p}, \qquad f'(a) \not\equiv 0 \pmod{p},$$

then for every $k \in \mathbb{N}$ there exists a unique residue class $x_k \pmod{p^k}$ with $x_k \equiv a \pmod{p}$ and $f(x_k) \equiv 0 \pmod{p^k}$.

Chapter 3

Geometry

Definition 3.1. Segments, Angles, and Directed Angles
For points A, B, write $\overline{AB}$ for the segment and AB for its length. The angle $\angle ABC$ has vertex B. For lines ℓ, m in the oriented Euclidean plane, $\angle(\ell, m)$ denotes their directed angle modulo π. In angle-chasing arguments, directed angles eliminate the need to distinguish many configurations involving interior and exterior angles.

Theorem 3.2. Triangle Inequality
The side lengths $a, b, c > 0$ of a nondegenerate triangle satisfy

$$a < b + c, \qquad b < c + a, \qquad c < a + b.$$

Conversely, any three positive numbers satisfying these inequalities are side lengths of a nondegenerate Euclidean triangle.

Theorem 3.3. Triangle Angle Sum and Exterior Angle Theorem
The interior angles of a nondegenerate Euclidean triangle satisfy

$$A + B + C = 180^\circ.$$

An exterior angle of a triangle equals the sum of its two remote interior angles.

Theorem 3.4. Congruence and Similarity Criteria
Two Euclidean triangles are congruent under SSS, SAS, or ASA (equivalently AAS). Two triangles are similar under AA, proportional corresponding sides (SSS similarity), or SAS similarity. For similar triangles with corresponding-side ratio $k > 0$, the ratios of their perimeters and areas are k and k^2, respectively.

Formula 3.5. Laws of Sines and Cosines
For $\triangle ABC$, write $a = BC$, $b = CA$, $c = AB$, and let R be its circumradius. Then

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R, \qquad a^2 = b^2 + c^2 - 2bc\cos A.$$

The analogous cosine-law identities follow by cyclic permutation.

Formula 3.6. Regular Polygons and Circles
For a regular n-gon ($n \geq 3$) with side length a, circumradius R, and apothem r,

$$a = 2R\sin\frac{\pi}{n}, \qquad r = R\cos\frac{\pi}{n}, \qquad \text{Area} = \frac{na^2}{4\tan(\pi/n)}.$$

A circle of radius R has circumference $2\pi R$ and area πR^2.

Formula 3.7. Triangle Area Formulas
If $\Delta = [ABC]$ and $s = (a+b+c)/2$, then

$$\Delta = \frac{1}{2}bc\sin A = rs = \frac{abc}{4R}, \qquad \Delta^2 = s(s-a)(s-b)(s-c),$$

where r is the inradius. If h_a is the altitude to the side of length a, then $\Delta = ah_a/2$.

Theorem 3.8. Angle Bisector Theorem
If the internal bisector of $\angle BAC$ meets BC at D, then

$$\frac{BD}{DC} = \frac{AB}{AC}.$$

Conversely, if D is interior to BC and satisfies this ratio, then AD bisects $\angle BAC$. The external bisector has the analogous ratio for external division, when defined.

Theorem 3.9. Stewart's Theorem
Let $D \in \overline{BC}$, with $BD = m$, $DC = n$, $BC = a = m+n$, and $AB = c$, $AC = b$, $AD = d$. Then

$$b^2m + c^2n = a(d^2 + mn).$$

Formula 3.10. Median and Angle-Bisector Lengths
If m_a is the median from A and ℓ_a is the internal angle bisector from A, then

$$m_a^2 = \frac{2b^2 + 2c^2 - a^2}{4}, \qquad \ell_a^2 = bc\left(1 - \frac{a^2}{(b+c)^2}\right).$$

Theorem 3.11. Ceva's Theorem
Let D, E, F lie in the interiors of BC, CA, AB, respectively. The cevians AD, BE, CF are concurrent if and only if

$$\frac{BD}{DC}\frac{CE}{EA}\frac{AF}{FB} = 1.$$

For points on the extended side lines, the corresponding directed-ratio formulation applies.

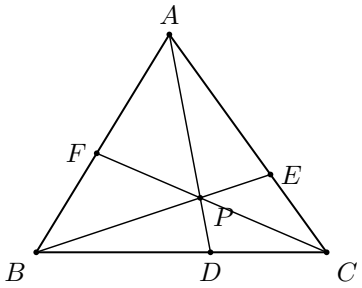


Figure 3.1: Three concurrent cevians in $\triangle ABC$.

Theorem 3.12. Menelaus' Theorem
Let D, E, F lie on the lines BC, CA, AB, respectively, with none a vertex and no two of D, E, F coinciding. Suppose an odd number of D, E, F lie outside their corresponding closed sides. Then D, E, F are collinear if and only if

$$\frac{BD}{DC}\frac{CE}{EA}\frac{AF}{FB} = 1,$$

where the numerator and denominator are ordinary (unsigned) lengths. In a collinear configuration with no vertex intersections, the number of exterior points is necessarily odd.

Theorem 3.13. Trigonometric Ceva
For interior cevians AD, BE, CF of $\triangle ABC$,

$$\frac{\sin\angle BAD}{\sin\angle DAC}\frac{\sin\angle CBE}{\sin\angle EBA}\frac{\sin\angle ACF}{\sin\angle FCB} = 1$$

if and only if AD, BE, CF are concurrent.

Definition 3.14. Pedal and Orthic Triangles
Let P be a point in the plane of a nondegenerate triangle ABC. The feet of the perpendiculars from P to the lines BC, CA, and AB form the pedal triangle of P when they are distinct and noncollinear. The pedal triangle of the orthocenter H of an acute triangle is called its orthic triangle.

Definition 3.15. Barycentric Coordinates
For a nondegenerate triangle with position vectors $\mathbf{a}, \mathbf{b}, \mathbf{c}$, a point P has barycentric coordinates $(\alpha : \beta : \gamma)$ when

$$\mathbf{p} = \frac{\alpha\mathbf{a} + \beta\mathbf{b} + \gamma\mathbf{c}}{\alpha + \beta + \gamma}, \qquad \alpha + \beta + \gamma \neq 0.$$

The coordinates are homogeneous: multiplying all three by a common nonzero scalar does not change the point.

Definition 3.16. Classical Triangle Centers
The *circumcenter* O is the intersection of the perpendicular bisectors; the *incenter* I is the intersection of the internal angle bisectors; the *centroid* G is the intersection of the medians; and the *orthocenter* H is the intersection of the altitudes. The circumcenter and orthocenter need not lie inside the triangle.

Definition 3.17. Trilinear Coordinates
For a reference triangle ABC with side lengths $a = BC$, $b = CA$, $c = AB$, the trilinear coordinates $(x : y : z)$ represent a point by its signed distances to the sidelines BC, CA, AB, up to common nonzero scale. The corresponding barycentric coordinates are

$$(ax : by : cz).$$

In particular, the incenter has trilinears $(1 : 1 : 1)$ and barycentrics $(a : b : c)$.

Theorem 3.18. Centroid Ratio
The three medians of a triangle meet at G. If D is the midpoint of BC, then $AG : GD = 2 : 1$. With position vectors $\mathbf{a}, \mathbf{b}, \mathbf{c}$ for the vertices,

$$\mathbf{g} = \frac{\mathbf{a} + \mathbf{b} + \mathbf{c}}{3}.$$

Theorem 3.19. Euler Line
For a triangle with circumcenter O, centroid G, and orthocenter H,

$$\overrightarrow{OH} = 3\overrightarrow{OG}.$$

Consequently, unless the three points coincide, they are collinear and $OG : GH = 1 : 2$.

Theorem 3.20. Orthocenter Reflection Lemma
Let H be the orthocenter of $\triangle ABC$. The reflection of H across each sideline lies on the circumcircle of $\triangle ABC$. The reflection of H across the midpoint of any side also lies on that circumcircle.

Theorem 3.21. Nine-Point Circle
The midpoints of the sides, the feet of the altitudes, and the midpoints of the segments joining the orthocenter to the vertices lie on one circle. Its center is the midpoint of OH, and its radius is $R/2$.

Formula 3.22. Euler's Incenter–Circumcenter Formula
If O, I are the circumcenter and incenter of a triangle, then

$$OI^2 = R(R - 2r).$$

In particular, $R \geq 2r$, with equality exactly for an equilateral triangle.

Definition 3.23. Excenters and Excircles
The three external angle bisectors, together with the appropriate internal bisector, determine the excenters I_a, I_b, I_c. The A-excircle has radius r_a and is tangent to BC and the extensions of AB, AC.

Formula 3.24. Exradii and Contact Lengths
For the exradii r_a, r_b, r_c of $\triangle ABC$,

$$\Delta = r_a(s - a) = r_b(s - b) = r_c(s - c).$$

If the incircle touches BC, CA, AB at D, E, F, respectively, then

$$BD = BF = s - b, \quad CD = CE = s - c, \quad AE = AF = s - a.$$

Definition 3.25. Isogonal Lines and Isogonal Conjugates
Two lines through A are *isogonal* in $\angle BAC$ if they are reflections of each other in its angle bisector. Two points P, Q not on the sidelines of $\triangle ABC$ are *isogonal conjugates* when AP, AQ are isogonal at A, and likewise at B and C.

Definition 3.26. Isotomic Conjugates
Two cevians from a vertex are isotomic if their intersections with the opposite sideline are reflections of each other about the midpoint of that side. The cevians isotomic to the three cevians through a point P are concurrent at its isotomic conjugate, when defined. In barycentric coordinates relative to ABC,

$$(x : y : z) \longmapsto \left(\frac{1}{x} : \frac{1}{y} : \frac{1}{z}\right),$$

for $xyz \neq 0$. The isogonal conjugation is similarly

$$(x : y : z) \longmapsto \left(\frac{a^2}{x} : \frac{b^2}{y} : \frac{c^2}{z}\right).$$

Theorem 3.27. Symmedian Lemma
Let $K \in BC$ and suppose AK is the isogonal of the median from A. Then

$$\frac{BK}{KC} = \frac{AB^2}{AC^2}.$$

The three symmedians are concurrent at the symmedian point (Lemoine point).

Proposition 3.28. Barycentric Coordinates of Triangle Centers
Relative to $\triangle ABC$, the centroid and incenter have barycentric coordinates

$$G = (1 : 1 : 1), \qquad I = (a : b : c),$$

where $a = BC$, $b = CA$, $c = AB$. The symmedian point has coordinates $(a^2 : b^2 : c^2)$.

Theorem 3.29. Gergonne and Nagel Points
The cevians from the vertices to the points of tangency of the incircle with the opposite sides are concurrent at the Gergonne point*. The cevians to the points of tangency of the corresponding excircles with the opposite sides are concurrent at the* Nagel point*.*

Theorem 3.30. Feuerbach's Theorem
The nine-point circle of a non-equilateral triangle is internally tangent to its incircle and externally tangent to each of its three excircles. For an equilateral triangle, the nine-point circle and incircle coincide.

Theorem 3.31. Fermat–Torricelli Point
If all angles of $\triangle ABC$ are less than 120°, there is a unique point P minimizing $PA+PB+PC$, characterized by

$$\angle APB = \angle BPC = \angle CPA = 120^\circ.$$

If one angle is at least 120°, its vertex minimizes that sum.

Formula 3.32. Half-Angle Relations in a Triangle
For $s = (a+b+c)/2$ and inradius r,

$$\tan\frac{A}{2} = \frac{r}{s-a}, \qquad \sin\frac{A}{2} = \sqrt{\frac{(s-b)(s-c)}{bc}},$$

with cyclic analogues. Moreover,

$$r = 4R\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}.$$

Theorem 3.33. Morley's Trisector Theorem
The three points obtained by intersecting pairs of adjacent internal angle trisectors of any nondegenerate triangle form an equilateral triangle.

Theorem 3.34. Napoleon's Theorem
Construct equilateral triangles externally on the three sides of an arbitrary triangle. Their centers are the vertices of an equilateral triangle. The corresponding statement also holds when all three equilateral triangles are constructed internally.

Theorem 3.35. Van Aubel's Theorem
If squares are constructed externally on the sides of a quadrilateral, the segments joining centers of squares on opposite sides have equal lengths and are perpendicular.

Theorem 3.36. Directed-Angle Criterion for Concyclicity
Let A, B, C be noncollinear points, and let D be distinct from them. The four points are concyclic if and only if

$$\angle(\overleftrightarrow{AB}, \overleftrightarrow{AC}) \equiv \angle(\overleftrightarrow{DB}, \overleftrightarrow{DC}) \pmod{\pi},$$

where the directed angles use the lines determined by the indicated pairs of points.

Theorem 3.37. Inscribed Angle and Tangent–Chord Theorems
An inscribed angle subtending an arc has half the measure of the corresponding central angle. The angle between a tangent to a circle at A and a chord AB equals the inscribed angle subtending chord AB, with directed angles interpreted modulo π.

Theorem 3.38. Cyclic Quadrilateral Criterion
A convex quadrilateral $ABCD$ is cyclic if and only if

$$\angle ABC + \angle ADC = 180^\circ.$$

Equivalently, its opposite angles are supplementary. With directed angles, cyclicity may be characterized without assuming convexity.

Formula 3.39. Equation of a Circle and Radical Axis
In Cartesian coordinates a circle with center (a, b) and radius $r > 0$ has equation

$$(x - a)^2 + (y - b)^2 = r^2.$$

For two circles written in the form

$$x^2 + y^2 + D_i x + E_i y + F_i = 0 \quad (i = 1, 2),$$

their radical axis, when the centers differ, is the line

$$(D_1 - D_2)x + (E_1 - E_2)y + F_1 - F_2 = 0.$$

This is the locus of points having equal power with respect to the two circles.

Theorem 3.40. Power of a Point
For a circle ω with center O and radius R, define

$$\mathrm{Pow}_\omega(P) = PO^2 - R^2.$$

If a line through P intersects the circle at A, B, let t_A, t_B be the signed coordinates of A, B along that line, measured from P in a common orientation. Then

$$t_A t_B = \mathrm{Pow}_\omega(P).$$

In particular, for external secants, the ordinary positive lengths satisfy $PA \cdot PB = PC \cdot PD$. For a tangent PT,

$$PT^2 = \mathrm{Pow}_\omega(P).$$

Theorem 3.41. Radical Axis and Radical Center
For two distinct nonconcentric circles ω_1, ω_2, the locus of points satisfying

$$\mathrm{Pow}_{\omega_1}(P) = \mathrm{Pow}_{\omega_2}(P)$$

is a line perpendicular to their line of centers, called the radical axis*. For three circles whose centers are not collinear, the pairwise radical axes are concurrent at their* radical center*.*

Definition 3.42. Coaxal Circles
A family of circles is coaxal if every pair of distinct nonconcentric members has the same radical axis. A circle with equation

$$x^2 + y^2 + Dx + Ey + F = 0$$

belongs to the coaxal family determined by two distinct circles $S_1 = 0$ and $S_2 = 0$ precisely when its equation can be written as $S_1 + t(S_2 - S_1) = 0$ for a real parameter t, whenever the resulting real locus is a nondegenerate circle. Any three circles in a coaxal family have a common radical axis.

Theorem 3.43. Ptolemy's Theorem and Inequality
For a convex cyclic quadrilateral $ABCD$,

$$AC \cdot BD = AB \cdot CD + BC \cdot DA.$$

For any four points in the Euclidean plane, the corresponding inequality

$$AC \cdot BD \leq AB \cdot CD + BC \cdot DA$$

holds. For four distinct vertices in convex order, equality characterizes cyclicity.

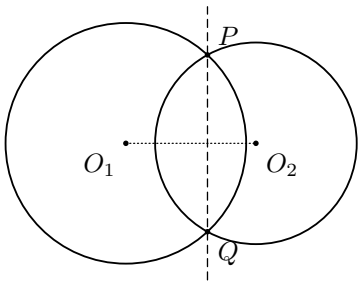


Figure 3.2: The radical axis of two intersecting circles is their common chord line.

Formula 3.44. Brahmagupta's Formula
For a convex cyclic quadrilateral with side lengths a, b, c, d and semiperimeter $s = (a+b+c+d)/2$, its area K satisfies

$$K = \sqrt{(s-a)(s-b)(s-c)(s-d)}.$$

Theorem 3.45. Pitot's Theorem
If a convex quadrilateral $ABCD$ has an incircle tangent to all four sides, then

$$AB + CD = BC + DA.$$

This equality alone does not characterize tangential quadrilaterals without additional hypotheses.

Theorem 3.46. Varignon's Theorem
The midpoints of the sides of any quadrilateral form a parallelogram. Its area is half the area of the original quadrilateral when the original quadrilateral is simple.

Theorem 3.47. Simson Line
A point P lies on the circumcircle of a triangle ABC if and only if its perpendicular projections onto the three side lines AB, BC, CA are collinear, provided the triangle is nondegenerate. Their common line is called the Simson line *of P.*

Theorem 3.48. Reim's Theorem
Let two circles meet at distinct points P, Q. Suppose a line through P meets the circles again at A, B, and a line through Q meets them again at C, D, with A, C on the first circle and B, D on the second. Then $AC \parallel BD$.

Theorem 3.49. Miquel's Theorem
Let $D \in BC$, $E \in CA$, and $F \in AB$ be points distinct from the vertices of $\triangle ABC$. The three circumcircles of AEF, BFD, and CDE pass through a common point (their Miquel point*), apart from possible pairwise shared vertices.*

Definition 3.50. Direct Similarity and Spiral Similarity
An orientation-preserving similarity of the complex plane has the form

$$z \longmapsto az + b, \qquad a \in \mathbb{C} \setminus \{0\},\ b \in \mathbb{C}.$$

If $a \neq 1$, it has a unique fixed point $z_0 = b/(1-a)$ and is a rotation followed by a homothety centered at z_0. If $a = 1$, it is a translation. The former case is commonly called a *spiral similarity.*

Definition 3.51. Homothety
For a point O and a real scalar $k \neq 0$, the homothety with center O and ratio k maps a point X to X' satisfying

$$\overrightarrow{OX'} = k\overrightarrow{OX}.$$

It multiplies all distances by $|k|$ and areas by k^2.

Definition 3.52. Inversion in a Circle
Fix a circle with center O and radius $r > 0$. Inversion sends $P \neq O$ to P' on the ray OP with $OP \cdot OP' = r^2$; it is an involution. A line through O maps to itself (with O omitted), and a line not through O maps to a circle through O (with O omitted). A circle through O maps to a line not through O; a circle not through O maps to another circle not through O.

Proposition 3.53. Inversion Distance Identity
Under inversion with center O and radius r, for $P, Q \neq O$ with images P', Q',

$$P'Q' = \frac{r^2}{OP \cdot OQ} PQ.$$

Inversion preserves the magnitude of angles between intersecting smooth curves, but reverses their orientation.

Theorem 3.54. Butterfly Theorem
Let M be the midpoint of a chord PQ of a circle. Let chords AB and CD pass through M, with A, C on one side of PQ and B, D on the other. If AD and BC meet the line PQ at X and Y, respectively, then M is the midpoint of XY.

Proposition 3.55. Circles Orthogonal to an Inversion Circle
A circle ω not passing through the center O of an inversion circle is invariant as a set under that inversion if and only if it is orthogonal to the inversion circle, meaning that their tangent lines at either intersection point are perpendicular.

Definition 3.56. Projective Plane and Homogeneous Coordinates
The real projective plane $\mathbb{RP}^2$ consists of one-dimensional linear subspaces of $\mathbb{R}^3$. A point is represented by nonzero homogeneous coordinates $[x : y : z]$, with nonzero scalar multiples identified. The affine plane embeds by $(x, y) \mapsto [x : y : 1]$, while $z = 0$ is the line at infinity. Two distinct projective lines meet at a unique projective point.

Proposition 3.57. Projective Collinearity Criterion
Three points $[x_i : y_i : z_i] \in \mathbb{RP}^2$, $i = 1, 2, 3$, are collinear if and only if

$$\det \begin{pmatrix} x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \\ x_3 & y_3 & z_3 \end{pmatrix} = 0.$$

A projective line has an equation $ax + by + cz = 0$ for some nonzero (a, b, c), determined up to a nonzero scalar.

Definition 3.58. Cross Ratio
For four distinct collinear points A, B, C, D on a projective line, their cross ratio in an affine coordinate t is

$$(A, B; C, D) = \frac{(t_C - t_A)(t_D - t_B)}{(t_C - t_B)(t_D - t_A)},$$

with the usual limiting interpretation when one point is at infinity. Projective transformations preserve the cross ratio.

Definition 3.59. Harmonic Range and Harmonic Division
Four distinct points A, B, C, D on a projective line form a harmonic range if their cross ratio satisfies

$$(A, B; C, D) = -1.$$

Equivalently, C, D are harmonic conjugates with respect to A, B.

Definition 3.60. Complete Quadrangle and Complete Quadrilateral
A complete quadrangle consists of four points with no three collinear and the six joining lines. Its three diagonal points are intersections of pairs of opposite sides. Dually, a complete quadrilateral consists of four lines with no three concurrent, their six intersections, and the three lines joining opposite intersections.

Theorem 3.61. Harmonic Property of a Complete Quadrangle
For four points A, B, C, D in general position, let $X = AB \cap CD$, $Y = AC \cap BD$, and $Z = AD \cap BC$ in the projective plane. The line YZ meets AB at T. Then the four collinear points $(A, B; X, T)$ form a harmonic range:

$$(A, B; X, T) = -1.$$

Theorem 3.62. Desargues' Theorem
In the real projective plane, if two triangles have corresponding vertices joined by three concurrent lines, then the intersections of corresponding sides are collinear. Conversely, if the three side intersections are collinear, then the lines joining corresponding vertices are concurrent in the projective plane.

Theorem 3.63. Pappus' Theorem
Let A, B, C lie on one line and A', B', C' on another, with all pairwise intersections below defined. Then the three points

$$AB' \cap A'B, \quad AC' \cap A'C, \quad BC' \cap B'C$$

are collinear in the real projective plane.

Definition 3.64. Conic in the Projective Plane
A nondegenerate projective conic is the zero set of a homogeneous quadratic form in $[x : y : z] \in \mathbb{RP}^2$ whose associated symmetric 3×3 matrix is nonsingular. In affine coordinates, ellipses, parabolas, and hyperbolas are the standard nonsingular real conics.

Definition 3.65. Pole and Polar with Respect to a Conic
Let a nondegenerate conic in the real projective plane have homogeneous equation $\mathbf{x}^{\mathrm{T}}\mathbf{Q}\mathbf{x} = 0$, where $\mathbf{Q}$ is an invertible symmetric 3×3 matrix. The polar of a point $[\mathbf{p}]$ is the line

$$\mathbf{p}^{\mathrm{T}}\mathbf{Q}\mathbf{x} = 0.$$

Every projective line has a unique pole. For a point P on the conic, its polar is its tangent. Polarity satisfies La Hire's reciprocity: P lies on the polar of Q if and only if Q lies on the polar of P.

Theorem 3.66. La Hire's Theorem
For a nondegenerate projective conic, the pole of the line through two distinct points P, Q is the intersection of the polars of P, Q, provided those polars are distinct. Dually, the polar of the intersection of two distinct lines is the line through their poles.

Theorem 3.67. Pascal's Theorem
For six distinct points A, B, C, D, E, F on a nondegenerate conic, the three intersections of opposite sides

$$AB \cap DE, \qquad BC \cap EF, \qquad CD \cap FA$$

are collinear in the projective plane, allowing points at infinity for parallel sides.

Theorem 3.68. Brianchon's Theorem
For a nondegenerate conic, let $A_1A_2A_3A_4A_5A_6$ be a hexagon whose six sides are tangent to the conic, with consecutive sides meeting at the named vertices. The three principal diagonals

$$A_1A_4, \qquad A_2A_5, \qquad A_3A_6$$

are concurrent in the projective plane.

Formula 3.69. Coordinate and Shoelace Area Formulas
For $A = (x_1, y_1)$, $B = (x_2, y_2)$, $C = (x_3, y_3)$,

$$[ABC] = \frac{1}{2}\left|\det\begin{pmatrix} x_2 - x_1 & y_2 - y_1 \\ x_3 - x_1 & y_3 - y_1 \end{pmatrix}\right|.$$

For a simple polygon with cyclically ordered vertices (x_i, y_i), $1 \le i \le n$,

$$\text{Area} = \frac{1}{2}\left|\sum_{i=1}^{n}(x_i y_{i+1} - x_{i+1} y_i)\right|, \quad (x_{n+1}, y_{n+1}) = (x_1, y_1).$$

Definition 3.70. Eccentricity and Focus–Directrix Property
A conic with focus F, directrix line ℓ, and eccentricity $e > 0$ is the locus of points P satisfying

$$PF = e\, d(P, \ell).$$

For $0 < e < 1$ it is an ellipse, for $e = 1$ a parabola, and for $e > 1$ a hyperbola, in the nondegenerate cases. The circle is the special case of an ellipse with equal semiaxes and eccentricity 0.

Formula 3.71. Standard Equations of Conic Sections
For $a > b > 0$ and $c^2 = a^2 - b^2$, an ellipse centered at the origin with major axis on the x-axis is

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, \qquad F_{\pm} = (\pm c, 0), \qquad e = \frac{c}{a}.$$

For $a, b > 0$ and $c^2 = a^2 + b^2$, the corresponding hyperbola is

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1, \qquad F_{\pm} = (\pm c, 0), \qquad e = \frac{c}{a},$$

with asymptotes $y = \pm(b/a)x$. A parabola with vertex at the origin, focus $(p, 0)$, and directrix $x = -p$, where $p \neq 0$, is $y^2 = 4px$.

Formula 3.72. Tangents to Standard Conics
For a point (x_0, y_0) on the indicated conic, its tangent line is

$$\frac{x x_0}{a^2} + \frac{y y_0}{b^2} = 1 \quad \text{for } \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1,$$

$$\frac{x x_0}{a^2} - \frac{y y_0}{b^2} = 1 \quad \text{for } \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1,$$

$$y y_0 = 2p(x + x_0) \quad \text{for } y^2 = 4px \ (p \neq 0).$$

The tangent at (x_0, y_0) to $x^2 + y^2 = r^2$ is $x x_0 + y y_0 = r^2$.

Formula 3.73. Complex Coordinates on the Unit Circle
For distinct points $a, b, c, d \in \mathbb{C}$ with $|a| = |b| = |c| = |d| = 1$,

$$\frac{a - b}{c - d} = \frac{ab}{cd}\overline{\left(\frac{a - b}{c - d}\right)}.$$

If a, b, c, d are four distinct points on the unit circle, then their complex cross ratio

$$\frac{(a - c)(b - d)}{(a - d)(b - c)}$$

is real.

Theorem 3.74. Pick's Theorem
For a simple lattice polygon in $\mathbb{R}^2$, with I lattice points in its interior and B lattice points on its boundary,

$$\text{Area} = I + \frac{B}{2} - 1.$$

Theorem 3.75. Helly's Theorem in the Plane
Let $\mathcal{F}$ be a finite family of convex subsets of $\mathbb{R}^2$ with at least three members. If every three members have a common point, then all members have a common point.

Theorem 3.76. Carathéodory's Theorem in the Plane
If x belongs to the convex hull of a set $S \subseteq \mathbb{R}^2$, then x is a convex combination of at most three points of S.

Theorem 3.77. Isoperimetric Inequality in the Plane
Let a simple closed rectifiable curve in the plane have length L and enclose area A. Then

$$L^2 \geq 4\pi A,$$

with equality if and only if the curve is a circle.

Definition 3.78. Lines and Planes in Three Dimensions
Let $\mathbf{p}, \mathbf{v}, \mathbf{n} \in \mathbb{R}^3$, with $\mathbf{v} \neq \mathbf{0}$ and $\mathbf{n} \neq \mathbf{0}$. The line through $\mathbf{p}$ with direction $\mathbf{v}$ is

$$\mathbf{x} = \mathbf{p} + t\mathbf{v} \quad (t \in \mathbb{R}),$$

while the plane through $\mathbf{p}$ with normal $\mathbf{n}$ is

$$\mathbf{n} \cdot (\mathbf{x} - \mathbf{p}) = 0.$$

Two planes are perpendicular if their normal vectors are perpendicular; they are parallel if their normal vectors are proportional.

Formula 3.79. Distances and Angles in Space
Let Π be the plane $ax + by + cz + d = 0$, with $(a, b, c) \neq (0, 0, 0)$. For $P = (x_0, y_0, z_0)$,

$$\operatorname{dist}(P, \Pi) = \frac{|ax_0 + by_0 + cz_0 + d|}{\sqrt{a^2 + b^2 + c^2}}.$$

For nonzero direction vectors $\mathbf{u}, \mathbf{v}$, the smaller angle $\theta \in [0, \pi/2]$ between their unoriented lines satisfies

$$\cos\theta = \frac{|\mathbf{u} \cdot \mathbf{v}|}{\|\mathbf{u}\|\|\mathbf{v}\|}.$$

The distance from $\mathbf{p}$ to the line $\mathbf{a} + t\mathbf{v}$ is $\|(\mathbf{p} - \mathbf{a}) \times \mathbf{v}\|/\|\mathbf{v}\|$.

Formula 3.80. Vector Products and Tetrahedron Volume
For $\mathbf{u}, \mathbf{v}, \mathbf{w} \in \mathbb{R}^3$, the area of the parallelogram spanned by $\mathbf{u}, \mathbf{v}$ is $\|\mathbf{u} \times \mathbf{v}\|$, and the volume of the corresponding parallelepiped is

$$|\mathbf{u} \cdot (\mathbf{v} \times \mathbf{w})|.$$

For tetrahedron $ABCD$ with position vectors $\mathbf{a}, \mathbf{b}, \mathbf{c}, \mathbf{d}$, its volume is

$$\frac{1}{6}\left|\det\begin{pmatrix} b_1 - a_1 & c_1 - a_1 & d_1 - a_1 \\ b_2 - a_2 & c_2 - a_2 & d_2 - a_2 \\ b_3 - a_3 & c_3 - a_3 & d_3 - a_3 \end{pmatrix}\right|.$$

Formula 3.81. Basic Solid Geometry
A right circular cylinder of radius r and height h has volume $\pi r^2 h$; a right circular cone has volume $\pi r^2 h/3$; and a ball of radius r has volume $4\pi r^3/3$ and boundary area $4\pi r^2$.

Theorem 3.82. Euler's Polyhedron Formula
For a convex polyhedron with V vertices, E edges, and F faces,

$$V - E + F = 2.$$

The same identity holds for a connected planar graph in a plane embedding when the unbounded face is included.

Theorem 3.83. Cavalieri's Principle
Two measurable plane regions having equal cross-sectional lengths at almost every level in a fixed direction have equal area, provided their areas are finite. Similarly, solids with equal cross-sectional areas at almost every height have equal volume when the volumes are finite.

Chapter 4

Combinatorics and Probability

Theorem 4.1. Addition and Multiplication Principles
If mutually exclusive classes of objects have respective cardinalities $a_1, \ldots, a_k$, their union has cardinality $a_1 + \cdots + a_k$. If an object is determined by k independent choices with a_i possibilities at step i, the number of objects is $a_1 \cdots a_k$.

Definition 4.2. Factorials, Permutations, and Combinations
For $n \in \mathbb{N}_0$, let $n! = \prod_{k=1}^{n} k$, with $0! = 1$. If $0 \leq r \leq n$, the number of ordered selections of r distinct elements from an n-element set is

$$_nP_r = \frac{n!}{(n-r)!},$$

and the number of r-element subsets is

$$\binom{n}{r} = \frac{n!}{r!(n-r)!}.$$

Formula 4.3. Multiset Combinations and Stars and Bars
For $n \geq 1$ and $r \geq 0$, the number of multisets of cardinality r chosen from n types is

$$\left(\!\binom{n}{r}\!\right) = \binom{n+r-1}{r}.$$

Equivalently, this is the number of nonnegative integer solutions of $x_1 + \cdots + x_n = r$. The number of positive integer solutions for $r \geq n$ is $\binom{r-1}{n-1}$.

Theorem 4.4. Binomial and Multinomial Theorems
For $n \in \mathbb{N}_0$,

$$(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^k y^{n-k}.$$

If $n_1, \ldots, n_m \in \mathbb{N}_0$ sum to n, the corresponding multinomial coefficient is

$$\binom{n}{n_1, \ldots, n_m} = \frac{n!}{n_1! \cdots n_m!},$$

and $(x_1 + \cdots + x_m)^n$ expands as the sum of these coefficients times $x_1^{n_1} \cdots x_m^{n_m}$.

Formula 4.5. Basic Binomial Identities
For $0 \leq k \leq n$,

$$\binom{n}{k} = \binom{n}{n-k}, \qquad \binom{n}{k} + \binom{n}{k-1} = \binom{n+1}{k},$$

where coefficients with inadmissible lower indices are zero. Furthermore,

$$\sum_{k=0}^{n}\binom{n}{k}=2^n,\quad \sum_{k=0}^{n}(-1)^k\binom{n}{k}=0\quad (n\geq 1),$$

$$\sum_{k=0}^{n}k\binom{n}{k}=n2^{n-1}\quad (n\geq 1).$$

Theorem 4.6. Vandermonde's Identity
For nonnegative integers r, s, n,

$$\sum_{k=0}^{n}\binom{r}{k}\binom{s}{n-k}=\binom{r+s}{n}.$$

In particular,

$$\sum_{k=0}^{n}\binom{n}{k}^2=\binom{2n}{n}.$$

Theorem 4.7. Binomial Inversion
For sequences (a_n) and (b_n) over a field of characteristic zero,

$$b_n=\sum_{k=0}^{n}\binom{n}{k}a_k\quad\Longleftrightarrow\quad a_n=\sum_{k=0}^{n}(-1)^{n-k}\binom{n}{k}b_k.$$

Theorem 4.8. Cauchy–Davenport Theorem
Let A, B be nonempty subsets of the cyclic group $\mathbb{Z}/p\mathbb{Z}$ for a prime p. Their sumset $A+B=\{a+b \mid a\in A,\ b\in B\}$ satisfies

$$|A+B|\geq \min(p,\ |A|+|B|-1).$$

Theorem 4.9. Pigeonhole Principle
If N objects are placed in $k\geq 1$ boxes, some box contains at least $\lceil N/k\rceil$ objects. In particular, placing $k+1$ objects in k boxes forces at least two into the same box.

Theorem 4.10. Inclusion–Exclusion Principle
For finite sets $A_1,\ldots,A_n$,

$$\left|\bigcup_{i=1}^{n}A_i\right|=\sum_{\varnothing\neq S\subseteq[n]}(-1)^{|S|+1}\left|\bigcap_{i\in S}A_i\right|.$$

Formula 4.11. Counting Surjections
For $n\geq 0$ and $k\geq 1$, the number of surjections from an n-element set to a k-element set is

$$\sum_{j=0}^{k}(-1)^j\binom{k}{j}(k-j)^n=k!\,S(n,k),$$

where $S(n,k)$ is the Stirling number of the second kind and $0^0=1$ in the finite counting sum.

Definition 4.12. Derangement
A derangement is a permutation with no fixed point. Write D_n or $!n$ for the number of derangements of n elements; here D_n is not the dihedral group, the notation being determined by context.

Formula 4.13. Derangement Formula
For $n \in \mathbb{N}_0$,

$$D_n = n! \sum_{k=0}^{n} \frac{(-1)^k}{k!}.$$

For $n \geq 2$, $D_n = (n-1)(D_{n-1} + D_{n-2})$, with $D_0 = 1$ and $D_1 = 0$. For $n \geq 1$, D_n is the nearest integer to $n!/e$.

Definition 4.14. Rook Polynomial
For a finite set B of forbidden cells on an $n \times n$ board, let $r_k(B)$ count placements of k pairwise nonattacking rooks entirely in B, with $r_0(B) = 1$. The rook polynomial is

$$R_B(x) = \sum_{k=0}^{n} r_k(B) x^k.$$

Formula 4.15. Forbidden Positions and Rook Numbers
The number of permutations $\sigma \in S_n$ whose selected cells $(i, \sigma(i))$ avoid a set B of forbidden cells is

$$\sum_{k=0}^{n} (-1)^k r_k(B)(n-k)!.$$

For the diagonal forbidden board, $r_k(B) = \binom{n}{k}$, giving the derangement formula.

Definition 4.16. Stirling Numbers and Bell Numbers
The unsigned Stirling number $c(n,k)$ counts permutations of $[n]$ with k cycles. The Stirling number $S(n,k)$ counts partitions of $[n]$ into k nonempty blocks. The Bell number

$$B_n = \sum_{k=0}^{n} S(n,k)$$

counts all set partitions of $[n]$.

Formula 4.17. Stirling Number Recurrences
For $n, k \geq 1$,

$$c(n,k) = c(n-1,k-1) + (n-1)c(n-1,k),$$

$$S(n,k) = S(n-1,k-1) + kS(n-1,k).$$

The boundary conditions are $c(0,0) = S(0,0) = 1$ and zero for inadmissible indices.

Definition 4.18. Latin Squares
A Latin square of order n is an $n \times n$ array of n symbols, each occurring exactly once in each row and each column.

Definition 4.19. Block Designs and Steiner Triple Systems
A 2-(v,k,λ) design is a v-element set together with a family of k-element blocks in which each pair of distinct points is contained in exactly λ blocks. A Steiner triple system is a 2-$(v,3,1)$ design. For $v \geq 3$, such a system exists if and only if $v \equiv 1$ or $3 \pmod 6$.

Definition 4.20. Catalan Numbers
For $n \in \mathbb{N}_0$, the nth Catalan number is

$$C_n = \frac{1}{n+1}\binom{2n}{n} = \binom{2n}{n} - \binom{2n}{n+1}.$$

It counts, among other objects, correctly matched parenthesis strings of length $2n$ and lattice paths from $(0,0)$ to (n,n) that never rise above the diagonal.

Formula 4.21. Catalan Recurrence and Generating Function
The Catalan numbers satisfy

$$C_0 = 1, \qquad C_{n+1} = \sum_{k=0}^{n} C_k C_{n-k}.$$

Their ordinary generating function is

$$C(x) = \sum_{n\geq 0} C_n x^n = \frac{1-\sqrt{1-4x}}{2x},$$

with the constant term understood by formal-power-series expansion.

Definition 4.22. Integer Partitions and Compositions
A partition of $n \in \mathbb{N}_0$ is a weakly decreasing sequence of positive integers summing to n, with the empty partition for $n = 0$. A composition of $n \in \mathbb{N}$ is an ordered sequence of positive integers summing to n. The number of compositions of n is 2^{n-1}.

Theorem 4.23. Euler's Partition Generating Function
If $p(n)$ is the number of integer partitions of n with $p(0) = 1$, then, as a formal power series,

$$\sum_{n\geq 0} p(n)x^n = \prod_{k\geq 1} \frac{1}{1-x^k}.$$

The number of partitions into distinct parts equals the number of partitions into odd parts.

Theorem 4.24. Reflection Principle for Lattice Paths
For nonnegative integers a, b with $a \geq b$, the number of lattice paths from $(0,0)$ to (a,b) using right and up steps and never going above $y = x$ is

$$\binom{a+b}{b} - \binom{a+b}{b-1} = \frac{a-b+1}{a+1}\binom{a+b}{b}.$$

For $a = b = n$, this equals C_n.

Formula 4.25. Ballot Theorem
Suppose candidate A receives a votes and candidate B receives b votes, with $a > b \geq 0$. Among all $\binom{a+b}{a}$ orders of the votes, the fraction for which A is strictly ahead of B after every vote is

$$\frac{a-b}{a+b}.$$

Consequently, the number of such orders is $\frac{a-b}{a+b}\binom{a+b}{a}$.

Definition 4.26. Recurrence Relation and Generating Function
A recurrence relation specifies sequence terms in terms of earlier terms. The ordinary generating function of $(a_n)_{n\geq 0}$ is

$$A(x) = \sum_{n\geq 0} a_n x^n,$$

viewed either formally or as an analytic series where it converges. The exponential generating function is $\sum_{n\geq 0} a_n x^n/n!$.

Formula 4.27. Geometric and Binomial Series
As formal power series, for $m \in \mathbb{N}$,

$$\frac{1}{1-x} = \sum_{n\geq 0} x^n, \qquad \frac{1}{(1-x)^m} = \sum_{n\geq 0} \binom{n+m-1}{m-1} x^n.$$

The same identities hold analytically when $|x| < 1$.

Theorem 4.28. Generating Functions for Linear Recurrences
A sequence satisfying a linear recurrence with constant coefficients for all sufficiently large indices has a rational ordinary generating function. Conversely, the coefficients of any rational formal power series satisfy a constant-coefficient linear recurrence for all sufficiently large indices.

Formula 4.29. Coefficient Extraction and Convolution
For formal power series $A(x) = \sum_{n\geq 0} a_n x^n$ and $B(x) = \sum_{n\geq 0} b_n x^n$,

$$[x^n](A(x)B(x)) = \sum_{k=0}^{n} a_k b_{n-k}.$$

More generally, $[x^n]x^k A(x) = a_{n-k}$ for $n \geq k$.

Formula 4.30. Exponential Generating Functions
For a sequence $(a_n)_{n\geq 0}$, define its exponential generating function by

$$A(x) = \sum_{n\geq 0} a_n \frac{x^n}{n!}.$$

If $B(x) = \sum_{n\geq 0} b_n x^n/n!$, then

$$n![x^n](A(x)B(x)) = \sum_{k=0}^{n} \binom{n}{k} a_k b_{n-k}.$$

This is the labeled-product convolution, in contrast with ordinary generating functions.

Theorem 4.31. Exponential Formula for Labeled Structures
Let $\mathcal{C}$ be a class of labeled connected structures, with c_n structures on an n-element label set. Suppose $\mathcal{A}$ consists of all finite unordered sets of $\mathcal{C}$-structures on pairwise disjoint nonempty label sets, including the empty set, and this decomposition is unique. If a_n counts objects of $\mathcal{A}$ on $[n]$, then, as formal power series,

$$\sum_{n\geq 0} a_n \frac{x^n}{n!} = \exp\left(\sum_{n\geq 1} c_n \frac{x^n}{n!}\right).$$

Theorem 4.32. Lagrange Inversion, Basic Form
Let $\phi(t)$ be a formal power series over a field of characteristic zero with $\phi(0) \neq 0$. If $T(x)$ is the unique formal series with $T(0) = 0$ satisfying

$$T = x\phi(T),$$

then for $n \geq 1$,

$$[x^n]T(x) = \frac{1}{n}[t^{n-1}]\phi(t)^n.$$

Theorem 4.33. Burnside's Lemma
If a finite group G acts on a finite set X, the number of orbits is

$$|G\backslash X| = \frac{1}{|G|}\sum_{g\in G} |\operatorname{Fix}(g)|,$$

where $\operatorname{Fix}(g) = \{x \in X \mid g \cdot x = x\}$.

Definition 4.34. Cycle Index of a Permutation Group
Let a finite group G act on n positions. Write $c_j(g)$ for the number of cycles of length j in the permutation induced by $g \in G$. The cycle index polynomial is

$$Z_G(z_1, \dots, z_n) = \frac{1}{|G|} \sum_{g \in G} \prod_{j=1}^{n} z_j^{c_j(g)}.$$

In particular, $\sum_{j=1}^{n} j c_j(g) = n$.

Theorem 4.35. Pólya Enumeration Theorem
Let a finite group G permute an n-element set, and suppose each position receives one of $q \in \mathbb{N}$ colors. If $c(g)$ denotes the number of cycles of g in its permutation of the positions, the number of colorings up to G-symmetry is

$$\frac{1}{|G|} \sum_{g \in G} q^{c(g)}.$$

Proposition 4.36. Weighted Pólya Enumeration
Let G act on n positions, with cycle-index polynomial Z_G. If the available colors have respective weights $w_1, \dots, w_q$, the generating polynomial of colorings up to G-symmetry, weighted by color multiplicities, is

$$Z_G\left(\sum_{i=1}^{q} w_i,\ \sum_{i=1}^{q} w_i^2,\ \dots,\ \sum_{i=1}^{q} w_i^n \right).$$

The coefficient of $\prod_i w_i^{m_i}$ counts the orbits in which color i occurs exactly m_i times. Substituting $w_i = 1$ yields $Z_G(q, \dots, q)$.

Definition 4.37. Graphs and Degrees
A finite simple undirected graph is $G = (V, E)$, where $E \subseteq \binom{V}{2}$. Its vertex degree $\deg(v)$ is the number of incident edges. A subgraph is determined by subsets of the vertices and edges; an induced subgraph on $S \subseteq V$ contains all edges of G with both endpoints in S.

Definition 4.38. Walks, Trails, Connectivity, and Components
A walk is a sequence of adjacent vertices, allowing repetition. A trail is a walk with no repeated edges, and a simple path has no repeated vertices. A graph is connected if every pair of vertices can be joined by a path. A connected component is a maximal connected subgraph. A bridge is an edge whose deletion increases the number of connected components; a cut vertex is a vertex whose deletion increases that number.

Theorem 4.39. Handshaking Lemma
For any finite undirected graph,

$$\sum_{v \in V(G)} \deg(v) = 2|E(G)|.$$

Consequently, the number of vertices of odd degree is even.

Theorem 4.40. Erdős–Gallai Theorem
Nonnegative integers $d_1 \geq \cdots \geq d_n$ are the degrees of a simple graph on n labeled vertices if and only if $\sum_i d_i$ is even and, for every $1 \leq k \leq n$,

$$\sum_{i=1}^{k} d_i \leq k(k-1) + \sum_{i=k+1}^{n} \min(d_i, k).$$

Definition 4.41. Trees and Spanning Trees
A tree is a connected graph with no cycles. A spanning tree of a connected graph is a subgraph that is a tree and contains every vertex.

Theorem 4.42. Equivalent Characterizations of a Tree
For a finite simple graph G with $n \geq 1$ vertices, any two of the following properties imply the third: G is connected; G is acyclic; G has exactly $n-1$ edges. In particular, a tree on n vertices has $n-1$ edges and there is a unique simple path between any two of its vertices.

Theorem 4.43. Cayley's Formula
For $n \geq 2$, the number of labeled trees on vertex set $[n]$ is

$$n^{n-2}.$$

For $n = 1$, there is exactly one such tree.

Theorem 4.44. Prüfer Codes
Labeled trees on vertex set $[n]$, for $n \geq 2$, are in bijection with sequences in $[n]^{n-2}$ (Prüfer codes). The degree of vertex i is one plus its number of appearances in the code.

Definition 4.45. Bipartite Graph
A graph $G = (V, E)$ is bipartite if V can be partitioned into disjoint sets U, W so that every edge joins a vertex of U to a vertex of W.

Theorem 4.46. Bipartite Graph Criterion
A finite graph is bipartite if and only if it has no cycle of odd length.

Definition 4.47. Vertex Coloring and Chromatic Number
A proper vertex coloring of a graph G assigns colors to the vertices so that adjacent vertices receive distinct colors. The *chromatic number* $\chi(G)$ is the smallest number of colors in a proper coloring.

Theorem 4.48. Greedy Coloring Bound
Every finite simple graph G satisfies

$$\chi(G) \leq \Delta(G) + 1,$$

where $\Delta(G)$ is its maximum vertex degree. Every bipartite graph with at least one edge has chromatic number 2.

Definition 4.49. Chromatic Polynomial
For a finite graph G, the chromatic polynomial $\chi_G(q)$ is the number of proper vertex colorings using q labeled colors, as a polynomial in q. It has degree $|V(G)|$ and leading coefficient 1. If G contains a loop, $\chi_G(q) = 0$.

Theorem 4.50. Deletion–Contraction Formula
For any nonloop edge e of a finite graph G,

$$\chi_G(q) = \chi_{G-e}(q) - \chi_{G/e}(q),$$

where $G - e$ deletes e and G/e contracts e. Multiple edges arising under contraction do not change the proper coloring count. For the edgeless graph on n vertices, $\chi_G(q) = q^n$.

Theorem 4.51. Dirac's and Ore's Hamiltonian Criteria
Let G be a finite simple graph with $n \geq 3$ vertices. If every vertex has degree at least $n/2$, then G has a Hamiltonian cycle (Dirac). More generally, if $\deg(u) + \deg(v) \geq n$ for every pair of distinct nonadjacent vertices u, v, then G has a Hamiltonian cycle (Ore).

Theorem 4.52. Eulerian Trail and Circuit Criteria
A finite connected graph has an Eulerian circuit exactly when every vertex has even degree. It has an Eulerian trail with two distinct endpoints exactly when precisely two vertices have odd degree.

Theorem 4.53. Euler's Formula for Planar Graphs
If a finite connected graph has a plane embedding with V vertices, E edges, and F faces, including the unbounded face, then

$$V - E + F = 2.$$

For a simple planar graph with $V \geq 3$,

$$E \leq 3V - 6.$$

If it is also bipartite, then $E \leq 2V - 4$.

Theorem 4.54. Edge Bounds for Simple Planar Graphs
A simple planar graph with $v \geq 3$ vertices and e edges satisfies $e \leq 3v - 6$. If it is also bipartite, then $e \leq 2v - 4$. In particular, every simple planar graph has a vertex of degree at most 5.

Theorem 4.55. Kuratowski's Theorem
A finite graph is planar if and only if it contains no subdivision of K_5 or $K_{3,3}$ as a subgraph.

Definition 4.56. Matching and Vertex Cover
A matching is a set of edges no two of which share a vertex. A matching is perfect if every vertex is incident with a matching edge. A vertex cover is a set of vertices incident with every edge.

Theorem 4.57. Hall's Marriage Theorem
A finite bipartite graph with vertex classes U, W has a matching saturating every vertex of U if and only if

$$|N(S)| \geq |S| \qquad \text{for every } S \subseteq U,$$

where $N(S)$ is the set of neighbors of vertices in S.

Theorem 4.58. Kőnig's Theorem
In every finite bipartite graph, the maximum size of a matching equals the minimum size of a vertex cover.

Theorem 4.59. Tutte's Perfect Matching Theorem
A finite undirected graph G has a perfect matching if and only if, for every $S \subseteq V(G)$, the graph $G - S$ has at most $|S|$ connected components of odd order.

Theorem 4.60. Stable Marriage Theorem
For two finite equally sized sets of applicants, each ranking all members of the opposite set strictly, there exists a perfect matching with no blocking pair. A blocking pair consists of two unmatched partners who both prefer one another to their assigned partners.

Definition 4.61. Flow Networks and Cuts
Let $G = (V, E)$ be a finite directed graph with source s, sink t, and capacities $c(e) \geq 0$. A feasible flow assigns $0 \leq f(e) \leq c(e)$ to each edge, with

$$\sum_{(u,v)\in E} f(u,v) = \sum_{(v,w)\in E} f(v,w) \qquad (v \notin \{s,t\}).$$

Its value is the net outflow at s. An s–t cut is a partition $V = S \sqcup T$ with $s \in S$ and $t \in T$; its capacity is $\sum_{u\in S,\, v\in T} c(u,v)$ over edges directed from S to T.

Theorem 4.62. Max-Flow Min-Cut Theorem
In a finite directed capacitated network with nonnegative edge capacities and distinguished source s and sink t, the maximum possible value of an s–t flow equals the minimum capacity of an s–t cut.

Theorem 4.63. Menger's Theorem, Vertex Form
For distinct nonadjacent vertices u, v in a finite graph, the maximum number of internally vertex-disjoint u–v paths equals the minimum number of vertices whose removal separates u and v.

Theorem 4.64. Matrix–Tree Theorem
Let G be a finite connected graph with adjacency matrix $\mathbf{A}$ and degree matrix $\mathbf{D}$. Set $\mathbf{L} = \mathbf{D} - \mathbf{A}$. The number of spanning trees of G equals any principal cofactor of $\mathbf{L}$.

Definition 4.65. Partially Ordered Sets, Chains, and Antichains
A partially ordered set (poset) is a set X with a reflexive, antisymmetric, transitive relation $\preceq$. A chain is a subset whose elements are pairwise comparable; an antichain is a subset whose distinct elements are pairwise incomparable.

Theorem 4.66. Dilworth's Theorem
In a finite poset, the maximum cardinality of an antichain equals the minimum number of chains in a partition of the poset.

Theorem 4.67. Sperner's Theorem
For a family $\mathcal{A}$ of subsets of $[n]$, no one of which properly contains another,

$$|\mathcal{A}| \leq \binom{n}{\lfloor n/2 \rfloor}.$$

Theorem 4.68. Lubell–Yamamoto–Meshalkin Inequality
If $\mathcal{A}$ is an antichain in the Boolean lattice $2^{[n]}$, then

$$\sum_{A \in \mathcal{A}} \frac{1}{\binom{n}{|A|}} \leq 1.$$

Definition 4.69. Finite Impartial Games and Nim-Sum
A finite impartial normal-play game has finitely many positions, no infinite sequence of legal moves, and the same available moves for both players; the player with no legal move loses. The nim-sum $a \oplus b$ is the bitwise exclusive-or of the binary expansions of nonnegative integers a, b.

Theorem 4.70. Sprague–Grundy Theorem
For a finite impartial normal-play game, define the Grundy value recursively by

$$g(P) = \operatorname{mex}\{g(Q) \mid P \to Q\},$$

where mex *is the least nonnegative integer absent from a set and terminal positions have value* 0*. A position is losing for the player to move if and only if $g(P) = 0$. The Grundy value of a disjunctive sum of games is the nim-sum of their Grundy values.*

Theorem 4.71. Bouton's Theorem for Nim
For a position in ordinary Nim with heaps of sizes $a_1, \ldots, a_k \in \mathbb{N}_0$, in which a move removes a positive number of objects from exactly one heap and the player making the last move wins, the position is losing if and only if

$$a_1 \oplus \cdots \oplus a_k = 0.$$

Theorem 4.72. Ramsey's Theorem
For positive integers $r, k_1, \ldots, k_m$, there exists N such that every m-coloring of the r-element subsets of $[N]$ contains a set of size k_i all of whose r-element subsets have color i, for at least one index i. In the graph case, $R(s,t)$ is the least N such that every red–blue edge coloring of K_N has a red K_s or a blue K_t.

Theorem 4.73. Erdős–Szekeres Monotone Subsequence Theorem
Every sequence of $(r-1)(s-1)+1$ distinct real numbers contains an increasing subsequence of length r or a decreasing subsequence of length s.

Theorem 4.74. Turán's Theorem
Among simple graphs on n vertices containing no complete graph K_{r+1}, the unique graph with the maximum number of edges (up to isomorphism) is the complete r-partite graph whose part sizes differ by at most one.

Definition 4.75. Probability Space and Conditional Probability
A probability space is a triple $(\Omega, \mathcal{F}, \mathbb{P})$, where $\mathcal{F}$ is a σ-algebra of subsets of Ω and $\mathbb{P}$ is a countably additive probability measure with $\mathbb{P}(\Omega) = 1$. For $\mathbb{P}(B) > 0$,

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}.$$

For a finite uniform space Ω, $\mathbb{P}(A) = |A|/|\Omega|$.

Theorem 4.76. Addition Rule and Union Bound
For events A, B,

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B).$$

For any finite or countable family of events,

$$\mathbb{P}\left(\bigcup_i A_i\right) \le \sum_i \mathbb{P}(A_i).$$

Theorem 4.77. Law of Total Probability and Bayes' Theorem
Let $(B_i)_{i=1}^m$ be a partition of Ω into events of positive probability. For any event A,

$$\mathbb{P}(A) = \sum_{i=1}^{m} \mathbb{P}(A \mid B_i)\mathbb{P}(B_i).$$

If $\mathbb{P}(A) > 0$, then

$$\mathbb{P}(B_j \mid A) = \frac{\mathbb{P}(A \mid B_j)\mathbb{P}(B_j)}{\sum_{i=1}^{m} \mathbb{P}(A \mid B_i)\mathbb{P}(B_i)}.$$

Definition 4.78. Independence
Events A, B are independent when $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$. A family of events is mutually independent if this product identity holds for every finite nonempty subfamily. Random variables X, Y are independent if the events $\{X \in U\}$ and $\{Y \in V\}$ are independent for all Borel sets U, V.

Definition 4.79. Random Variable and Expectation
A real-valued random variable X is a measurable map from Ω to $\mathbb{R}$. Its expectation, when defined, is denoted $\mathbb{E}[X]$. For a discrete random variable with countable range and $\sum_x |x|\mathbb{P}(X = x) < \infty$,

$$\mathbb{E}[X] = \sum_x x\mathbb{P}(X = x).$$

Definition 4.80. Distribution of a Random Variable
The distribution of X is specified by $\mathbb{P}(X \in B)$ for Borel sets B. For a discrete random variable, its probability mass function is $p_X(x) = \mathbb{P}(X = x)$; its cumulative distribution function is $F_X(x) = \mathbb{P}(X \le x)$.

Definition 4.81. Geometric Probability
For a bounded measurable planar region D of positive finite area, a point X chosen uniformly from D satisfies

$$\mathbb{P}(X \in A) = \frac{\text{Area}(A \cap D)}{\text{Area}(D)}$$

for every measurable A. In three dimensions, volume replaces area. The probability model and sampling measure must be specified before applying such ratios.

Formula 4.82. Bernoulli, Binomial, and Geometric Distributions
If $X \sim \text{Bern}(p)$, then $\mathbb{P}(X = 1) = p$ and $\mathbb{E}[X] = p$, $\text{Var}(X) = p(1-p)$. If $X \sim \text{Bin}(n, p)$, then for $0 \le k \le n$,

$$\mathbb{P}(X = k) = \binom{n}{k} p^k (1-p)^{n-k}, \quad \mathbb{E}[X] = np, \quad \text{Var}(X) = np(1-p).$$

If $Y \sim \text{Geom}(p)$ counts the number of independent Bernoulli trials up to and including the first success, with $0 < p \le 1$, then

$$\mathbb{P}(Y = k) = (1-p)^{k-1} p \quad (k \in \mathbb{N}), \qquad \mathbb{E}[Y] = \frac{1}{p}.$$

Formula 4.83. Hypergeometric Distribution
If a sample of size n is drawn uniformly without replacement from N objects, of which K are designated successes, the number X of successes satisfies

$$\mathbb{P}(X = k) = \frac{\binom{K}{k}\binom{N-K}{n-k}}{\binom{N}{n}}$$

for integers k with $\max(0, n - (N-K)) \le k \le \min(n, K)$, and $\mathbb{E}[X] = nK/N$.

Formula 4.84. Poisson Distribution and Poisson Limit
For $\lambda > 0$, a Poisson random variable $X \sim \text{Poi}(\lambda)$ has

$$\mathbb{P}(X = k) = e^{-\lambda} \frac{\lambda^k}{k!}, \quad k \in \mathbb{N}_0, \qquad \mathbb{E}[X] = \text{Var}(X) = \lambda.$$

If $X_n \sim \text{Bin}(n, p_n)$ with $np_n \to \lambda$, then for every fixed $k \in \mathbb{N}_0$,

$$\mathbb{P}(X_n = k) \longrightarrow e^{-\lambda} \frac{\lambda^k}{k!}.$$

Formula 4.85. Uniform and Exponential Distributions
If X is uniform on $[a, b]$ with $a < b$, its density is $f_X(x) = 1/(b-a)$ for $a \le x \le b$ (and 0 otherwise), and $\mathbb{E}[X] = (a+b)/2$. If Y is exponential with rate $\lambda > 0$, its density is

$$f_Y(y) = \lambda e^{-\lambda y} \quad (y \ge 0), \qquad \mathbb{E}[Y] = \frac{1}{\lambda}, \quad \text{Var}(Y) = \frac{1}{\lambda^2}.$$

Definition 4.86. Finite Markov Chains
A finite-state, time-homogeneous Markov chain has a row-stochastic transition matrix $\mathbf{P} = (p_{ij})$ satisfying

$$\mathbb{P}(X_{n+1} = j \mid X_n = i, \dots, X_0) = p_{ij}.$$

A probability row vector $\boldsymbol{\pi}$ is stationary if $\boldsymbol{\pi}\mathbf{P} = \boldsymbol{\pi}$.

Theorem 4.87. Stationary Distributions of Finite Markov Chains
Every finite Markov chain has a stationary distribution. If its transition graph is irreducible, the stationary distribution is unique and strictly positive. If it is also aperiodic, then $\mathbb{P}(X_n = j \mid X_0 = i) \to \pi_j$ *for every pair of states* i, j.

Definition 4.88. Normal Distribution
A real random variable X has normal distribution $\mathcal{N}(\mu, \sigma^2)$, where $\sigma > 0$, if its density is

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \qquad x \in \mathbb{R}.$$

Its expectation is μ, its variance is σ^2, and $(X - \mu)/\sigma \sim \mathcal{N}(0, 1)$.

Definition 4.89. Conditional Expectation
For integrable X and a random variable Y, the conditional expectation $\mathbb{E}[X \mid Y]$ denotes an integrable function of Y whose expectation on each event determined by Y agrees with that of X. In the discrete case, when $\mathbb{P}(Y = y) > 0$,

$$\mathbb{E}[X \mid Y = y] = \sum_x x \mathbb{P}(X = x \mid Y = y),$$

provided the sum is absolutely convergent.

Theorem 4.90. Linearity of Expectation
For integrable random variables X, Y *and scalars* a, b,

$$\mathbb{E}[aX + bY] = a\mathbb{E}[X] + b\mathbb{E}[Y].$$

No independence assumption is needed. If $X = \sum_{i=1}^{n} \mathbb{1}_{A_i}$ *counts how many of the events* A_i *occur, then*

$$\mathbb{E}[X] = \sum_{i=1}^{n} \mathbb{P}(A_i).$$

Formula 4.91. Variance and Covariance
For square-integrable real random variables X, Y,

$$\operatorname{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2,$$

$$\operatorname{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y],$$

$$\operatorname{Var}(X + Y) = \operatorname{Var}(X) + \operatorname{Var}(Y) + 2\operatorname{Cov}(X, Y).$$

If X and Y are independent, then $\operatorname{Cov}(X, Y) = 0$.

Theorem 4.92. Law of Total Expectation and Total Variance
If X *is integrable and* Y *is a random variable, then*

$$\mathbb{E}[X] = \mathbb{E}[\mathbb{E}[X \mid Y]].$$

If X *is square-integrable, then*

$$\operatorname{Var}(X) = \mathbb{E}[\operatorname{Var}(X \mid Y)] + \operatorname{Var}(\mathbb{E}[X \mid Y]).$$

Theorem 4.93. Laws of Large Numbers
Let $X_1, X_2, \dots$ be independent identically distributed real random variables with $\mathbb{E}|X_1| < \infty$ and common mean μ. Then

$$\frac{1}{n}\sum_{k=1}^{n} X_k \longrightarrow \mu \quad a.s.,$$

and hence also in probability (strong and weak laws of large numbers).

Theorem 4.94. Classical Central Limit Theorem
Let $X_1, X_2, \dots$ be i.i.d. real random variables of common mean μ and variance $0 < \sigma^2 < \infty$. Then

$$\frac{\sum_{k=1}^{n} X_k - n\mu}{\sigma\sqrt{n}} \xrightarrow{\mathrm{d}} \mathcal{N}(0,1).$$

Theorem 4.95. Tail-Sum Formula
If X is a nonnegative integer-valued random variable, then, allowing the value $+\infty$,

$$\mathbb{E}[X] = \sum_{k=1}^{\infty} \mathbb{P}(X \geq k).$$

Theorem 4.96. Markov's Inequality
If $X \geq 0$ almost surely and $a > 0$, then

$$\mathbb{P}(X \geq a) \leq \frac{\mathbb{E}[X]}{a},$$

provided the right-hand side is finite.

Theorem 4.97. Chebyshev's Inequality
If X has finite variance and $t > 0$, then

$$\mathbb{P}(|X - \mathbb{E}[X]| \geq t) \leq \frac{\operatorname{Var}(X)}{t^2}.$$

Theorem 4.98. Jensen's Inequality for Expectations
For an integrable random variable X taking values in an interval I and a convex function $f\colon I \to \mathbb{R}$, whenever the relevant expectations exist,

$$f(\mathbb{E}[X]) \leq \mathbb{E}[f(X)].$$

Theorem 4.99. Chernoff Bounds for Binomial Random Variables
If $X \sim \operatorname{Bin}(n,p)$ has mean $\mu = np > 0$, then for $\delta > 0$,

$$\mathbb{P}(X \geq (1+\delta)\mu) \leq \left(\frac{e^{\delta}}{(1+\delta)^{1+\delta}}\right)^{\mu},$$

and for $0 < \delta < 1$,

$$\mathbb{P}(X \leq (1-\delta)\mu) \leq e^{-\mu\delta^2/2}.$$

Theorem 4.100. The First-Moment Method
If X is a nonnegative integer-valued random variable and $\mathbb{E}[X] < 1$, then $\mathbb{P}(X = 0) > 0$. More generally, if $\mathbb{E}[X] > 0$ and $X \geq 0$, then $\mathbb{P}(X > 0) > 0$.

Theorem 4.101. Second-Moment Inequality
For a real random variable X with finite second moment and $\mathbb{E}[X^2] > 0$,

$$\mathbb{P}(X \neq 0) \geq \frac{(\mathbb{E}[X])^2}{\mathbb{E}[X^2]}.$$

If $X \geq 0$ almost surely, then for $0 < \theta < 1$ the Paley–Zygmund inequality gives

$$\mathbb{P}(X > \theta\mathbb{E}[X]) \geq (1-\theta)^2 \frac{(\mathbb{E}[X])^2}{\mathbb{E}[X^2]}.$$

Definition 4.102. Filtration and Martingale
A filtration $(\mathcal{F}_n)_{n\geq 0}$ is an increasing sequence of sub-σ-algebras. An adapted process (M_n) is a martingale if each M_n is integrable and

$$\mathbb{E}[M_{n+1} \mid \mathcal{F}_n] = M_n \qquad \text{a.s.}$$

A stopping time τ satisfies $\{\tau \leq n\} \in \mathcal{F}_n$ for all n.

Theorem 4.103. Optional Stopping Theorem, Bounded Case
If $(M_n, \mathcal{F}_n)$ is a discrete-time martingale and τ is a bounded stopping time, then M_τ is integrable and

$$\mathbb{E}[M_\tau] = \mathbb{E}[M_0].$$

Formula 4.104. Uniform Random Permutations
If π is uniformly distributed over the permutations of $[n]$, then the expected number of fixed points is 1 for $n \geq 1$, and the expected number of cycles is

$$H_n = \sum_{k=1}^{n} \frac{1}{k}.$$

The probability that π is a derangement is $D_n/n!$.

Formula 4.105. Coupon Collector
If one independently samples from n equally likely types until all types have appeared, the expected number of samples is

$$nH_n = n \sum_{k=1}^{n} \frac{1}{k}.$$

Formula 4.106. Gambler's Ruin Probability
For a nearest-neighbor random walk on $\{0, 1, \ldots, N\}$ absorbed at $0, N$, with independent steps $+1$ of probability p and -1 of probability $q = 1 - p$, the probability of reaching N before 0 from $i \in \{0, \ldots, N\}$ is

$$\mathbb{P}_i(\text{hit } N \text{ before } 0) = \begin{cases} \dfrac{1 - (q/p)^i}{1 - (q/p)^N}, & p \neq q,\ 0 < p < 1, \\ \dfrac{i}{N}, & p = q = \frac{1}{2}. \end{cases}$$

At $p = 0$ or $p = 1$, the probability is determined by the deterministic walk.

Theorem 4.107. Lovász Local Lemma, Symmetric Form
Let $A_1, \ldots, A_n$ be events with a dependency graph of maximum degree at most d: each A_i is independent of the collection of all events corresponding to its nonneighbors. If $\mathbb{P}(A_i) \leq p$ for all i and

$$ep(d + 1) \leq 1,$$

then $\mathbb{P}(\bigcap_{i=1}^{n} A_i^c) > 0$.

Chapter 5

Calculus and Analysis

Definition 5.1. Sequences, Limits, and Convergence
A sequence (a_n) in $\mathbb{R}$ converges to $L \in \mathbb{R}$ if for every $\varepsilon > 0$ there is $N \in \mathbb{N}$ such that $|a_n - L| < \varepsilon$ for all $n \geq N$. The same definition with modulus applies in $\mathbb{C}$. A sequence is *Cauchy* if for every $\varepsilon > 0$ there is N such that $|a_m - a_n| < \varepsilon$ for all $m, n \geq N$.

Theorem 5.2. Monotone Convergence Theorem for Sequences
Every bounded monotone real sequence converges. In particular, an increasing sequence (a_n) bounded above converges to $\sup\{a_n \mid n \in \mathbb{N}\}$, and a decreasing sequence bounded below converges to its infimum.

Theorem 5.3. Cauchy Criterion and Bolzano–Weierstrass Theorem
A sequence in $\mathbb{R}^d$ converges if and only if it is Cauchy. Every bounded sequence in $\mathbb{R}^d$ has a convergent subsequence.

Theorem 5.4. Stolz–Cesàro Theorem
Let (b_n) be strictly increasing with $b_n \to \infty$. If

$$\lim_{n\to\infty} \frac{a_{n+1} - a_n}{b_{n+1} - b_n} = L \in \overline{\mathbb{R}},$$

then $a_n / b_n \to L$. The assertion holds for extended real limits L.

Theorem 5.5. Squeeze Theorem
If $a_n \leq b_n \leq c_n$ for all sufficiently large n and $a_n, c_n \to L$, then $b_n \to L$. The analogous result holds for functions approaching a limit point.

Definition 5.6. Limit and Continuity of a Real Function
Let $f\colon D \to \mathbb{R}$ and let a be a limit point of $D \subseteq \mathbb{R}$. We write $\lim_{x\to a} f(x) = L$ if for every $\varepsilon > 0$ there exists $\delta > 0$ such that $x \in D$ and $0 < |x - a| < \delta$ imply $|f(x) - L| < \varepsilon$. If $a \in D$, the function is continuous at a when $\lim_{x\to a} f(x) = f(a)$.

Theorem 5.7. Sequential Characterization of Limits
For $f\colon D \to \mathbb{R}$ and a limit point a of D, one has $\lim_{x\to a} f(x) = L$ if and only if $f(x_n) \to L$ whenever (x_n) lies in $D \setminus \{a\}$ and $x_n \to a$.

Formula 5.8. Standard Limits
For real $x \to 0$,

$$\frac{\sin x}{x} \to 1, \quad \frac{\tan x}{x} \to 1, \quad \frac{e^x - 1}{x} \to 1, \quad \frac{\ln(1+x)}{x} \to 1,$$

$$\frac{1 - \cos x}{x^2} \to \frac{1}{2}, \qquad \frac{(1+x)^\alpha - 1}{x} \to \alpha \quad (\alpha \in \mathbb{R}).$$

For real t, $(1 + t/n)^n \to e^t$ as $n \to \infty$.

Theorem 5.9. Intermediate Value Theorem
If $f\colon [a,b] \to \mathbb{R}$ is continuous and y lies between $f(a)$ and $f(b)$, there exists $c \in [a,b]$ such that $f(c) = y$. In particular, a continuous function changing sign on an interval has a zero there.

Theorem 5.10. Extreme Value Theorem
A real-valued continuous function on a nonempty compact subset of $\mathbb{R}^n$ is bounded and attains its maximum and minimum.

Definition 5.11. Derivative and Differentiability
For $f\colon I \to \mathbb{R}$ and an interior point $a \in I$, the derivative is

$$f'(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h},$$

when this limit exists. Differentiability at a implies continuity there.

Formula 5.12. Differentiation Rules
For differentiable f, g and a scalar c,

$$(cf+g)' = cf' + g', \quad (fg)' = f'g + fg', \quad \left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2} \quad (g \neq 0),$$

$$(f \circ g)' = (f' \circ g)g'.$$

If f is locally invertible with $f'(a) \neq 0$, then $(f^{-1})'(f(a)) = 1/f'(a)$.

Formula 5.13. Elementary Derivatives
On their natural real domains,

$$\frac{\mathrm{d}}{\mathrm{d}x} x^{\alpha} = \alpha x^{\alpha-1} \quad (x > 0), \qquad \frac{\mathrm{d}}{\mathrm{d}x} e^{x} = e^{x}, \qquad \frac{\mathrm{d}}{\mathrm{d}x} \ln x = \frac{1}{x} \quad (x > 0),$$

$$\frac{\mathrm{d}}{\mathrm{d}x} \sin x = \cos x, \qquad \frac{\mathrm{d}}{\mathrm{d}x} \cos x = -\sin x, \qquad \frac{\mathrm{d}}{\mathrm{d}x} \arctan x = \frac{1}{1+x^2}.$$

Here $\frac{\mathrm{d}}{\mathrm{d}x}$ denotes differentiation with respect to x.

Theorem 5.14. Rolle's and Mean Value Theorems
If $f\colon [a,b] \to \mathbb{R}$ is continuous on $[a,b]$ and differentiable on (a,b), then $f(a) = f(b)$ implies $f'(c) = 0$ for some $c \in (a,b)$. More generally, some $c \in (a,b)$ satisfies

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

Theorem 5.15. Darboux's Theorem for Derivatives
If f is differentiable on an interval, its derivative has the intermediate value property, even when f' is not continuous.

Theorem 5.16. Cauchy's Mean Value Theorem
If f, g are continuous on $[a,b]$ and differentiable on (a,b), then for some $c \in (a,b)$,

$$(f(b) - f(a))g'(c) = (g(b) - g(a))f'(c).$$

A quotient form requires the relevant denominator to be nonzero.

Theorem 5.17. L'Hôpital's Rule
Suppose f and g are differentiable on a deleted neighborhood of a, with $g'(x) \neq 0$ there, and either $f(x), g(x) \to 0$ or $|f(x)|, |g(x)| \to \infty$ as $x \to a$. If $f'(x)/g'(x) \to L \in \overline{\mathbb{R}}$, then $f(x)/g(x) \to L$. One-sided and infinite-endpoint versions hold under the corresponding hypotheses.

Theorem 5.18. Taylor's Theorem with Lagrange Remainder
If f has $n+1$ continuous derivatives on the closed interval between a and x, then some ξ strictly between a and x satisfies

$$f(x) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k + \frac{f^{(n+1)}(\xi)}{(n+1)!}(x-a)^{n+1}.$$

If $|f^{(n+1)}| \le M$ throughout the interval, the remainder has modulus at most $M|x-a|^{n+1}/(n+1)!$.

Theorem 5.19. Taylor's Formula with Integral Remainder
If $f \in C^{n+1}([a,b])$ and $x \in [a,b]$, then

$$f(x) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k + \frac{1}{n!}\int_a^x (x-t)^n f^{(n+1)}(t)\ \mathrm{d}t.$$

In particular, if $|f^{(n+1)}(t)| \le M$ on $[a,b]$, the absolute remainder is at most $M|x-a|^{n+1}/(n+1)!$.

Formula 5.20. Maclaurin Expansions
Within their indicated real or complex disks of convergence,

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}, \quad \sin x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}, \quad \cos x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!},$$

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n \quad (|x|<1), \qquad \ln(1+x) = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}x^n}{n} \quad (|x|<1),$$

$$(1+x)^{\alpha} = \sum_{n=0}^{\infty} \binom{\alpha}{n} x^n \quad (|x|<1,\ \alpha \in \mathbb{R}).$$

The exponential and trigonometric expansions converge for all complex x.

Theorem 5.21. Convexity and the Second Derivative
For a twice differentiable function f on an interval, $f'' \ge 0$ implies convexity. A differentiable convex function satisfies

$$f(y) \ge f(x) + f'(x)(y-x).$$

If $f'' > 0$ on an interval, f is strictly convex there.

Definition 5.22. Darboux Sums and Riemann Integrability
For a bounded $f\colon [a,b] \to \mathbb{R}$ and a partition $P\colon a = x_0 < \cdots < x_n = b$, let m_i and M_i be the infimum and supremum of f on $[x_{i-1}, x_i]$. The lower and upper Darboux sums are

$$L(f,P) = \sum_{i=1}^{n} m_i(x_i - x_{i-1}), \quad U(f,P) = \sum_{i=1}^{n} M_i(x_i - x_{i-1}).$$

The function is Riemann integrable exactly when for every $\varepsilon > 0$ some partition satisfies $U(f,P) - L(f,P) < \varepsilon$.

Definition 5.23. Riemann Integral
A bounded function $f\colon [a,b] \to \mathbb{R}$ is Riemann integrable if the upper and lower Darboux integrals agree; their common value is $\int_a^b f(x)\ \mathrm{d}x$. Every continuous function on $[a,b]$ is Riemann integrable.

Theorem 5.24. Mean Value Theorem for Integrals
If f is continuous on $[a,b]$ with $a<b$, then for some $c\in[a,b]$,

$$\int_a^b f(x)\,\mathrm{d}x = f(c)(b-a).$$

More generally, if g is integrable and nonnegative on $[a,b]$, there exists $c\in[a,b]$ such that

$$\int_a^b f(x)g(x)\,\mathrm{d}x = f(c)\int_a^b g(x)\,\mathrm{d}x.$$

Theorem 5.25. Fundamental Theorem of Calculus
For continuous $f\colon[a,b]\to\mathbb{R}$, the function $F(x)=\int_a^x f(t)\,\mathrm{d}t$ satisfies $F'(x)=f(x)$ for $x\in(a,b)$. If $F'=f$ on $[a,b]$ (with the appropriate one-sided derivatives at the endpoints), then

$$\int_a^b f(x)\,\mathrm{d}x = F(b)-F(a).$$

Formula 5.26. Integration by Parts and Substitution
For continuously differentiable $u,v\colon[a,b]\to\mathbb{R}$,

$$\int_a^b u(x)v'(x)\,\mathrm{d}x = [u(x)v(x)]_a^b - \int_a^b u'(x)v(x)\,\mathrm{d}x.$$

For continuously differentiable $\varphi\colon[\alpha,\beta]\to[a,b]$ and continuous f,

$$\int_\alpha^\beta f(\varphi(t))\varphi'(t)\,\mathrm{d}t = \int_{\varphi(\alpha)}^{\varphi(\beta)} f(x)\,\mathrm{d}x.$$

Formula 5.27. Useful Elementary Antiderivatives
On intervals where the expressions are defined,

$$\int x^\alpha\,\mathrm{d}x = \frac{x^{\alpha+1}}{\alpha+1}+C \qquad (\alpha\neq-1,\ x>0),$$

$$\int\frac{\mathrm{d}x}{x} = \ln|x|+C, \qquad \int\frac{\mathrm{d}x}{1+x^2} = \arctan x + C,$$

$$\int\frac{\mathrm{d}x}{\sqrt{1-x^2}} = \arcsin x + C \qquad (-1<x<1),$$

$$\int e^{ax}\,\mathrm{d}x = \frac{e^{ax}}{a}+C \qquad (a\neq0).$$

Formula 5.28. Symmetry and Periodicity of Integrals
If f is integrable on $[a,b]$, then

$$\int_a^b f(x)\,\mathrm{d}x = \int_a^b f(a+b-x)\,\mathrm{d}x.$$

If f is even, $\int_{-a}^a f(x)\,\mathrm{d}x = 2\int_0^a f(x)\,\mathrm{d}x$; if f is odd, the symmetric integral is zero. If f is periodic with period $T>0$ and integrable on every compact interval, then $\int_a^{a+T} f(x)\,\mathrm{d}x$ is independent of a.

Theorem 5.29. Leibniz Rule for Differentiation under the Integral Sign
If f and $\partial f/\partial t$ are continuous on $[a,b]\times I$ with I an open interval, then

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_a^b f(x,t)\,\mathrm{d}x = \int_a^b \frac{\partial f}{\partial t}(x,t)\,\mathrm{d}x.$$

For differentiable endpoints $a(t),b(t)$, the corresponding formula is

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_{a(t)}^{b(t)} f(x,t)\,\mathrm{d}x = f(b(t),t)b'(t) - f(a(t),t)a'(t) + \int_{a(t)}^{b(t)}\frac{\partial f}{\partial t}(x,t)\,\mathrm{d}x,$$

under the analogous continuity hypotheses.

Definition 5.30. Improper Integrals
For a locally integrable function $f\colon [a,\infty) \to \mathbb{R}$, the improper integral is defined by

$$\int_a^\infty f(x)\,\mathrm{d}x = \lim_{R\to\infty} \int_a^R f(x)\,\mathrm{d}x,$$

provided the limit exists and is finite. At a singular endpoint a, one instead takes a limit as the lower integration endpoint tends to a from the right. For a two-sided improper integral, the separate one-sided improper integrals must converge; a Cauchy principal value alone does not imply improper convergence.

Theorem 5.31. Integral Comparison and Convergence
Suppose $0 \le f(x) \le g(x)$ for all sufficiently large x, with f, g locally integrable on $[a,\infty)$. Convergence of $\int_a^\infty g(x)\,\mathrm{d}x$ implies convergence of $\int_a^\infty f(x)\,\mathrm{d}x$. The integral

$$\int_1^\infty \frac{\mathrm{d}x}{x^p}$$

converges if and only if $p > 1$; the integral $\int_0^1 x^{-p}\,\mathrm{d}x$ converges if and only if $p < 1$.

Formula 5.32. Gaussian Integral
For $a > 0$,

$$\int_{-\infty}^{\infty} e^{-ax^2}\,\mathrm{d}x = \sqrt{\frac{\pi}{a}}, \qquad \int_0^\infty e^{-ax^2}\,\mathrm{d}x = \frac{1}{2}\sqrt{\frac{\pi}{a}}.$$

Consequently $\int_{-\infty}^{\infty} x^{2m+1} e^{-ax^2}\,\mathrm{d}x = 0$ for $m \in \mathbb{N}_0$, and even moments can be obtained by differentiating with respect to a.

Formula 5.33. Logarithmic Moments and Dirichlet's Integral
For $a > -1$ and $m \in \mathbb{N}_0$,

$$\int_0^1 x^a (\ln x)^m\,\mathrm{d}x = \frac{(-1)^m m!}{(a+1)^{m+1}}.$$

The conditionally convergent improper integral

$$\int_0^\infty \frac{\sin x}{x}\,\mathrm{d}x = \frac{\pi}{2}$$

is known as Dirichlet's integral.

Formula 5.34. Frullani's Integral
If $a, b > 0$ and $f\colon (0,\infty) \to \mathbb{R}$ is continuously differentiable with finite limits at 0^+ and ∞, and the improper integral exists, then

$$\int_0^\infty \frac{f(ax) - f(bx)}{x}\,\mathrm{d}x = (f(0^+) - f(\infty)) \ln \frac{b}{a}.$$

Definition 5.35. Gamma and Beta Functions
For $s > 0$ and $x, y > 0$, respectively,

$$\Gamma(s) = \int_0^\infty t^{s-1} e^{-t}\,\mathrm{d}t, \qquad B(x,y) = \int_0^1 t^{x-1}(1-t)^{y-1}\,\mathrm{d}t.$$

They satisfy $\Gamma(s+1) = s\Gamma(s)$ and $\Gamma(n+1) = n!$ for $n \in \mathbb{N}_0$.

Formula 5.36. Beta–Gamma Identity and Euler Reflection Formula
For $x, y > 0$,

$$B(x,y) = \frac{\Gamma(x)\Gamma(y)}{\Gamma(x+y)}.$$

For $z \in \mathbb{C} \setminus \mathbb{Z}$ where both sides are defined through meromorphic continuation,

$$\Gamma(z)\Gamma(1-z) = \frac{\pi}{\sin(\pi z)}.$$

In particular, $\Gamma(1/2) = \sqrt{\pi}$.

Formula 5.37. Stirling's Formula
As $n \to \infty$ through positive integers,

$$n! \sim \sqrt{2\pi n}\left(\frac{n}{e}\right)^n, \qquad \ln(n!) = n\ln n - n + \frac{1}{2}\ln(2\pi n) + O(1/n).$$

Here $a_n \sim b_n$ means $a_n/b_n \to 1$.

Definition 5.38. Infinite Series and Absolute Convergence
The series $\sum_{n=1}^{\infty} a_n$ converges when its partial sums converge. It converges absolutely if $\sum_{n=1}^{\infty} |a_n| < \infty$; absolute convergence implies convergence for real or complex series.

Theorem 5.39. Comparison and Integral Tests
If $0 \le a_n \le b_n$ for all sufficiently large n and $\sum b_n$ converges, then $\sum a_n$ converges. If $f\colon [1,\infty) \to [0,\infty)$ is decreasing, then $\sum_{n=1}^{\infty} f(n)$ converges if and only if $\int_1^{\infty} f(x)\,\mathrm{d}x$ converges. In particular, $\sum_{n=1}^{\infty} n^{-p}$ converges exactly when $p > 1$.

Theorem 5.40. Ratio and Root Tests
For a series $\sum a_n$, let $L = \limsup_{n\to\infty} |a_n|^{1/n}$. If $L < 1$, the series converges absolutely; if $L > 1$, it diverges. If the ratio $|a_{n+1}/a_n|$ exists eventually and tends to L, the same conclusions hold for $L < 1$ and $L > 1$. Neither test decides the case $L = 1$.

Theorem 5.41. Alternating Series Test and Remainder
If $a_n \ge 0$ decreases to zero, the series $\sum_{n=1}^{\infty}(-1)^{n-1}a_n$ converges. Its remainder after N terms satisfies

$$\left|\sum_{n=N+1}^{\infty}(-1)^{n-1}a_n\right| \le a_{N+1}.$$

Theorem 5.42. Cauchy Condensation Test
If (a_n) is nonincreasing and nonnegative, then

$$\sum_{n=1}^{\infty} a_n < \infty \quad\Longleftrightarrow\quad \sum_{k=0}^{\infty} 2^k a_{2^k} < \infty.$$

Theorem 5.43. Dirichlet's and Abel's Tests for Series
If the partial sums of (a_n) are bounded and (b_n) is a real monotone sequence tending to zero, then $\sum a_n b_n$ converges (Dirichlet). If $\sum a_n$ converges and (b_n) is a bounded monotone real sequence, then $\sum a_n b_n$ converges (Abel).

Formula 5.44. Summation by Parts
For scalar sequences (a_n) and (b_n), let $A_n = \sum_{k=1}^{n} a_k$. Then

$$\sum_{k=1}^{n} a_k b_k = A_n b_n + \sum_{k=1}^{n-1} A_k (b_k - b_{k+1}).$$

This is the discrete analogue of integration by parts.

Formula 5.45. Euler–Maclaurin Summation Formula
For $f \in C^2([m,n])$ with integers $m < n$, the exact formula

$$\sum_{k=m}^{n} f(k) = \int_m^n f(x)\,\mathrm{d}x + \frac{f(m)+f(n)}{2} + \frac{B_2}{2!}\left(f'(n) - f'(m)\right) - \frac{1}{2!}\int_m^n B_2(\{x\})f''(x)\,\mathrm{d}x$$

holds, where $B_2 = 1/6$ and $B_2(t) = t^2 - t + 1/6$ is the second Bernoulli polynomial. Higher-order forms involve $B_{2j}(f^{(2j-1)}(n) - f^{(2j-1)}(m))/(2j)!$ and a periodic Bernoulli remainder.

Definition 5.46. Power Series and Radius of Convergence
For the power series $\sum_{n=0}^{\infty} a_n(z-z_0)^n$, its radius of convergence $R \in [0,\infty]$ is characterized by absolute convergence when $|z - z_0| < R$ and divergence when $|z - z_0| > R$.

Theorem 5.47. Abel's Theorem for Power Series
Suppose $\sum_{n=0}^{\infty} a_n$ converges to S. Then

$$\lim_{r\to 1^-} \sum_{n=0}^{\infty} a_n r^n = S.$$

The power series on the left converges absolutely for $0 \le r < 1$.

Theorem 5.48. Cauchy–Hadamard Formula
The radius of convergence of $\sum_{n=0}^{\infty} a_n(z-z_0)^n$ is

$$\frac{1}{R} = \limsup_{n\to\infty} |a_n|^{1/n},$$

with the conventions $1/0 = \infty$ and $1/\infty = 0$.

Theorem 5.49. Differentiation and Integration of Power Series
Inside its disk of convergence, a power series may be differentiated and integrated term by term; the resulting series have the same radius of convergence. In particular,

$$\frac{\mathrm{d}}{\mathrm{d}z}\sum_{n=0}^{\infty} a_n(z-z_0)^n = \sum_{n=1}^{\infty} n a_n (z-z_0)^{n-1}.$$

Definition 5.50. Metric Spaces, Open Sets, and Compactness
A metric space (X,d) consists of a set X and a metric d. A subset $U \subseteq X$ is open if every $x \in U$ has some ball $B_r(x) \subseteq U$. A set $K \subseteq X$ is compact if every open cover of K has a finite subcover. A metric space is complete if every Cauchy sequence converges in it.

Theorem 5.51. Heine–Borel Theorem
A subset of $\mathbb{R}^n$ is compact if and only if it is closed and bounded. Every sequence in a compact metric space has a convergent subsequence with limit in that space.

Definition 5.52. Connectedness
A topological space is connected if it cannot be expressed as the union of two disjoint nonempty open subsets. A subset of $\mathbb{R}$ is connected exactly when it is an interval. A continuous image of a connected space is connected.

Theorem 5.53. Continuous Images of Compact Sets
The continuous image of a compact topological space is compact. A continuous bijection from a compact space onto a Hausdorff space has a continuous inverse.

Definition 5.54. Uniform Continuity
A function $f\colon (X, d_X) \to (Y, d_Y)$ is uniformly continuous if for every $\varepsilon > 0$ there exists $\delta > 0$ such that

$$d_X(x, y) < \delta \quad \Longrightarrow \quad d_Y(f(x), f(y)) < \varepsilon$$

for all $x, y \in X$.

Theorem 5.55. Heine–Cantor Theorem
Every continuous map from a compact metric space into a metric space is uniformly continuous.

Definition 5.56. Pointwise and Uniform Convergence
For functions $f_n\colon E \to \mathbb{R}$, pointwise convergence to f means $f_n(x) \to f(x)$ for every $x \in E$. Uniform convergence means

$$\sup_{x \in E} |f_n(x) - f(x)| \longrightarrow 0$$

when the supremum is finite for all sufficiently large n.

Theorem 5.57. Weierstrass Approximation Theorem
Every continuous real-valued function on a compact interval $[a, b]$ can be uniformly approximated by real polynomials: for every $\varepsilon > 0$ there is a polynomial p such that

$$\sup_{x \in [a,b]} |f(x) - p(x)| < \varepsilon.$$

Theorem 5.58. Dini's Theorem
If K is compact and $f_n\colon K \to \mathbb{R}$ are continuous functions increasing pointwise to a continuous function f, then $f_n \to f$ uniformly. The corresponding conclusion holds for a pointwise decreasing sequence.

Theorem 5.59. Uniform Limit Theorem
If f_n is continuous on a metric space E for every n and $f_n \to f$ uniformly on E, then f is continuous on E.

Theorem 5.60. Weierstrass M-Test
Let $f_n\colon E \to \mathbb{C}$ and suppose $|f_n(x)| \le M_n$ for every $x \in E$, where $M_n \ge 0$ and $\sum_n M_n < \infty$. Then $\sum_n f_n$ converges absolutely and uniformly on E.

Theorem 5.61. Termwise Integration for Uniformly Convergent Series
If (f_n) is a sequence of Riemann-integrable functions on $[a, b]$ converging uniformly to f, then f is Riemann integrable and

$$\lim_{n \to \infty} \int_a^b f_n(x)\, \mathrm{d}x = \int_a^b f(x)\, \mathrm{d}x.$$

The same conclusion holds for uniformly convergent series by considering their partial sums.

Theorem 5.62. Termwise Differentiation
Suppose f_n is continuously differentiable on $[a, b]$, the derivatives f_n' converge uniformly to g there, and $f_n(x_0)$ converges for some $x_0 \in [a, b]$. Then f_n converges uniformly to a differentiable function f with $f' = g$ on (a, b).

Theorem 5.63. Banach Fixed-Point Theorem
Let (X, d) be a nonempty complete metric space and let $T\colon X \to X$ satisfy $d(Tx, Ty) \le q\,d(x, y)$ for all $x, y \in X$, where $0 \le q < 1$. Then T has a unique fixed point x^*. For any starting point x_0, the iterates $x_{n+1} = T(x_n)$ converge to x^*.

Definition 5.64. Equicontinuity and Uniform Boundedness
A family $\mathcal{F}$ of functions from a metric space X to a metric space Y is equicontinuous at $x \in X$ if, for every $\varepsilon > 0$, some $\delta > 0$ works simultaneously for all $f \in \mathcal{F}$:

$$d_X(x, y) < \delta \implies d_Y(f(x), f(y)) < \varepsilon.$$

For scalar-valued functions, $\mathcal{F}$ is uniformly bounded if $\sup_{f\in\mathcal{F}} \sup_{x\in X} |f(x)| < \infty$.

Theorem 5.65. Arzelà–Ascoli Theorem
A uniformly bounded, equicontinuous family of real- or complex-valued continuous functions on a compact metric space has compact closure in the uniform norm. Equivalently, every sequence in such a family has a uniformly convergent subsequence.

Theorem 5.66. Stone–Weierstrass Theorem, Real Form
Let K be a compact Hausdorff space. If a subalgebra $\mathcal{A} \subseteq C(K, \mathbb{R})$ contains the constants and separates points of K, then $\mathcal{A}$ is dense in $C(K, \mathbb{R})$ in the uniform norm.

Definition 5.67. Nowhere Dense Set
A subset A of a topological space X is nowhere dense if the interior of its closure is empty:

$$\operatorname{int}(\overline{A}) = \varnothing.$$

A set is meager (of first category) if it is a countable union of nowhere dense sets.

Theorem 5.68. Baire Category Theorem
A nonempty complete metric space is not a countable union of nowhere dense sets. Equivalently, the intersection of countably many open dense subsets is dense.

Definition 5.69. Limit Superior and Limit Inferior
For a real sequence (a_n), its extended-real limit superior and limit inferior are

$$\limsup_{n\to\infty} a_n = \lim_{n\to\infty} \sup_{k\geq n} a_k, \quad \liminf_{n\to\infty} a_n = \lim_{n\to\infty} \inf_{k\geq n} a_k.$$

If both values are finite and equal to L, then $a_n \to L$.

Definition 5.70. Measurable Functions and Lebesgue Integration
On a measure space $(X, \mathcal{A}, \mu)$, a real-valued measurable function has measurable preimages of Borel sets. A measurable function is integrable if $\int_X |f| \, \mathrm{d}\mu < \infty$; statements holding outside a null set hold *almost everywhere* (a.e.).

Theorem 5.71. Monotone Convergence Theorem for Integrals
If (f_n) is an increasing sequence of nonnegative measurable functions converging pointwise almost everywhere to f, then

$$\int_X f_n \, \mathrm{d}\mu \longrightarrow \int_X f \, \mathrm{d}\mu,$$

allowing the value $+\infty$.

Theorem 5.72. Fatou's Lemma
For nonnegative measurable functions (f_n),

$$\int_X \liminf_{n\to\infty} f_n \, \mathrm{d}\mu \leq \liminf_{n\to\infty} \int_X f_n \, \mathrm{d}\mu.$$

Theorem 5.73. Dominated Convergence Theorem
Suppose $f_n \to f$ almost everywhere, each f_n is measurable, and $|f_n| \leq g$ almost everywhere for an integrable function g. Then f is integrable and

$$\lim_{n\to\infty} \int_X f_n \, \mathrm{d}\mu = \int_X f \, \mathrm{d}\mu.$$

Theorem 5.74. Tonelli's and Fubini's Theorems
On the product of two σ-finite measure spaces, a nonnegative measurable function may be integrated by iterated integration in either order (Tonelli), possibly with value $+\infty$. If the function is absolutely integrable on the product, both iterated integrals exist and equal the product-space integral (Fubini).

Definition 5.75. Lebesgue L^p Spaces
For $1 \le p < \infty$, $L^p(X, \mu)$ is the space of measurable functions f (identified up to equality almost everywhere) for which

$$\|f\|_p = \left(\int_X |f|^p \, \mathrm{d}\mu \right)^{1/p} < \infty.$$

The space $L^\infty(X, \mu)$ consists of essentially bounded measurable functions, with norm given by the essential supremum. The space L^2 has the inner product $\langle f, g \rangle = \int_X \overline{f} g \, \mathrm{d}\mu$.

Theorem 5.76. Hölder's Inequality for Integrals
On a measure space $(X, \mathcal{A}, \mu)$, if $1 \le p, q \le \infty$ satisfy $1/p + 1/q = 1$ and $f \in L^p$, $g \in L^q$, then $fg \in L^1$ and

$$\int_X |fg| \, \mathrm{d}\mu \le \|f\|_{L^p} \|g\|_{L^q}.$$

The endpoints use $1/\infty = 0$. The case $p = q = 2$ is integral Cauchy–Schwarz.

Theorem 5.77. Minkowski's Inequality for Integrals
For $1 \le p \le \infty$ and $f, g \in L^p(X, \mu)$,

$$\|f + g\|_{L^p} \le \|f\|_{L^p} + \|g\|_{L^p}.$$

Consequently the usual L^p spaces (modulo equality almost everywhere) are normed spaces.

Theorem 5.78. Jensen's Inequality for Integrals
Let μ be a probability measure on X, let $f \colon X \to I$ be integrable, and let $\varphi \colon I \to \mathbb{R}$ be convex on an interval I. If $\varphi \circ f$ is integrable and $\int_X f \, \mathrm{d}\mu \in I$, then

$$\varphi\left(\int_X f \, \mathrm{d}\mu \right) \le \int_X \varphi(f) \, \mathrm{d}\mu.$$

Definition 5.79. Differentiability in Several Variables
For $f \colon U \subseteq \mathbb{R}^n \to \mathbb{R}^m$ with U open, f is differentiable at $\mathbf{a} \in U$ if there is a linear map $Df(\mathbf{a})$ such that

$$f(\mathbf{a} + \mathbf{h}) = f(\mathbf{a}) + Df(\mathbf{a})\mathbf{h} + o(\|\mathbf{h}\|).$$

Its matrix in standard coordinates is the Jacobian $\mathbf{J}_f(\mathbf{a})$. For scalar f, the derivative is represented by the gradient $\nabla f(\mathbf{a})$.

Definition 5.80. Gradient, Directional Derivative, and Hessian
For $f \colon U \subseteq \mathbb{R}^n \to \mathbb{R}$ differentiable, its gradient is

$$\nabla f = (\partial_1 f, \ldots, \partial_n f)^{\mathrm{T}}.$$

For a vector $\mathbf{v} \in \mathbb{R}^n$, the directional derivative is $D_{\mathbf{v}} f(\mathbf{a}) = \nabla f(\mathbf{a}) \cdot \mathbf{v}$. If the second partial derivatives exist, the Hessian $\operatorname{Hess} f$ is the matrix with (i, j)-entry $\partial_i \partial_j f$.

Theorem 5.81. Multivariable Chain Rule
If $f \colon U \to V$ and $g \colon V \to W$ are differentiable at $\mathbf{a}$ and $f(\mathbf{a})$, respectively, then

$$D(g \circ f)(\mathbf{a}) = Dg(f(\mathbf{a}))Df(\mathbf{a}).$$

Theorem 5.82. Clairaut's Theorem
If the mixed second partial derivatives $\partial^2 f/(\partial x_i \partial x_j)$ and $\partial^2 f/(\partial x_j \partial x_i)$ are continuous in a neighborhood of $\mathbf{a}$, then they are equal at $\mathbf{a}$.

Theorem 5.83. Multivariable Taylor's Theorem, Second Order
If $f\colon U \subseteq \mathbb{R}^n \to \mathbb{R}$ is twice continuously differentiable near $\mathbf{a}$, then as $\mathbf{h} \to \mathbf{0}$,

$$f(\mathbf{a}+\mathbf{h}) = f(\mathbf{a}) + \nabla f(\mathbf{a}) \cdot \mathbf{h} + \frac{1}{2}\mathbf{h}^{\mathrm{T}}(\operatorname{Hess} f)(\mathbf{a})\mathbf{h} + o(\|\mathbf{h}\|^2).$$

Theorem 5.84. Inverse Function Theorem
If $f\colon U \subseteq \mathbb{R}^n \to \mathbb{R}^n$ is continuously differentiable and $\det Df(\mathbf{a}) \neq 0$, then f restricts to a continuously differentiable bijection between suitable neighborhoods of $\mathbf{a}$ and $f(\mathbf{a})$, and

$$D(f^{-1})(f(\mathbf{a})) = (Df(\mathbf{a}))^{-1}.$$

Theorem 5.85. Implicit Function Theorem
Let $F\colon U \subseteq \mathbb{R}^{n+m} \to \mathbb{R}^m$ be continuously differentiable, with $F(\mathbf{a},\mathbf{b}) = \mathbf{0}$. If the $m \times m$ partial derivative matrix $D_{\mathbf{y}}F(\mathbf{a},\mathbf{b})$ is invertible, then locally there is a unique continuously differentiable function $\mathbf{y} = g(\mathbf{x})$ satisfying $F(\mathbf{x}, g(\mathbf{x})) = \mathbf{0}$ and $g(\mathbf{a}) = \mathbf{b}$.

Theorem 5.86. Second Derivative Test
Suppose $f\colon U \subseteq \mathbb{R}^n \to \mathbb{R}$ is twice continuously differentiable and $\nabla f(\mathbf{a}) = \mathbf{0}$. If $\operatorname{Hess} f(\mathbf{a})$ is positive definite, $\mathbf{a}$ is a strict local minimum; if it is negative definite, a strict local maximum; if it is indefinite, $\mathbf{a}$ is a saddle point. A singular positive or negative semidefinite Hessian alone need not decide the extremum.

Theorem 5.87. Lagrange Multipliers
Let $f, g_1, \ldots, g_k$ be continuously differentiable near a constrained local extremum $\mathbf{a}$ with $g_i(\mathbf{a}) = 0$. If $\nabla g_1(\mathbf{a}), \ldots, \nabla g_k(\mathbf{a})$ are linearly independent, then some scalars λ_i satisfy

$$\nabla f(\mathbf{a}) = \sum_{i=1}^{k} \lambda_i \nabla g_i(\mathbf{a}).$$

Theorem 5.88. Euler's Theorem for Homogeneous Functions
If $f\colon U \subseteq \mathbb{R}^n \to \mathbb{R}$ is differentiable and homogeneous of degree d on an open cone U, then

$$\sum_{i=1}^{n} x_i \frac{\partial f}{\partial x_i}(\mathbf{x}) = df(\mathbf{x}).$$

Theorem 5.89. Fubini's Theorem for Continuous Functions
If f is continuous on the rectangle $[a,b] \times [c,d]$, then

$$\int_a^b \int_c^d f(x,y)\ \mathrm{d}y\ \mathrm{d}x = \int_c^d \int_a^b f(x,y)\ \mathrm{d}x\ \mathrm{d}y.$$

Theorem 5.90. Change of Variables in Multiple Integrals
Let $\varphi\colon U \to V$ be a continuously differentiable diffeomorphism between open subsets of $\mathbb{R}^n$. For integrable f with suitable compact support in V,

$$\int_V f(\mathbf{y})\ \mathrm{d}\mathbf{y} = \int_U f(\varphi(\mathbf{x}))|\det D\varphi(\mathbf{x})|\ \mathrm{d}\mathbf{x}.$$

Formula 5.91. Polar and Spherical Jacobians
The plane polar substitution $x = r\cos\theta$, $y = r\sin\theta$ has area element $\mathrm{d}A = r\ \mathrm{d}r\ \mathrm{d}\theta$ for $r \geq 0$. With spherical coordinates

$$(x, y, z) = (r\sin\phi\cos\theta, r\sin\phi\sin\theta, r\cos\phi),$$

where $r \geq 0$ and $0 \leq \phi \leq \pi$, the volume element is $\mathrm{d}V = r^2\sin\phi\ \mathrm{d}r\ \mathrm{d}\phi\ \mathrm{d}\theta$.

Theorem 5.92. Green's Theorem
Let $D \subseteq \mathbb{R}^2$ be a bounded plane region with positively oriented, piecewise smooth, simple closed boundary and let P, Q have continuous first partial derivatives on a neighborhood of $\overline{D}$. Then

$$\oint_{\partial D} P\ \mathrm{d}x + Q\ \mathrm{d}y = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) \mathrm{d}A.$$

Theorem 5.93. Stokes' and Divergence Theorems
For a sufficiently smooth oriented surface $S \subseteq \mathbb{R}^3$ with positively oriented boundary and a smooth vector field $\mathbf{F}$,

$$\oint_{\partial S} \mathbf{F}\cdot\mathrm{d}\mathbf{r} = \iint_S (\nabla\times\mathbf{F})\cdot\mathbf{n}\ \mathrm{d}S.$$

For a bounded solid region Ω with piecewise smooth outward-oriented boundary,

$$\iint_{\partial\Omega} \mathbf{F}\cdot\mathbf{n}\ \mathrm{d}S = \iiint_\Omega (\nabla\cdot\mathbf{F})\ \mathrm{d}V.$$

Definition 5.94. Holomorphic and Entire Functions
For an open set $U \subseteq \mathbb{C}$, a function $f\colon U \to \mathbb{C}$ is *holomorphic* if its complex derivative exists at every point of U. It is *entire* if $U = \mathbb{C}$. A meromorphic function on U is holomorphic except at isolated poles.

Theorem 5.95. Cauchy–Riemann Equations
If $f = u + iv$ is complex differentiable at $z = x + iy$ and u, v have the corresponding real partial derivatives, then

$$u_x = v_y, \qquad u_y = -v_x.$$

Conversely, if u, v have continuous first partial derivatives on an open set and satisfy these equations, then f is holomorphic there.

Formula 5.96. Complex Exponential, Argument, and Logarithm
For $z = x + iy$,

$$e^z = e^x(\cos y + i\sin y), \qquad e^{2\pi i} = 1.$$

For $z \neq 0$, let $-\pi < \operatorname{Arg} z \leq \pi$. The principal logarithm is

$$\operatorname{Log} z = \ln|z| + i\operatorname{Arg} z,$$

and the multivalued logarithm is $\log z = \{\operatorname{Log} z + 2\pi i k \mid k \in \mathbb{Z}\}$. The principal logarithm is holomorphic on $\mathbb{C}\setminus(-\infty, 0]$.

Definition 5.97. Complex Contours and Winding Number
A contour is a piecewise continuously differentiable parametrized curve $\gamma\colon [a, b] \to \mathbb{C}$. For continuous f on the image of γ,

$$\int_\gamma f(z)\ \mathrm{d}z = \int_a^b f(\gamma(t))\gamma'(t)\ \mathrm{d}t.$$

For a closed contour and w outside its image, the winding number is

$$\operatorname{Ind}(\gamma, w) = \frac{1}{2\pi i}\int_\gamma \frac{\mathrm{d}z}{z - w} \in \mathbb{Z}.$$

A positively oriented simple closed contour traverses its bounded interior counterclockwise.

Theorem 5.98. Cauchy–Goursat Theorem
If $U \subseteq \mathbb{C}$ is simply connected and f is holomorphic on U, then $\oint_\gamma f(z)\,\mathrm{d}z = 0$ for every closed piecewise smooth curve γ in U.

Theorem 5.99. Cauchy Integral Formula
Let f be holomorphic in a neighborhood of the closure of a bounded domain $D \subseteq \mathbb{C}$ whose boundary ∂D is a positively oriented, piecewise smooth, simple closed curve. For $a \in D$ and $n \in \mathbb{N}_0$,

$$f^{(n)}(a) = \frac{n!}{2\pi i} \oint_{\partial D} \frac{f(z)}{(z-a)^{n+1}}\,\mathrm{d}z.$$

Theorem 5.100. Cauchy Estimates
If f is holomorphic on a neighborhood of the closed disk $\overline{B_R(a)}$ and $|f(z)| \le M$ on its boundary, then for every $n \in \mathbb{N}_0$,

$$|f^{(n)}(a)| \le \frac{n!M}{R^n}.$$

Theorem 5.101. Schwarz Lemma
If f is holomorphic on the unit disk, $|f(z)| \le 1$ there, and $f(0) = 0$, then

$$|f(z)| \le |z|, \qquad |f'(0)| \le 1.$$

If equality holds for some nonzero z in the disk, or if $|f'(0)| = 1$, then $f(z) = e^{i\theta}z$ for some real θ.

Theorem 5.102. Liouville's Theorem
Every bounded entire function is constant.

Theorem 5.103. Maximum Modulus Principle
If f is holomorphic and nonconstant on a connected open set U, then $|f|$ has no local maximum in U. Consequently, if f is continuous on $\overline{D}$ and holomorphic on a bounded domain D, then $\max_{\overline{D}} |f| = \max_{\partial D} |f|$.

Theorem 5.104. Identity Theorem
If two holomorphic functions on a connected open set U agree on a subset having a limit point in U, then they agree everywhere on U. In particular, the zeros of a nonzero holomorphic function are isolated.

Theorem 5.105. Taylor and Laurent Expansions
A holomorphic function on a disk $B_R(a)$ has a Taylor series converging to it throughout that disk:

$$f(z) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(z-a)^n.$$

A holomorphic function on an annulus $r < |z-a| < R$ has a unique Laurent expansion $\sum_{n\in\mathbb{Z}} c_n(z-a)^n$ converging locally uniformly there.

Definition 5.106. Zeros, Poles, and Residues
An isolated zero of f has order m when $f(z) = (z-a)^m g(z)$ with $g(a) \neq 0$ holomorphic. A pole has order m when $1/f$ has a zero of order m. The residue $\operatorname{Res}(f; a)$ is the coefficient of $(z-a)^{-1}$ in the Laurent series of f around a.

Formula 5.107. Residues at Poles
For a simple pole a of f, $\operatorname{Res}(f; a) = \lim_{z\to a}(z-a)f(z)$. For a pole of order m,

$$\operatorname{Res}(f; a) = \frac{1}{(m-1)!} \lim_{z\to a} \frac{\mathrm{d}^{m-1}}{\mathrm{d}z^{m-1}}\big((z-a)^m f(z)\big).$$

If $g(a) = 0$, $g'(a) \neq 0$, and h is holomorphic near a, then $\operatorname{Res}(h/g; a) = h(a)/g'(a)$.

Theorem 5.108. Residue Theorem
If f is meromorphic near the closure of a bounded domain D with positively oriented, piecewise smooth simple boundary, and has no pole on ∂D, then

$$\oint_{\partial D} f(z)\,\mathrm{d}z = 2\pi i \sum_{a\in D} \operatorname{Res}(f;a),$$

where the sum runs over the poles in D.

Theorem 5.109. Rouché's Theorem
If f, g are holomorphic in a neighborhood of the closure of a bounded domain D with piecewise smooth simple boundary and $|g(z)| < |f(z)|$ on ∂D, then f and $f+g$ have the same number of zeros in D, counted with multiplicity.

Theorem 5.110. Argument Principle
Let D be a bounded domain with positively oriented piecewise smooth simple closed boundary. If f is meromorphic in a neighborhood of $\overline{D}$, with no zeros or poles on ∂D, then

$$\frac{1}{2\pi i}\oint_{\partial D} \frac{f'(z)}{f(z)}\,\mathrm{d}z = N - P,$$

where N and P count zeros and poles in D with multiplicity.

Definition 5.111. Möbius Transformations
A Möbius transformation of the extended complex plane is

$$T(z) = \frac{az+b}{cz+d}, \qquad ad - bc \neq 0.$$

It maps generalized circles (circles or lines in $\mathbb{C}$, interpreted on the Riemann sphere) to generalized circles and preserves cross ratios.

Theorem 5.112. Fundamental Theorem of Algebra via Complex Analysis
Every nonconstant polynomial with complex coefficients has a complex root. Equivalently, the field $\mathbb{C}$ is algebraically closed.

Definition 5.113. Fourier Coefficients and Partial Sums
For a 2π-periodic integrable complex-valued function f, its complex Fourier coefficients and symmetric partial sums are

$$\widehat{f}(n) = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x)e^{-inx}\,\mathrm{d}x, \qquad S_N f(x) = \sum_{n=-N}^{N} \widehat{f}(n)e^{inx}.$$

The Fejèr means are $\sigma_N f = (N+1)^{-1}\sum_{k=0}^{N} S_k f$.

Theorem 5.114. Fejèr's Theorem
If f is continuous and 2π-periodic, then its Fejèr means $\sigma_N f$ converge uniformly to f on $\mathbb{R}$.

Theorem 5.115. Dirichlet–Jordan Theorem
If a 2π-periodic function f has bounded variation on $[-\pi,\pi]$, its Fourier series converges at every x to

$$\frac{f(x^-)+f(x^+)}{2}.$$

In particular, the Fourier series converges to $f(x)$ at continuity points.

Formula 5.116. Fourier Orthogonality and Parseval's Identity
For integers m, n,

$$\frac{1}{2\pi}\int_{-\pi}^{\pi} e^{i(m-n)x}\,\mathrm{d}x = \begin{cases}1, & m=n,\\ 0, & m\neq n.\end{cases}$$

If $f \in L^2([-\pi,\pi])$ and $\widehat{f}(n) = (2\pi)^{-1}\int_{-\pi}^{\pi} f(x)e^{-inx}\,\mathrm{d}x$, then

$$\frac{1}{2\pi}\int_{-\pi}^{\pi} |f(x)|^2\,\mathrm{d}x = \sum_{n\in\mathbb{Z}} |\widehat{f}(n)|^2.$$

Theorem 5.117. Riemann–Lebesgue Lemma
If $f \in L^1([a,b])$, then

$$\lim_{|t|\to\infty}\int_a^b f(x)e^{itx}\,\mathrm{d}x = 0.$$

In particular, the Fourier coefficients $\widehat{f}(n)$ of any integrable 2π-periodic function satisfy $\widehat{f}(n)\to 0$ as $|n|\to\infty$.

Theorem 5.118. Gronwall's Inequality, Integral Form
Suppose $u\colon [0,T]\to[0,\infty)$ is continuous and $u(t)\le A + B\int_0^t u(s)\,\mathrm{d}s$ for constants $A, B \ge 0$. Then

$$u(t) \le Ae^{Bt} \qquad (0\le t\le T).$$

Definition 5.119. Ordinary Differential Equation and Initial Value Problem
An ordinary differential equation (ODE) of order m relates an unknown function $y(x)$ to its derivatives through order m. An initial value problem (IVP) prescribes at a point x_0 the values

$$y(x_0),\quad y'(x_0),\quad \ldots,\quad y^{(m-1)}(x_0).$$

Formula 5.120. Separable First-Order Differential Equations
If $y' = f(x)g(y)$ and $g(y)\neq 0$ along a solution, then

$$\int \frac{\mathrm{d}y}{g(y)} = \int f(x)\,\mathrm{d}x + C.$$

Constant solutions at zeros of g must be checked separately and need not arise from division by $g(y)$.

Theorem 5.121. Existence and Uniqueness for First-Order ODEs
For $y' = F(x,y)$ with F continuous near (x_0,y_0) and locally Lipschitz in y there, the IVP $y(x_0)=y_0$ has a unique local solution. In particular, continuity of both F and $\partial F/\partial y$ on a neighborhood is sufficient.

Formula 5.122. First-Order Linear Differential Equations
For continuous p, q on an interval, every solution of $y' + p(x)y = q(x)$ satisfies

$$y(x) = e^{-P(x)}\left(C + \int_{x_0}^{x} e^{P(t)}q(t)\,\mathrm{d}t\right), \qquad P(x) = \int_{x_0}^{x} p(t)\,\mathrm{d}t.$$

Formula 5.123. Second-Order Linear ODEs with Constant Coefficients
Consider $ay'' + by' + cy = 0$ with real constants $a\neq 0, b, c$, and let r_1, r_2 be the roots of $ar^2 + br + c = 0$. If the roots are distinct and real, the real solutions are

$$y(x) = C_1e^{r_1x} + C_2e^{r_2x}.$$

If there is a repeated real root r, they are $(C_1 + C_2x)e^{rx}$. For complex conjugate roots $\alpha \pm i\beta$ with $\beta > 0$, they are

$$y(x) = e^{\alpha x}(C_1\cos\beta x + C_2\sin\beta x).$$

For a nonhomogeneous equation $ay'' + by' + cy = h(x)$, every solution is a particular solution plus a solution of the homogeneous equation.

Theorem 5.124. Variation of Parameters for Second-Order ODEs
For $y'' + p(x)y' + q(x)y = r(x)$ on an interval, with continuous p, q, r, let y_1, y_2 be linearly independent solutions of the homogeneous equation, and $W = y_1y_2' - y_1'y_2$. Then $W \neq 0$ throughout the interval and a particular solution is

$$y_p(x) = -y_1(x) \int^x \frac{y_2(t)r(t)}{W(t)} \, \mathrm{d}t + y_2(x) \int^x \frac{y_1(t)r(t)}{W(t)} \, \mathrm{d}t.$$

Theorem 5.125. Wronskian and Abel's Identity
Let y_1, y_2 solve $y'' + p(x)y' + q(x)y = 0$ on an interval with continuous p, q. Their Wronskian

$$W(x) = y_1(x)y_2'(x) - y_1'(x)y_2(x)$$

satisfies $W(x) = W(x_0)\exp(-\int_{x_0}^x p(t) \, \mathrm{d}t)$. The solutions are linearly independent if and only if $W(x_0) \neq 0$.

Chapter 6

Linear Algebra

Definition 6.1. Vector Space and Subspace
A vector space V over a field F is an abelian group under addition equipped with scalar multiplication satisfying the vector-space axioms. A subset $U \subseteq V$ is a subspace when it contains $\mathbf{0}$ and is closed under addition and scalar multiplication.

Definition 6.2. Linear Combinations, Span, and Independence
For vectors $\mathbf{v}_1, \dots, \mathbf{v}_k \in V$, their span is

$$\operatorname{span}(\mathbf{v}_1, \dots, \mathbf{v}_k) = \left\{ \sum_{i=1}^{k} c_i \mathbf{v}_i \;\middle|\; c_i \in F \right\}.$$

They are linearly independent if $\sum_i c_i \mathbf{v}_i = \mathbf{0}$ implies $c_1 = \cdots = c_k = 0$.

Definition 6.3. Basis and Dimension
A basis of V is a linearly independent set spanning V. If V has a finite basis, all bases have the same number of vectors, called $\dim V$. By convention, $\dim\{\mathbf{0}\} = 0$.

Theorem 6.4. Basis Extension Theorem
Every linearly independent list in a finite-dimensional vector space extends to a basis, and every finite spanning list contains a basis. Consequently, any $n+1$ vectors in an n-dimensional space are linearly dependent.

Theorem 6.5. Dimension Formula for Subspaces
For subspaces U, W of a finite-dimensional vector space,

$$\dim(U + W) = \dim U + \dim W - \dim(U \cap W).$$

Their sum is direct exactly when $U \cap W = \{\mathbf{0}\}$.

Definition 6.6. Direct Sum and Quotient Space
For subspaces $U, W \subseteq V$, the sum $U + W$ is a direct sum $U \oplus W$ if every element has a unique expression $\mathbf{u} + \mathbf{w}$. For $U \subseteq V$, the quotient vector space V/U consists of cosets $\mathbf{v} + U$; if V is finite-dimensional, $\dim(V/U) = \dim V - \dim U$.

Definition 6.7. Linear Map, Kernel, and Image
A map $T\colon V \to W$ between vector spaces over F is linear if $T(a\mathbf{u} + b\mathbf{v}) = aT(\mathbf{u}) + bT(\mathbf{v})$. Its kernel and image are

$$\ker T = \{\mathbf{v} \in V \mid T(\mathbf{v}) = \mathbf{0}\}, \qquad \operatorname{im} T = \{T(\mathbf{v}) \mid \mathbf{v} \in V\}.$$

Theorem 6.8. Rank–Nullity Theorem
For a linear map $T\colon V \to W$ with finite-dimensional domain,

$$\dim V = \dim \ker T + \dim \operatorname{im} T.$$

In particular, a linear endomorphism of a finite-dimensional space is injective if and only if it is surjective.

Theorem 6.9. First Isomorphism Theorem for Vector Spaces
If $T\colon V \to W$ is linear, then $V/\ker T \cong \operatorname{im} T$ as vector spaces.

Definition 6.10. Matrix Representation of a Linear Map
For ordered bases $\mathcal{B}$ of V and $\mathcal{C}$ of W, the matrix $[T]_{\mathcal{C}\leftarrow\mathcal{B}}$ of $T\colon V \to W$ is characterized by

$$[T(\mathbf{v})]_{\mathcal{C}} = [T]_{\mathcal{C}\leftarrow\mathcal{B}}[\mathbf{v}]_{\mathcal{B}}.$$

Columns represent images of the basis vectors of $\mathcal{B}$ in coordinates relative to $\mathcal{C}$.

Theorem 6.11. Similarity and Change of Basis
If $\mathbf{A}$ and $\mathbf{B}$ represent a linear operator on the same finite-dimensional space in two bases, then

$$\mathbf{B} = \mathbf{P}^{-1}\mathbf{A}\mathbf{P}$$

for an invertible change-of-coordinates matrix $\mathbf{P}$. Similar matrices have the same characteristic polynomial, trace, determinant, and eigenvalues with multiplicities.

Definition 6.12. Elementary Row Operations and Echelon Forms
Elementary row operations interchange two rows, multiply a row by a nonzero scalar, or add a multiple of one row to another. A row-echelon matrix has all zero rows at the bottom and successive leading entries strictly to the right. In reduced row-echelon form (RREF), every leading entry is 1 and is the only nonzero entry in its column. Every matrix over a field has a unique RREF under elementary row operations.

Definition 6.13. Rank, Column Space, and Null Space
For $\mathbf{A} \in F^{m\times n}$, let $\operatorname{Col}(\mathbf{A}) \subseteq F^m$ be the column space and

$$\operatorname{Null}(\mathbf{A}) = \{\mathbf{x} \in F^n \mid \mathbf{A}\mathbf{x} = \mathbf{0}\}.$$

The rank $\operatorname{rk}(\mathbf{A})$ is $\dim \operatorname{Col}(\mathbf{A})$. The nullity is $\dim \operatorname{Null}(\mathbf{A})$, and $\operatorname{rk}(\mathbf{A}) + \dim \operatorname{Null}(\mathbf{A}) = n$.

Theorem 6.14. Row Rank Equals Column Rank
For every matrix over a field, the maximum number of linearly independent rows equals the maximum number of linearly independent columns. Both equal $\operatorname{rk}(\mathbf{A})$.

Theorem 6.15. Rank Factorization
If $\mathbf{A} \in F^{m\times n}$ has rank r, there exist matrices $\mathbf{B} \in F^{m\times r}$ and $\mathbf{C} \in F^{r\times n}$, each of rank r, such that

$$\mathbf{A} = \mathbf{B}\mathbf{C}.$$

Conversely, if $\mathbf{B}, \mathbf{C}$ have these sizes and both have rank r, then $\mathbf{B}\mathbf{C}$ has rank r.

Theorem 6.16. Smith Normal Form
For every integer matrix $\mathbf{A} \in \mathbb{Z}^{m\times n}$ of rank r, there exist unimodular matrices $\mathbf{U} \in \mathrm{GL}_m(\mathbb{Z})$ and $\mathbf{V} \in \mathrm{GL}_n(\mathbb{Z})$ such that

$$\mathbf{U}\mathbf{A}\mathbf{V} = \operatorname{diag}(d_1, \dots, d_r, 0, \dots, 0),$$

where d_i are positive integers with $d_1 \mid d_2 \mid \cdots \mid d_r$. The d_i are uniquely determined and are called the invariant factors.

Theorem 6.17. Ranks of Products and Sums
For compatible matrices over a field,

$$\operatorname{rk}(\mathbf{AB}) \le \min(\operatorname{rk}\mathbf{A}, \operatorname{rk}\mathbf{B}), \qquad \operatorname{rk}(\mathbf{A}+\mathbf{B}) \le \operatorname{rk}\mathbf{A} + \operatorname{rk}\mathbf{B}.$$

If $\mathbf{A} \in F^{m\times n}$ *and* $\mathbf{B} \in F^{n\times p}$*, Sylvester's inequality states*

$$\operatorname{rk}(\mathbf{AB}) \ge \operatorname{rk}\mathbf{A} + \operatorname{rk}\mathbf{B} - n.$$

Theorem 6.18. Sylvester's and Frobenius' Rank Inequalities
If $\mathbf{A} \in F^{m\times n}$ *and* $\mathbf{B} \in F^{n\times p}$*, then*

$$\operatorname{rk}(\mathbf{AB}) \ge \operatorname{rk}(\mathbf{A}) + \operatorname{rk}(\mathbf{B}) - n.$$

For composable matrices $\mathbf{A}, \mathbf{B}, \mathbf{C}$*,*

$$\operatorname{rk}(\mathbf{ABC}) + \operatorname{rk}(\mathbf{B}) \ge \operatorname{rk}(\mathbf{AB}) + \operatorname{rk}(\mathbf{BC}).$$

Theorem 6.19. Rouché–Capelli Criterion
A linear system $\mathbf{Ax} = \mathbf{b}$ *is consistent if and only if*

$$\operatorname{rk}(\mathbf{A}) = \operatorname{rk}\begin{pmatrix}\mathbf{A} & \mathbf{b}\end{pmatrix}.$$

If consistent, its solution set is an affine translate of $\operatorname{Null}(\mathbf{A})$ *and has dimension* $n - \operatorname{rk}(\mathbf{A})$ *when* $\mathbf{A}$ *has* n *columns.*

Theorem 6.20. Invertible Matrix Theorem
For $\mathbf{A} \in F^{n\times n}$*, the following are equivalent:* $\mathbf{A}$ *is invertible;* $\operatorname{rk}(\mathbf{A}) = n$*;* $\ker\mathbf{A} = \{\mathbf{0}\}$*;* $\det\mathbf{A} \neq 0$*; and* 0 *is not an eigenvalue of* $\mathbf{A}$*.*

Formula 6.21. Inverse of a Two-by-Two Matrix
For $ad - bc \neq 0$,

$$\begin{pmatrix}a & b\\ c & d\end{pmatrix}^{-1} = \frac{1}{ad-bc}\begin{pmatrix}d & -b\\ -c & a\end{pmatrix}.$$

Definition 6.22. Determinant
For $\mathbf{A} = (a_{ij}) \in F^{n\times n}$, its determinant is

$$\det\mathbf{A} = \sum_{\sigma\in S_n} \operatorname{sgn}(\sigma) \prod_{i=1}^{n} a_{i,\sigma(i)}.$$

It is the unique alternating multilinear function of the columns normalized by $\det\mathbf{I}_n = 1$.

Theorem 6.23. Determinant Identities
For square $n \times n$ *matrices* $\mathbf{A}, \mathbf{B}$*,*

$$\det(\mathbf{AB}) = \det\mathbf{A}\det\mathbf{B}, \qquad \det(\mathbf{A}^{\mathrm{T}}) = \det\mathbf{A},$$

$$\det(c\mathbf{A}) = c^n \det\mathbf{A}.$$

If $\mathbf{A}$ *is invertible,* $\det(\mathbf{A}^{-1}) = (\det\mathbf{A})^{-1}$*. An upper or lower triangular matrix has determinant equal to the product of its diagonal entries.*

Formula 6.24. Cofactors and Laplace Expansion
Let M_{ij} be the determinant of the matrix obtained by deleting row i and column j, and let $C_{ij} = (-1)^{i+j} M_{ij}$. Then, for every fixed row i and column j,

$$\det \mathbf{A} = \sum_{k=1}^{n} a_{ik} C_{ik} = \sum_{k=1}^{n} a_{kj} C_{kj}.$$

Theorem 6.25. Cramer's Rule
If $\mathbf{A} \in F^{n\times n}$ is invertible and $\mathbf{A}\mathbf{x} = \mathbf{b}$, then

$$x_i = \frac{\det \mathbf{A}_i}{\det \mathbf{A}},$$

where $\mathbf{A}_i$ is obtained by replacing column i of $\mathbf{A}$ by $\mathbf{b}$.

Theorem 6.26. Cauchy–Binet Formula
For $\mathbf{A} \in F^{m\times n}$ and $\mathbf{B} \in F^{n\times m}$ with $m \le n$,

$$\det(\mathbf{A}\mathbf{B}) = \sum_{\substack{S \subseteq [n] \\ |S| = m}} \det(\mathbf{A}_S) \det(\mathbf{B}^S).$$

Here $\mathbf{A}_S$ consists of columns indexed by S in increasing order, and $\mathbf{B}^S$ consists of the corresponding rows. More generally, every $k \times k$ minor of a product is the sum of products of matching $k \times k$ minors.

Formula 6.27. Adjugate Identity
For any square matrix $\mathbf{A}$ over a commutative ring,

$$\mathbf{A}\,\operatorname{adj}(\mathbf{A}) = \operatorname{adj}(\mathbf{A})\mathbf{A} = (\det \mathbf{A})\mathbf{I}.$$

If $\det \mathbf{A}$ is invertible, $\mathbf{A}^{-1} = (\det \mathbf{A})^{-1} \operatorname{adj}(\mathbf{A})$.

Formula 6.28. Vandermonde Determinant
For scalars $x_1, \dots, x_n$ in a commutative ring,

$$\det \begin{pmatrix} 1 & x_1 & \cdots & x_1^{n-1} \\ 1 & x_2 & \cdots & x_2^{n-1} \\ \vdots & \vdots & & \vdots \\ 1 & x_n & \cdots & x_n^{n-1} \end{pmatrix} = \prod_{1 \le i < j \le n} (x_j - x_i).$$

Formula 6.29. Cauchy Determinant
If $x_i + y_j \neq 0$ for all $1 \le i, j \le n$, then

$$\det\left(\frac{1}{x_i + y_j}\right)_{i,j=1}^{n} = \frac{\displaystyle\prod_{1 \le i < j \le n} (x_j - x_i)(y_j - y_i)}{\displaystyle\prod_{i=1}^{n}\prod_{j=1}^{n} (x_i + y_j)}.$$

Theorem 6.30. Matrix Determinant Lemma
If $\mathbf{A} \in F^{n\times n}$ is invertible and $\mathbf{u}, \mathbf{v} \in F^n$, then

$$\det(\mathbf{A} + \mathbf{u}\mathbf{v}^{\mathrm{T}}) = \det \mathbf{A}\,(1 + \mathbf{v}^{\mathrm{T}}\mathbf{A}^{-1}\mathbf{u}).$$

Theorem 6.31. Sylvester's Determinant Identity
For $\mathbf{A} \in F^{m\times n}$ and $\mathbf{B} \in F^{n\times m}$,

$$\det(\mathbf{I}_m + \mathbf{AB}) = \det(\mathbf{I}_n + \mathbf{BA}).$$

Formula 6.32. Block Triangular Determinants
If $\mathbf{A}, \mathbf{D}$ are square, then

$$\det \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{0} & \mathbf{D} \end{pmatrix} = \det \mathbf{A} \det \mathbf{D}.$$

The analogous identity holds for lower block triangular matrices.

Formula 6.33. Schur Complement Determinant Formula
If $\mathbf{A}$ is invertible, then

$$\det \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{C} & \mathbf{D} \end{pmatrix} = \det \mathbf{A} \det(\mathbf{D} - \mathbf{C}\mathbf{A}^{-1}\mathbf{B}).$$

If $\mathbf{D}$ is invertible, the symmetric counterpart uses $\mathbf{A} - \mathbf{B}\mathbf{D}^{-1}\mathbf{C}$.

Formula 6.34. Sherman–Morrison Formula
If $\mathbf{A}$ is invertible and $1 + \mathbf{v}^{\mathrm{T}}\mathbf{A}^{-1}\mathbf{u} \neq 0$, then

$$(\mathbf{A} + \mathbf{u}\mathbf{v}^{\mathrm{T}})^{-1} = \mathbf{A}^{-1} - \frac{\mathbf{A}^{-1}\mathbf{u}\mathbf{v}^{\mathrm{T}}\mathbf{A}^{-1}}{1 + \mathbf{v}^{\mathrm{T}}\mathbf{A}^{-1}\mathbf{u}}.$$

Definition 6.35. Eigenvalues, Eigenvectors, and Characteristic Polynomial
An eigenvalue $\lambda \in F$ of $\mathbf{A} \in F^{n\times n}$ satisfies $\mathbf{A}\mathbf{v} = \lambda\mathbf{v}$ for some nonzero $\mathbf{v} \in F^n$. The eigenspace is $E_\lambda(\mathbf{A}) = \ker(\mathbf{A} - \lambda\mathbf{I})$. The characteristic polynomial is

$$\chi_{\mathbf{A}}(t) = \det(t\mathbf{I} - \mathbf{A}).$$

Over an algebraic closure, its roots are all eigenvalues counted with algebraic multiplicity.

Theorem 6.36. Trace, Determinant, and Eigenvalues
If $\lambda_1, \dots, \lambda_n$ are the eigenvalues of $\mathbf{A} \in \mathbb{C}^{n\times n}$ counted with multiplicity, then

$$\operatorname{tr} \mathbf{A} = \sum_{i=1}^{n} \lambda_i, \qquad \det \mathbf{A} = \prod_{i=1}^{n} \lambda_i, \qquad \operatorname{tr}(\mathbf{A}^k) = \sum_{i=1}^{n} \lambda_i^k \quad (k \in \mathbb{N}).$$

Theorem 6.37. Cyclic Property of the Trace
For matrices $\mathbf{A} \in F^{m\times n}$ and $\mathbf{B} \in F^{n\times m}$,

$$\operatorname{tr}(\mathbf{AB}) = \operatorname{tr}(\mathbf{BA}).$$

For a product of square matrices, the trace is invariant under cyclic permutations of factors, but not generally under arbitrary permutations.

Definition 6.38. Diagonalizability
A matrix $\mathbf{A} \in F^{n\times n}$ is diagonalizable over F if there are an invertible $\mathbf{P} \in F^{n\times n}$ and a diagonal $\mathbf{D}$ with $\mathbf{A} = \mathbf{P}\mathbf{D}\mathbf{P}^{-1}$. Equivalently, F^n has a basis consisting of eigenvectors of $\mathbf{A}$.

Definition 6.39. Invariant Subspace
A subspace $U \subseteq V$ is invariant under a linear operator $T\colon V \to V$ if $T(U) \subseteq U$. An eigenspace, a kernel of a polynomial in T, and an image of a polynomial in T are invariant subspaces.

Theorem 6.40. Simultaneous Diagonalization of Commuting Operators
Over an algebraically closed field, a finite family of pairwise commuting diagonalizable linear operators on a finite-dimensional vector space has a common eigenbasis. Equivalently, the matrices are simultaneously diagonalizable. In particular, diagonalizable matrices that commute can be diagonalized by the same change of basis.

Theorem 6.41. Criteria for Diagonalizability
A matrix $\mathbf{A} \in F^{n\times n}$ is diagonalizable over F if and only if its minimal polynomial splits into distinct linear factors over F. If $\mathbf{A}$ has n distinct eigenvalues in F, then it is diagonalizable over F.

Theorem 6.42. Cayley–Hamilton Theorem
Every square matrix satisfies its characteristic polynomial:

$$\chi_{\mathbf{A}}(\mathbf{A}) = \mathbf{0}.$$

Consequently, if $\mathbf{A}$ has size $n \times n$, every power $\mathbf{A}^k$ with $k \geq n$ can be expressed as a linear combination of $\mathbf{I}, \mathbf{A}, \ldots, \mathbf{A}^{n-1}$.

Definition 6.43. Minimal Polynomial
The minimal polynomial $m_{\mathbf{A}}(t)$ of a square matrix over a field F is the unique monic polynomial of least positive degree with $m_{\mathbf{A}}(\mathbf{A}) = \mathbf{0}$. It divides $\chi_{\mathbf{A}}(t)$, and every polynomial p satisfying $p(\mathbf{A}) = \mathbf{0}$ is divisible by $m_{\mathbf{A}}$.

Definition 6.44. Matrix Exponential
For every complex square matrix $\mathbf{A}$, define

$$e^{\mathbf{A}} = \sum_{k=0}^{\infty} \frac{\mathbf{A}^k}{k!}.$$

It satisfies $e^{(s+t)\mathbf{A}} = e^{s\mathbf{A}} e^{t\mathbf{A}}$ and

$$\det(e^{\mathbf{A}}) = e^{\operatorname{tr} \mathbf{A}}.$$

If $\mathbf{AB} = \mathbf{BA}$, then $e^{\mathbf{A}+\mathbf{B}} = e^{\mathbf{A}} e^{\mathbf{B}}$.

Theorem 6.45. Eigenvalues of Products
For $\mathbf{A} \in F^{m\times n}$ and $\mathbf{B} \in F^{n\times m}$, the matrices $\mathbf{AB}$ and $\mathbf{BA}$ have the same nonzero eigenvalues with the same algebraic multiplicities over an algebraic closure of F.

Formula 6.46. Matrix Geometric Series
For every square matrix $\mathbf{A}$ and $n \in \mathbb{N}$,

$$(\mathbf{I} - \mathbf{A}) \sum_{k=0}^{n-1} \mathbf{A}^k = \mathbf{I} - \mathbf{A}^n.$$

If $\mathbf{I} - \mathbf{A}$ is invertible, this sum is $(\mathbf{I} - \mathbf{A}^n)(\mathbf{I} - \mathbf{A})^{-1}$. If its spectral radius $\rho(\mathbf{A}) < 1$ over $\mathbb{C}$, then $\sum_{k=0}^{\infty} \mathbf{A}^k = (\mathbf{I} - \mathbf{A})^{-1}$.

Formula 6.47. Fibonacci Matrix Identity
For $\mathbf{Q} = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$ and $F_0 = 0$, $F_1 = 1$, $F_{n+1} = F_n + F_{n-1}$,

$$\mathbf{Q}^n = \begin{pmatrix} F_{n+1} & F_n \\ F_n & F_{n-1} \end{pmatrix} \qquad (n \geq 1).$$

In particular, $F_{n+1}F_{n-1} - F_n^2 = (-1)^n$.

Definition 6.48. Nilpotent, Idempotent, and Involutory Matrices
A square matrix $\mathbf{A}$ is nilpotent if $\mathbf{A}^k = \mathbf{0}$ for some $k \geq 1$, idempotent if $\mathbf{A}^2 = \mathbf{A}$, and involutory if $\mathbf{A}^2 = \mathbf{I}$. An idempotent matrix is diagonalizable with eigenvalues in $\{0, 1\}$; over a field of characteristic different from 2, an involutory matrix is diagonalizable with eigenvalues in $\{-1, 1\}$.

Proposition 6.49. Trace of a Commutator
For any square matrices $\mathbf{A}, \mathbf{B}$ over a field,

$$\operatorname{tr}(\mathbf{AB} - \mathbf{BA}) = 0.$$

If $\mathbf{A}$ and $\mathbf{B}$ commute, each eigenspace of $\mathbf{A}$ is invariant under $\mathbf{B}$.

Theorem 6.50. Jordan Canonical Form
Every complex square matrix is similar to a direct sum of Jordan blocks $\mathbf{J}_k(\lambda) = \lambda \mathbf{I}_k + \mathbf{N}_k$, where $\mathbf{N}_k$ has ones on its superdiagonal and zero elsewhere. The sizes and eigenvalues of the blocks are unique up to their order.

Definition 6.51. Spectral Radius
For a square matrix $\mathbf{A}$ over $\mathbb{C}$, its spectrum $\sigma(\mathbf{A})$ is the set of eigenvalues. Its spectral radius is

$$\rho(\mathbf{A}) = \max_{\lambda \in \sigma(\mathbf{A})} |\lambda|.$$

Over $\mathbb{R}$, the spectrum and spectral radius are understood after extending scalars to $\mathbb{C}$.

Theorem 6.52. Gershgorin Circle Theorem
Every eigenvalue of $\mathbf{A} = (a_{ij}) \in \mathbb{C}^{n \times n}$ lies in the union of the disks

$$\bigcup_{i=1}^{n} \left\{ z \in \mathbb{C} \;\middle|\; |z - a_{ii}| \leq \sum_{j \neq i} |a_{ij}| \right\}.$$

Theorem 6.53. Perron–Frobenius Theorem, Positive Case
If $\mathbf{A} \in \mathbb{R}^{n \times n}$ has strictly positive entries, its spectral radius $r > 0$ is a simple eigenvalue with a strictly positive eigenvector. Every other eigenvalue λ satisfies $|\lambda| < r$.

Definition 6.54. Inner Product and Norm
An inner product on a complex vector space V is conjugate-linear in its first argument, linear in its second, Hermitian symmetric, and positive definite. Its induced norm is $\|\mathbf{v}\| = \sqrt{\langle \mathbf{v}, \mathbf{v} \rangle}$. On $\mathbb{C}^n$, the standard inner product is $\langle \mathbf{u}, \mathbf{v} \rangle = \mathbf{u}^{\mathrm{H}} \mathbf{v}$.

Definition 6.55. Adjoint and Orthogonal Complement
For a linear operator T on a finite-dimensional real or complex inner product space, its adjoint T^* is defined by

$$\langle T\mathbf{u}, \mathbf{v} \rangle = \langle \mathbf{u}, T^* \mathbf{v} \rangle.$$

In an orthonormal basis, its matrix is $\mathbf{A}^{\mathrm{H}}$ (or $\mathbf{A}^{\mathrm{T}}$ over $\mathbb{R}$). For a subspace U, its orthogonal complement is

$$U^{\perp} = \{ \mathbf{v} \mid \langle \mathbf{u}, \mathbf{v} \rangle = 0 \text{ for every } \mathbf{u} \in U \}.$$

Definition 6.56. Matrix Norms
For $\mathbf{A} \in \mathbb{C}^{m \times n}$, the Frobenius norm is

$$\|\mathbf{A}\|_{\mathrm{F}} = \left(\sum_{i=1}^{m} \sum_{j=1}^{n} |a_{ij}|^2 \right)^{1/2},$$

and the operator norm induced by the Euclidean norm is

$$\|\mathbf{A}\|_2 = \sup_{\|\mathbf{x}\| = 1} \|\mathbf{A}\mathbf{x}\|.$$

A matrix norm is submultiplicative if $\|\mathbf{AB}\| \leq \|\mathbf{A}\| \|\mathbf{B}\|$ whenever the product is defined. Every norm induced by a vector norm is submultiplicative for compatible matrices.

Theorem 6.57. Cauchy–Schwarz Inequality
For vectors $\mathbf{u}, \mathbf{v}$ *in a real or complex inner product space,*

$$|\langle \mathbf{u}, \mathbf{v} \rangle|^2 \le \langle \mathbf{u}, \mathbf{u} \rangle \langle \mathbf{v}, \mathbf{v} \rangle.$$

Equality holds if and only if $\mathbf{u}$ *and* $\mathbf{v}$ *are linearly dependent.*

Theorem 6.58. Triangle and Parallelogram Identities
For vectors in an inner product space,

$$\|\mathbf{u} + \mathbf{v}\| \le \|\mathbf{u}\| + \|\mathbf{v}\|,$$

$$\|\mathbf{u} + \mathbf{v}\|^2 + \|\mathbf{u} - \mathbf{v}\|^2 = 2\|\mathbf{u}\|^2 + 2\|\mathbf{v}\|^2.$$

Formula 6.59. Polarization Identity
For a real inner product space,

$$\langle \mathbf{u}, \mathbf{v} \rangle = \frac{1}{4}(\|\mathbf{u} + \mathbf{v}\|^2 - \|\mathbf{u} - \mathbf{v}\|^2).$$

For a complex inner product linear in the second argument,

$$\langle \mathbf{u}, \mathbf{v} \rangle = \frac{1}{4}(\|\mathbf{u} + \mathbf{v}\|^2 - \|\mathbf{u} - \mathbf{v}\|^2 - i\|\mathbf{u} + i\mathbf{v}\|^2 + i\|\mathbf{u} - i\mathbf{v}\|^2).$$

Theorem 6.60. Gram–Schmidt Orthogonalization
Every linearly independent list $(\mathbf{v}_1, \ldots, \mathbf{v}_k)$ *in an inner product space can be transformed into an orthonormal list spanning the same successive subspaces. Recursively,*

$$\mathbf{w}_j = \mathbf{v}_j - \sum_{i=1}^{j-1} \langle \mathbf{q}_i, \mathbf{v}_j \rangle \mathbf{q}_i, \qquad \mathbf{q}_j = \frac{\mathbf{w}_j}{\|\mathbf{w}_j\|}.$$

Theorem 6.61. Bessel's Inequality and Parseval's Identity
If $\mathbf{q}_1, \ldots, \mathbf{q}_k$ *are orthonormal, then for every* $\mathbf{v}$*,*

$$\sum_{i=1}^{k} |\langle \mathbf{q}_i, \mathbf{v} \rangle|^2 \le \|\mathbf{v}\|^2.$$

If the list is an orthonormal basis of a finite-dimensional inner product space, equality holds.

Formula 6.62. Orthogonal Projection
If $U \subseteq V$ has orthonormal basis $(\mathbf{q}_1, \ldots, \mathbf{q}_k)$, then

$$\operatorname{proj}_U \mathbf{v} = \sum_{i=1}^{k} \langle \mathbf{q}_i, \mathbf{v} \rangle \mathbf{q}_i.$$

For $\mathbf{A} \in \mathbb{R}^{m \times n}$ of full column rank, the orthogonal projector onto $\operatorname{Col}(\mathbf{A})$ is $\mathbf{P} = \mathbf{A}(\mathbf{A}^{\mathrm{T}}\mathbf{A})^{-1}\mathbf{A}^{\mathrm{T}}$; replace T with H over $\mathbb{C}$.

Theorem 6.63. Projection Theorem
For a subspace U *of a finite-dimensional inner product space, every* $\mathbf{v}$ *has a unique decomposition*

$$\mathbf{v} = \mathbf{u} + \mathbf{w}, \qquad \mathbf{u} \in U, \quad \mathbf{w} \in U^{\perp}.$$

The vector $\mathbf{u} = \operatorname{proj}_U \mathbf{v}$ *is the unique point of* U *minimizing the distance to* $\mathbf{v}$*.*

Theorem 6.64. Least-Squares Normal Equations
For $\mathbf{A} \in \mathbb{R}^{m\times n}$ and $\mathbf{b} \in \mathbb{R}^m$, a vector $\mathbf{x}$ minimizes $\|\mathbf{Ax}-\mathbf{b}\|^2$ if and only if

$$\mathbf{A}^{\mathrm{T}}\mathbf{Ax} = \mathbf{A}^{\mathrm{T}}\mathbf{b}.$$

Such minimizers always exist; the minimizing vector is unique exactly when $\mathbf{A}$ has full column rank. Over $\mathbb{C}$, use $\mathbf{A}^{\mathrm{H}}$.

Definition 6.65. Orthogonal and Unitary Matrices
A real square matrix $\mathbf{Q}$ is orthogonal if $\mathbf{Q}^{\mathrm{T}}\mathbf{Q} = \mathbf{I}$. A complex square matrix $\mathbf{U}$ is unitary if $\mathbf{U}^{\mathrm{H}}\mathbf{U} = \mathbf{I}$. These matrices preserve the standard inner product and Euclidean norm, and their eigenvalues have modulus 1.

Definition 6.66. Hermitian and Normal Matrices
A complex square matrix $\mathbf{A}$ is Hermitian if $\mathbf{A}^{\mathrm{H}} = \mathbf{A}$ and normal if $\mathbf{AA}^{\mathrm{H}} = \mathbf{A}^{\mathrm{H}}\mathbf{A}$. Real symmetric matrices are Hermitian; every Hermitian matrix is normal.

Theorem 6.67. Spectral Theorem for Real Symmetric Matrices
Every real symmetric matrix $\mathbf{A}$ can be orthogonally diagonalized:

$$\mathbf{A} = \mathbf{QDQ}^{\mathrm{T}},$$

where $\mathbf{Q}$ is real orthogonal and $\mathbf{D}$ is a real diagonal matrix. Eigenvectors belonging to distinct eigenvalues are orthogonal.

Theorem 6.68. Spectral Theorem for Normal Matrices
A complex square matrix $\mathbf{A}$ is normal if and only if it is unitarily diagonalizable:

$$\mathbf{A} = \mathbf{UDU}^{\mathrm{H}},$$

where $\mathbf{U}$ is unitary and $\mathbf{D}$ is diagonal. For Hermitian $\mathbf{A}$, the diagonal entries are real.

Definition 6.69. Positive Definite and Semidefinite Matrices
A Hermitian matrix $\mathbf{A}$ is positive definite, written $\mathbf{A} \succ 0$, if $\mathbf{x}^{\mathrm{H}}\mathbf{Ax} > 0$ for all nonzero $\mathbf{x}$. It is positive semidefinite, written $\mathbf{A} \succeq 0$, if $\mathbf{x}^{\mathrm{H}}\mathbf{Ax} \geq 0$ for all $\mathbf{x}$. Such matrices are characterized by all eigenvalues being positive or nonnegative, respectively.

Proposition 6.70. Principal Minors of Positive Semidefinite Matrices
A Hermitian matrix $\mathbf{A}$ is positive semidefinite if and only if every principal minor of $\mathbf{A}$ is nonnegative.

Definition 6.71. Bilinear and Quadratic Forms
A bilinear form on a real vector space is a map $B\colon V \times V \to \mathbb{R}$ that is linear in each argument. A real quadratic form on $\mathbb{R}^n$ has the form

$$q(\mathbf{x}) = \mathbf{x}^{\mathrm{T}}\mathbf{Ax}$$

for a unique symmetric matrix $\mathbf{A}$. Under the change of variables $\mathbf{x} = \mathbf{Py}$, the representing matrix changes by congruence to $\mathbf{P}^{\mathrm{T}}\mathbf{AP}$.

Theorem 6.72. Sylvester's Law of Inertia
Every real symmetric matrix is congruent to a diagonal matrix with entries in $\{-1, 0, 1\}$. The numbers of positive, negative, and zero diagonal entries are independent of the congruence transformation; equivalently, they are the numbers of positive, negative, and zero eigenvalues counted with multiplicity.

Theorem 6.73. Schur Product Theorem
If $\mathbf{A}, \mathbf{B} \in \mathbb{C}^{n\times n}$ are Hermitian positive semidefinite, then their Hadamard (entrywise) product

$$\mathbf{A} \circ \mathbf{B} = (a_{ij} b_{ij})$$

is Hermitian positive semidefinite.

Theorem 6.74. Sylvester's Criterion
A real symmetric or complex Hermitian matrix is positive definite if and only if all its leading principal minors are positive. It is positive semidefinite if and only if all its principal minors are nonnegative.

Theorem 6.75. Hadamard's Determinant Inequality
For vectors $\mathbf{v}_1, \dots, \mathbf{v}_n \in \mathbb{C}^n$ as columns of $\mathbf{A}$,

$$|\det \mathbf{A}| \le \prod_{i=1}^{n} \|\mathbf{v}_i\|.$$

Equality holds precisely when the nonzero column vectors are pairwise orthogonal; if any column is zero, equality holds automatically.

Theorem 6.76. Gram Determinant Inequality
If $\mathbf{v}_1, \dots, \mathbf{v}_k$ lie in a complex inner product space, the Gram matrix $\mathbf{G} = (\langle \mathbf{v}_i, \mathbf{v}_j \rangle)_{i,j}$ is positive semidefinite and

$$0 \le \det \mathbf{G} \le \prod_{i=1}^{k} \|\mathbf{v}_i\|^2.$$

It has positive determinant exactly when the vectors are linearly independent; if all are nonzero, equality in the upper bound holds exactly when they are pairwise orthogonal.

Theorem 6.77. Courant–Fischer Min–Max Principle
Let $\mathbf{A} \in \mathbb{C}^{n\times n}$ be Hermitian, with real eigenvalues $\lambda_1 \ge \cdots \ge \lambda_n$. Then for $1 \le k \le n$,

$$\lambda_k = \max_{\dim U = k} \min_{\substack{\mathbf{x} \in U \\ \|\mathbf{x}\|=1}} \mathbf{x}^{\mathrm{H}} \mathbf{A} \mathbf{x} = \min_{\dim W = n-k+1} \max_{\substack{\mathbf{x} \in W \\ \|\mathbf{x}\|=1}} \mathbf{x}^{\mathrm{H}} \mathbf{A} \mathbf{x}.$$

Theorem 6.78. Rayleigh Quotient Bounds
If $\mathbf{A}$ is Hermitian with smallest and largest eigenvalues $\lambda_{\min}$ and $\lambda_{\max}$, then for every nonzero $\mathbf{x}$,

$$\lambda_{\min} \le \frac{\mathbf{x}^{\mathrm{H}} \mathbf{A} \mathbf{x}}{\mathbf{x}^{\mathrm{H}} \mathbf{x}} \le \lambda_{\max}.$$

The endpoints are attained precisely by eigenvectors from the corresponding eigenspaces.

Theorem 6.79. Weyl's Eigenvalue Inequalities, Simple Form
For Hermitian matrices $\mathbf{A}, \mathbf{B} \in \mathbb{C}^{n\times n}$, with eigenvalues in decreasing order,

$$\lambda_k(\mathbf{A}) + \lambda_n(\mathbf{B}) \le \lambda_k(\mathbf{A} + \mathbf{B}) \le \lambda_k(\mathbf{A}) + \lambda_1(\mathbf{B}) \qquad (1 \le k \le n).$$

Theorem 6.80. Cauchy Eigenvalue Interlacing
Let $\mathbf{A} \in \mathbb{C}^{n\times n}$ be Hermitian with eigenvalues $\lambda_1 \ge \cdots \ge \lambda_n$, and let $\mathbf{B}$ be any $(n-1) \times (n-1)$ principal submatrix with eigenvalues $\mu_1 \ge \cdots \ge \mu_{n-1}$. Then

$$\lambda_i \ge \mu_i \ge \lambda_{i+1} \qquad (1 \le i \le n-1).$$

Theorem 6.81. Spectral Radius Bound
For every matrix norm subordinate to a vector norm and every square matrix $\mathbf{A}$*,*

$$\rho(\mathbf{A}) \leq \|\mathbf{A}\|.$$

For Hermitian matrices, the Euclidean operator norm equals $\max_i |\lambda_i|$*.*

Theorem 6.82. Neumann Series for Matrices
If $\mathbf{A}$ *is a square complex matrix with* $\rho(\mathbf{A}) < 1$*, then*

$$(\mathbf{I} - \mathbf{A})^{-1} = \sum_{k=0}^{\infty} \mathbf{A}^k,$$

with convergence in every matrix norm. In particular, the condition $\|\mathbf{A}\| < 1$ *for any submultiplicative matrix norm is sufficient.*

Theorem 6.83. Gelfand's Formula
For a complex square matrix and any submultiplicative matrix norm,

$$\rho(\mathbf{A}) = \lim_{k \to \infty} \|\mathbf{A}^k\|^{1/k}.$$

Definition 6.84. Singular Values
For $\mathbf{A} \in \mathbb{C}^{m \times n}$, the singular values $\sigma_1 \geq \cdots \geq \sigma_{\min(m,n)} \geq 0$ are the square roots of the eigenvalues of $\mathbf{A}^{\mathrm{H}}\mathbf{A}$, with zeros appended or omitted according to the smaller dimension. The nonzero singular values are independent of this padding convention. The largest singular value equals the Euclidean operator norm: $\sigma_1(\mathbf{A}) = \|\mathbf{A}\|_2$.

Theorem 6.85. Singular Value Decomposition
Every $\mathbf{A} \in \mathbb{C}^{m \times n}$ *admits a singular value decomposition*

$$\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^{\mathrm{H}},$$

where $\mathbf{U} \in \mathbb{C}^{m \times m}$ *and* $\mathbf{V} \in \mathbb{C}^{n \times n}$ *are unitary and* $\mathbf{\Sigma} \in \mathbb{R}^{m \times n}$ *is diagonal rectangular with nonnegative entries. The number of positive singular values equals* $\operatorname{rk} \mathbf{A}$*.*

Theorem 6.86. Eckart–Young–Mirsky Theorem
If $\mathbf{A} \in \mathbb{C}^{m \times n}$ *has singular values* $\sigma_1 \geq \cdots \geq \sigma_r > 0$ *and* $0 \leq k < r$*, then the truncated SVD* $\mathbf{A}_k$ *minimizes both* $\|\mathbf{A} - \mathbf{B}\|_{\mathrm{F}}$ *and* $\|\mathbf{A} - \mathbf{B}\|_2$ *over matrices* $\mathbf{B}$ *of rank at most* k*. The minimum errors are*

$$\left(\sum_{j=k+1}^{r} \sigma_j^2 \right)^{1/2} \quad \text{and} \quad \sigma_{k+1},$$

respectively.

Definition 6.87. Moore–Penrose Pseudoinverse
For $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^{\mathrm{H}}$ in a singular value decomposition, define

$$\mathbf{A}^{\dagger} = \mathbf{V}\mathbf{\Sigma}^{\dagger}\mathbf{U}^{\mathrm{H}},$$

where the nonzero diagonal entries of $\mathbf{\Sigma}$ are reciprocated and zeros remain zero. Then $\mathbf{A}\mathbf{A}^{\dagger}$ and $\mathbf{A}^{\dagger}\mathbf{A}$ are orthogonal projectors onto the column space and the row space (over $\mathbb{C}$, with the standard Hermitian inner product), respectively.

Theorem 6.88. QR Decomposition
Every $\mathbf{A} \in \mathbb{C}^{m\times n}$ with $m \geq n$ and linearly independent columns admits a reduced QR decomposition $\mathbf{A} = \mathbf{QR}$, where $\mathbf{Q} \in \mathbb{C}^{m\times n}$ has orthonormal columns and $\mathbf{R} \in \mathbb{C}^{n\times n}$ is invertible upper triangular. A full QR decomposition uses a square unitary matrix and a rectangular upper-trapezoidal factor.

Theorem 6.89. LU and PLU Decompositions
A square matrix $\mathbf{A}$ admits an LU factorization $\mathbf{A} = \mathbf{LU}$ whenever Gaussian elimination can be carried out without row exchanges, where $\mathbf{L}$ is unit lower triangular and $\mathbf{U}$ is upper triangular. For every square matrix over a field, some permutation matrix $\mathbf{P}$ yields $\mathbf{PA} = \mathbf{LU}$.

Theorem 6.90. Cholesky Decomposition
Every Hermitian positive definite matrix $\mathbf{A}$ has a unique factorization

$$\mathbf{A} = \mathbf{L}\mathbf{L}^{\mathrm{H}},$$

where $\mathbf{L}$ is lower triangular with positive real diagonal entries.

Theorem 6.91. Schur Triangularization
Every complex square matrix $\mathbf{A}$ is unitarily similar to an upper triangular matrix:

$$\mathbf{A} = \mathbf{Q}\mathbf{T}\mathbf{Q}^{\mathrm{H}},$$

where $\mathbf{Q}$ is unitary and the diagonal entries of $\mathbf{T}$ are the eigenvalues of $\mathbf{A}$, with multiplicity.

Theorem 6.92. Spectral Decomposition of a Normal Matrix
If $\mathbf{A}$ is normal with distinct eigenvalues $\lambda_1, \ldots, \lambda_s$, then

$$\mathbf{A} = \sum_{j=1}^{s} \lambda_j \mathbf{P}_j, \qquad \mathbf{I} = \sum_{j=1}^{s} \mathbf{P}_j,$$

where the $\mathbf{P}_j$ are pairwise orthogonal Hermitian projections onto the eigenspaces and $\mathbf{P}_i\mathbf{P}_j = \mathbf{0}$ for $i \neq j$.

Formula 6.93. Trace of Powers and Newton's Identities
Let $p_k = \operatorname{tr}(\mathbf{A}^k)$ and let e_j be the elementary symmetric polynomials in the eigenvalues of $\mathbf{A} \in \mathbb{C}^{n\times n}$. For $1 \leq k \leq n$,

$$p_k - e_1 p_{k-1} + e_2 p_{k-2} - \cdots + (-1)^k k e_k = 0.$$

Thus the characteristic polynomial coefficients are determined by $\operatorname{tr}\mathbf{A}, \ldots, \operatorname{tr}(\mathbf{A}^n)$.

Formula 6.94. Circulant Matrix Eigenvalues
For the $n \times n$ circulant matrix with first row $(c_0, c_1, \ldots, c_{n-1})$ and successive rows obtained by cyclic right shifts, let $\omega = e^{2\pi i/n}$. Its eigenvalues, counted with multiplicity, are

$$\lambda_k = \sum_{j=0}^{n-1} c_j \omega^{kj} \qquad (0 \leq k < n).$$

The eigenvectors can be taken from the discrete Fourier basis, with orientation chosen consistently with the row-shift convention.

Formula 6.95. Determinant of a Tridiagonal Matrix
For the $n\times n$ tridiagonal matrix with diagonal entries a_k, superdiagonal entries b_k, and subdiagonal entries c_k, let $D_0 = 1$, $D_1 = a_1$. Then

$$D_k = a_k D_{k-1} - b_{k-1}c_{k-1}D_{k-2} \qquad (2 \leq k \leq n),$$

and D_n is its determinant.

Formula 6.96. Kronecker Product
For matrices of compatible sizes, the Kronecker product satisfies

$$(\mathbf{A}\otimes\mathbf{B})(\mathbf{C}\otimes\mathbf{D})=(\mathbf{AC})\otimes(\mathbf{BD}),\qquad (\mathbf{A}\otimes\mathbf{B})^{\mathrm{T}}=\mathbf{A}^{\mathrm{T}}\otimes\mathbf{B}^{\mathrm{T}}.$$

If $\mathbf{A}\in F^{m\times m}$ and $\mathbf{B}\in F^{n\times n}$,

$$\det(\mathbf{A}\otimes\mathbf{B})=(\det\mathbf{A})^{n}(\det\mathbf{B})^{m}.$$

Theorem 6.97. Linear Recurrences and Companion Matrices
A scalar sequence satisfying a monic constant-coefficient recurrence of order d can be represented by iterating a $d\times d$ companion matrix. Consequently, if the characteristic polynomial splits into distinct roots $\lambda_1,\dots,\lambda_d$, then every solution is a linear combination of λ_i^n. A root of multiplicity m contributes terms $n^j\lambda^n$ for $0\le j<m$ when $\lambda\neq 0$.

Theorem 6.98. Determinantal Characterization of Linear Independence
Vectors $\mathbf{v}_1,\dots,\mathbf{v}_n\in F^n$ are linearly independent if and only if the matrix with these vectors as columns has nonzero determinant.

Part II

Problem-Solving Strategies